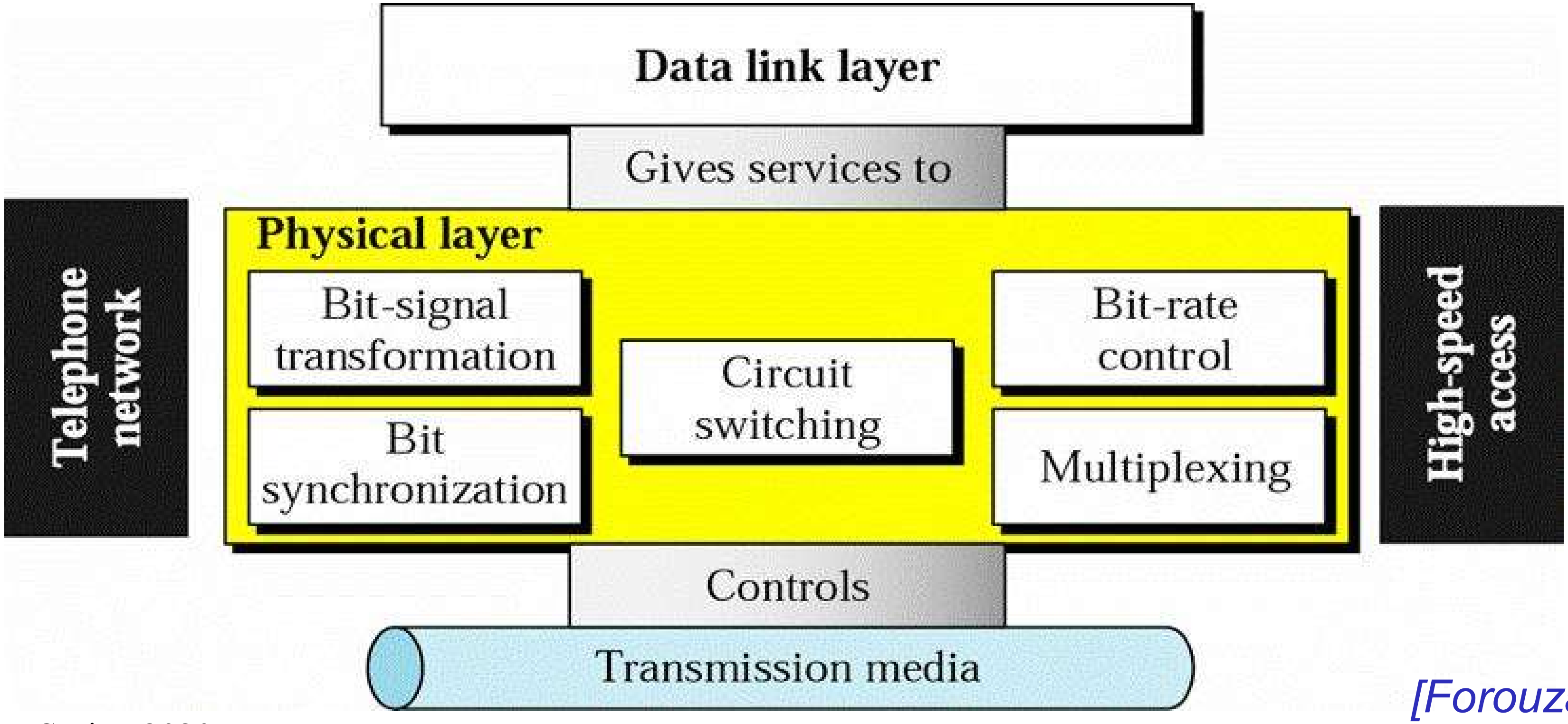
Chapter 2
The Physical Layer

Outline

- *Position and Functions of Physical Layer*
- Basic concepts on data communications
- Theoretical basis for data communication
- Transmission medium
- Digital modulation and encoding
- Multiplexing
- Physical network: PSTN
- Physical network: Cable TV system
- Circuit switching vs. packet switching
- Physical layer protocol: 10BaseT

Position and Function of Physical Layer

- The lowest layer in our protocol model.
- It defines the electrical, timing and other interfaces by which bits are sent as signals over channels.



[Forouzan]

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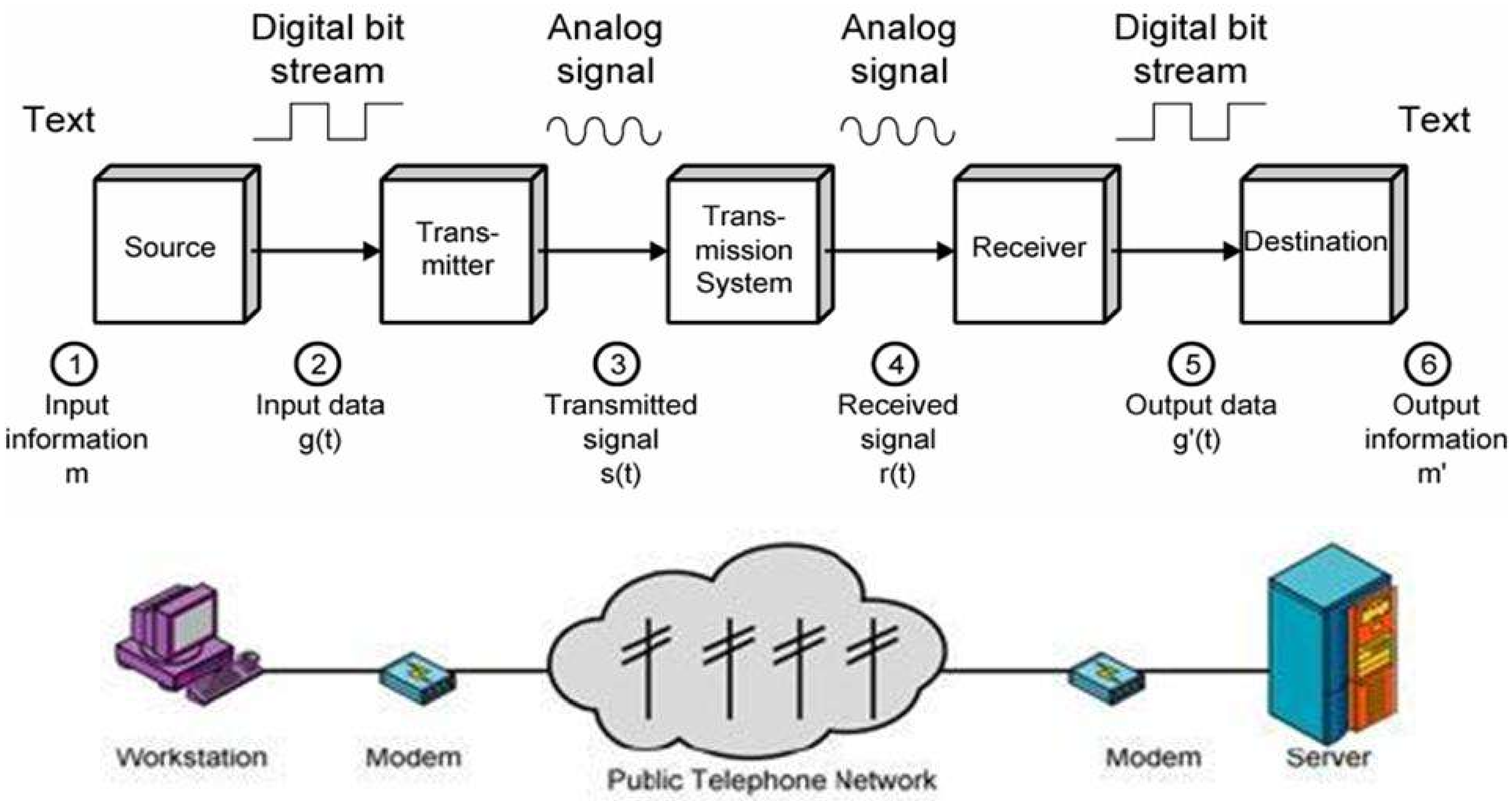
Simplified Data Communications Model



(b) Example

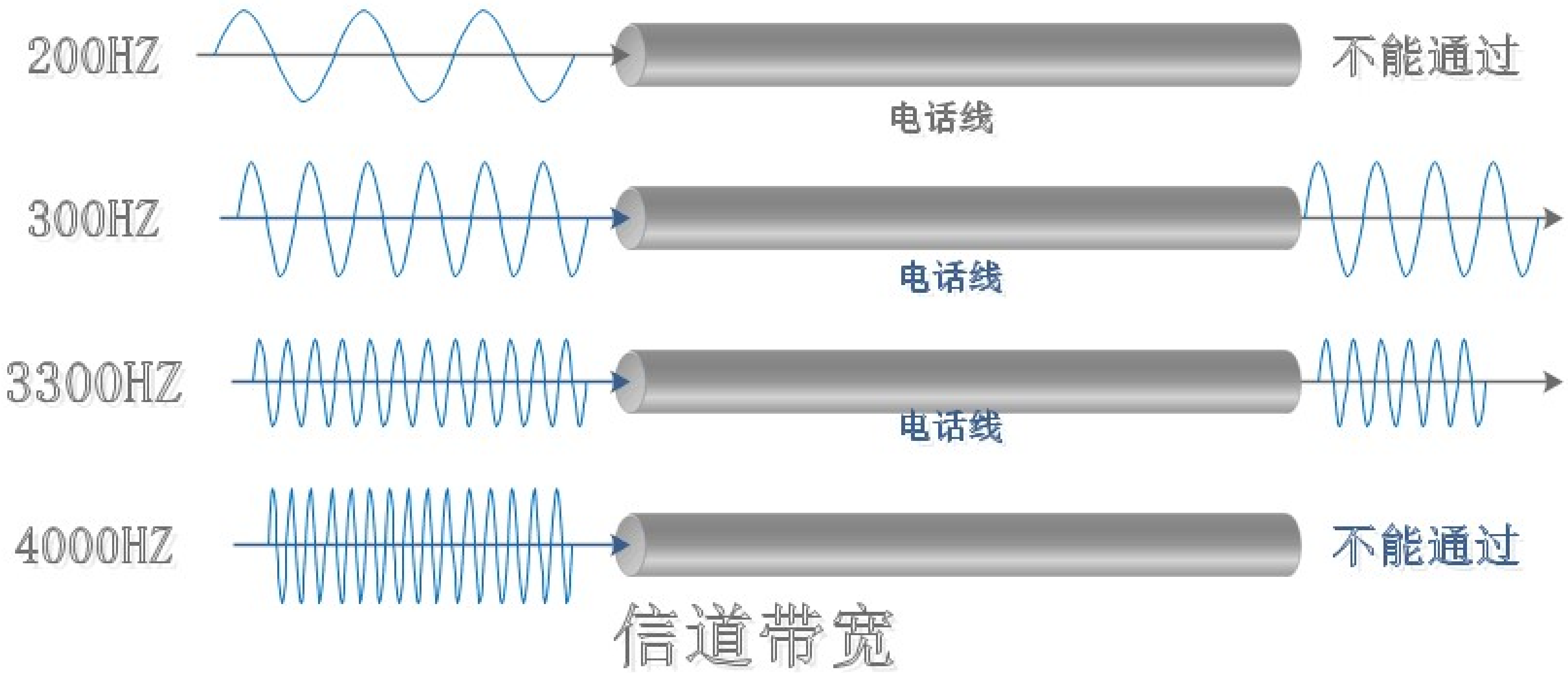
Fundamental Concepts (1)

- **Channel**：传送信息的媒体（介质）
- **Bandwidth**：The Width of the frequency range transmitted without being strongly attenuated (Hz)
- **Bit Rate**(bps)：The number of bits transmitted per second
- **Baud Rate**(Baud)：The number of signal units per second required to represent bits

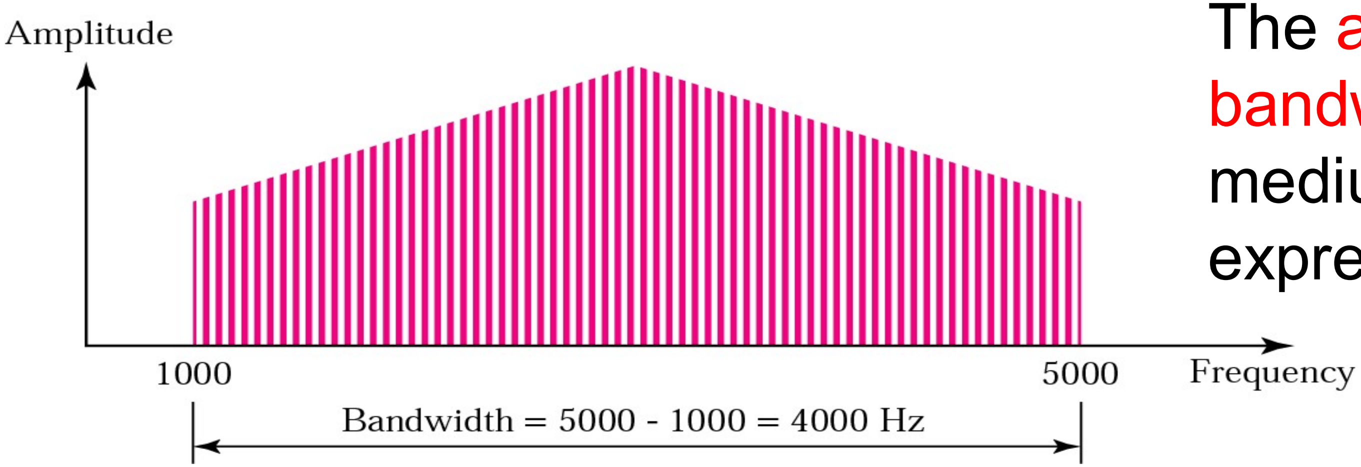
Baud Rate=1/T

T is the period of a signal unit

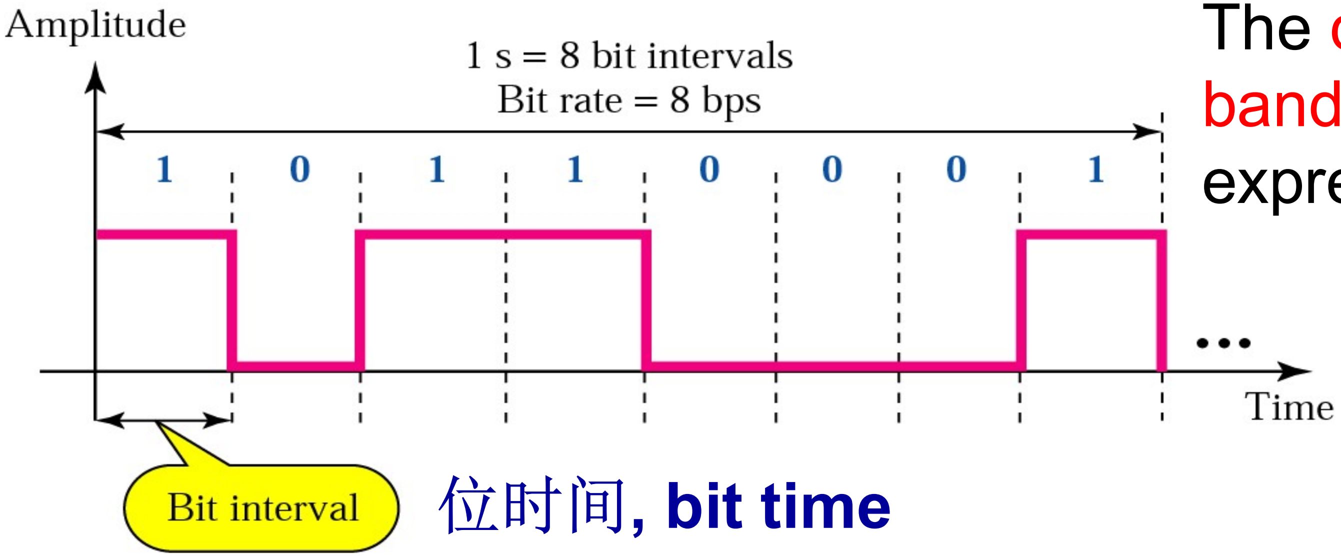
Bandwidth of medium



Bandwidth



The analog bandwidth of a medium is expressed in Hz

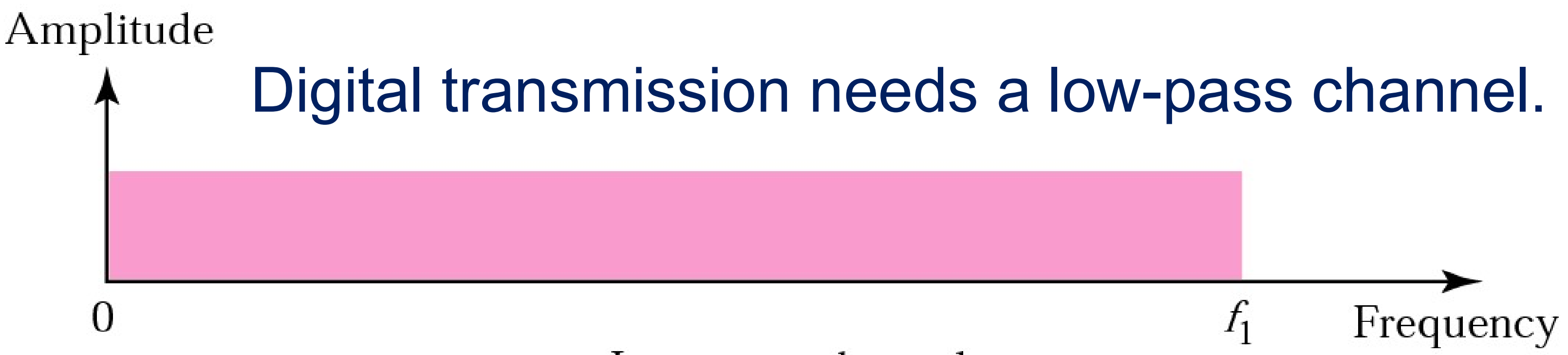


The digital bandwidth is expressed in bps

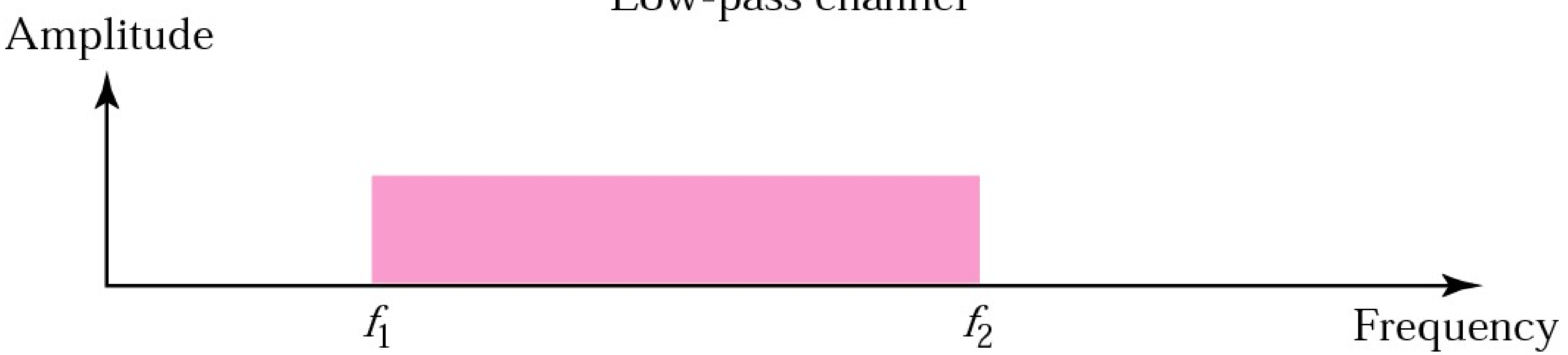
位时间, bit time

Low-pass and Band-pass

Digital transmission needs a low-pass channel.



Low-pass channel



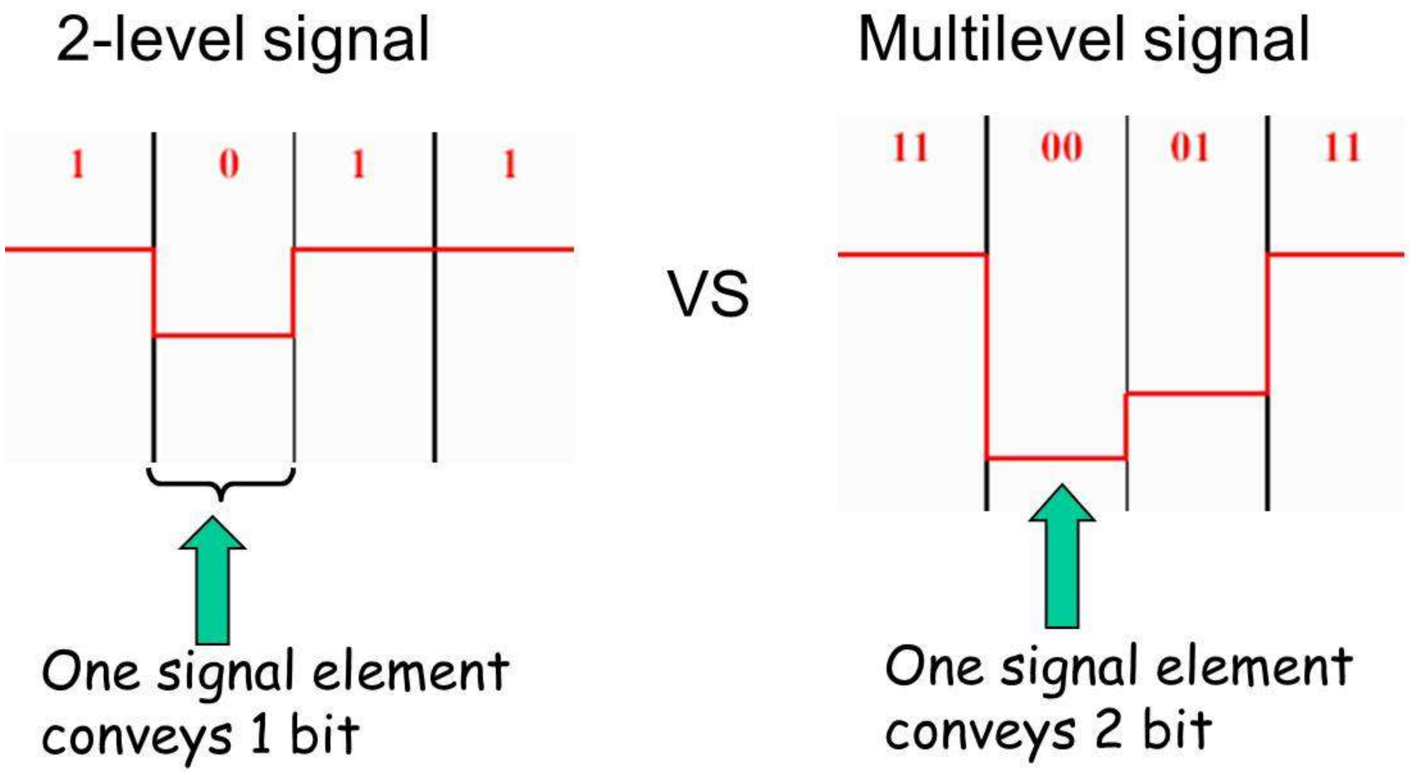
Band-pass channel

Analog transmission can use a band-pass channel.

Baud Rate vs. Bit Rate

Bit Rate = Baud Rate*(log$_2$V)

V是信号的有效状态数(电平级数)



Fundamental Concepts (2)

- 信道容量(Channel Capacity)：信道的最大数据率
- 吞吐量(Throughput)：网络容量的度量，表示单位时间内网络可以传送的数据位数（bps）
- 负载(Load):表示单位时间内注入(进入)网络的数据位数（bps）
- 传播速度(Propagation Speed)：通信线路上，信号单位时间内传送的距离（米/秒）
- 误码率BER(Bit Error Rate)：信道传输可靠性指标

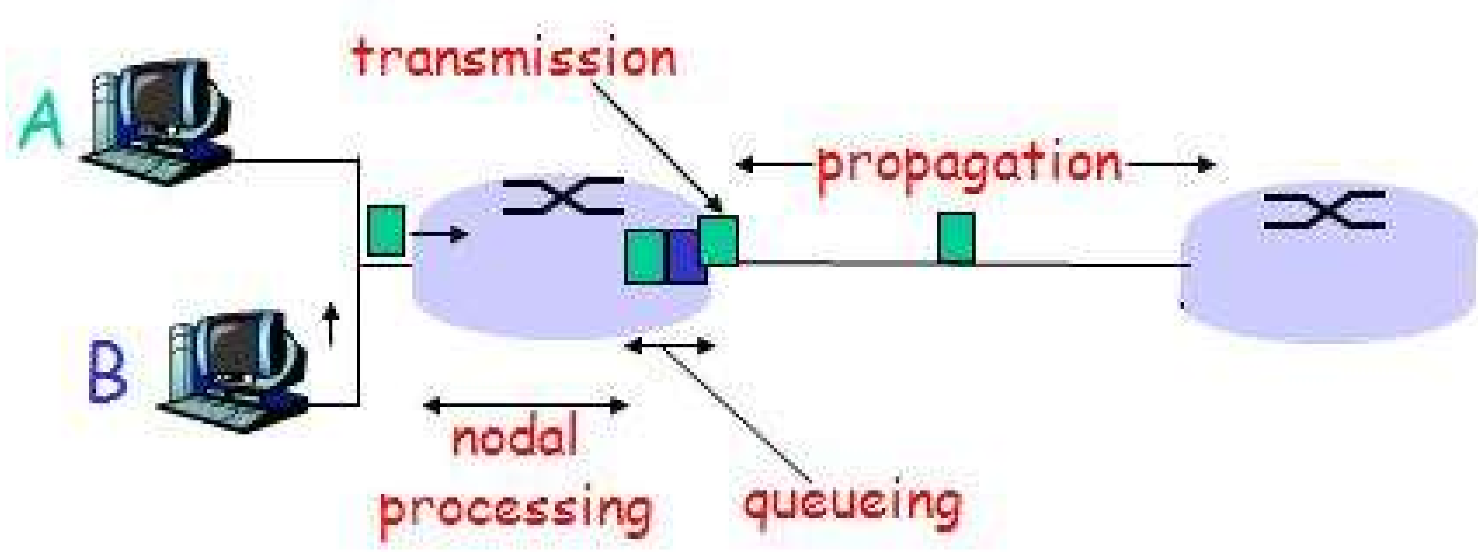
  P= 传送错的位数 /总的传送位数

Fundamental Concepts (3)

- 时延(Delay)：从向网络中发送数据块的第一位开始，到最后一位数据被接收所经历的时间
- 时延的组成：发送时延、传播时延、结点处理时延、排队时延
  - 发送时延(Transmission Delay)：设备发送一个数据块所需要的时间（数据块长度/信道带宽）
  - 传播时延(Propagation Delay)：信号通过传输介质的时间。
  - 节点处理时延(Nodal Processing Delay)：交换机/路由器检查数据、选路的时间
  - 排队时延(Queuing Delay)：在交换机/路由器中排队等待的时间

Sources of Delay (1)

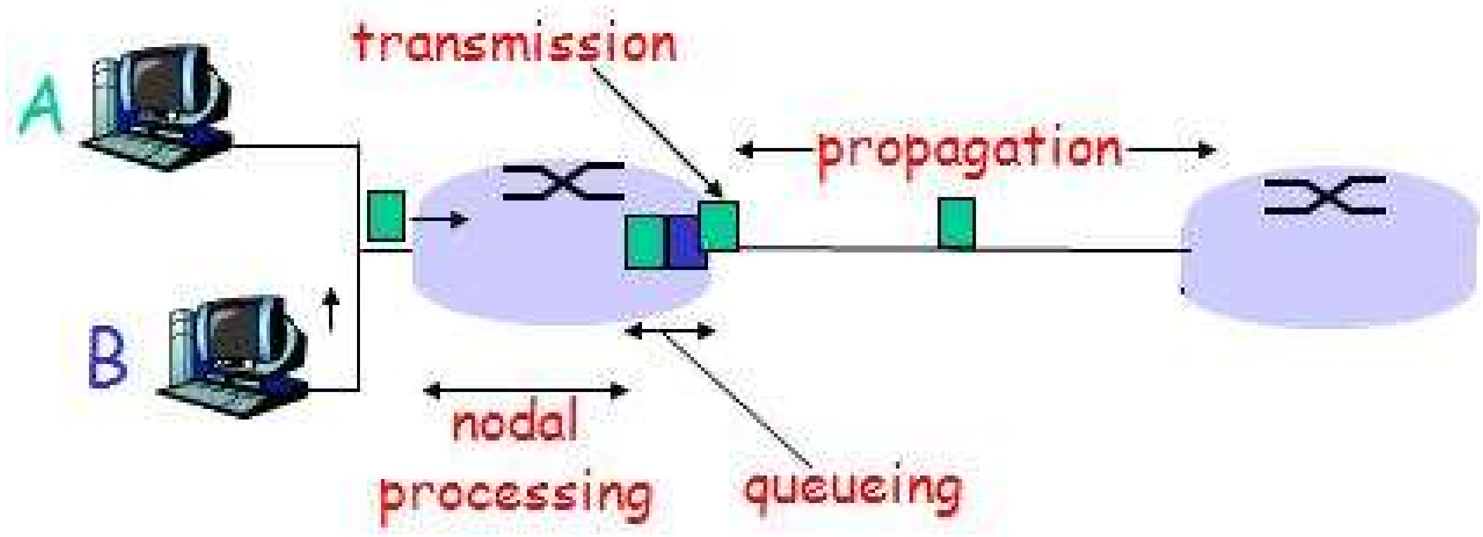
- 1. Transmission delay:
  - R=link bandwidth (bps)
  - L=packet length (bits)
  - time to send bits into link = L/R
- 2. Propagation delay:
  - d = length of physical link
  - s = propagation speed in medium (~2×10^8 m/sec)
  - propagation delay = d/s



[Kurose]

Sources of Delay (2)

- 3. Nodal processing delay:
  - check bit errors
  - determine output link
- 4. Queuing delay:
  - time waiting at output link for transmission
  - depends on congestion level of router

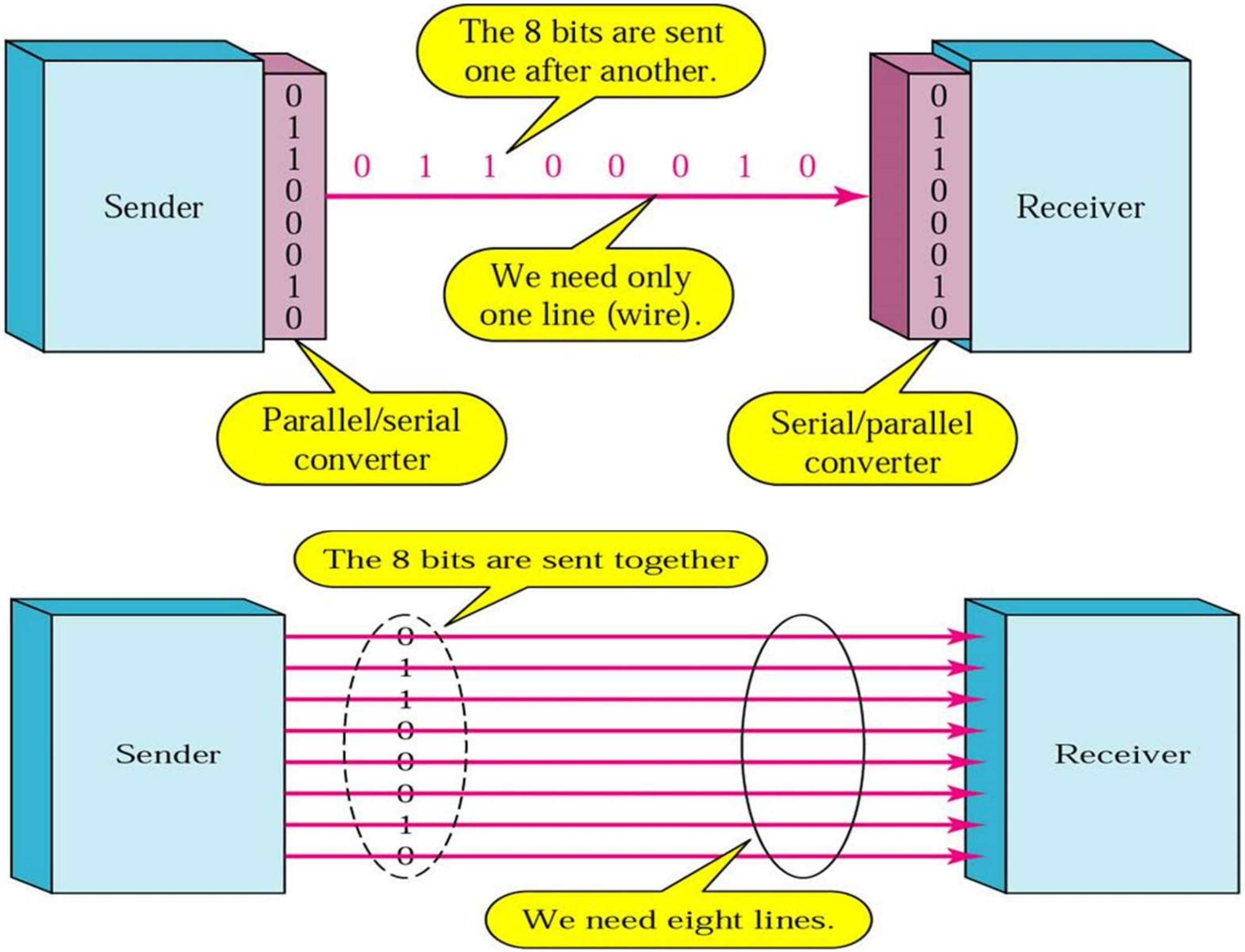


[Kurose]

Simplex, Half-duplex, Full-duplex

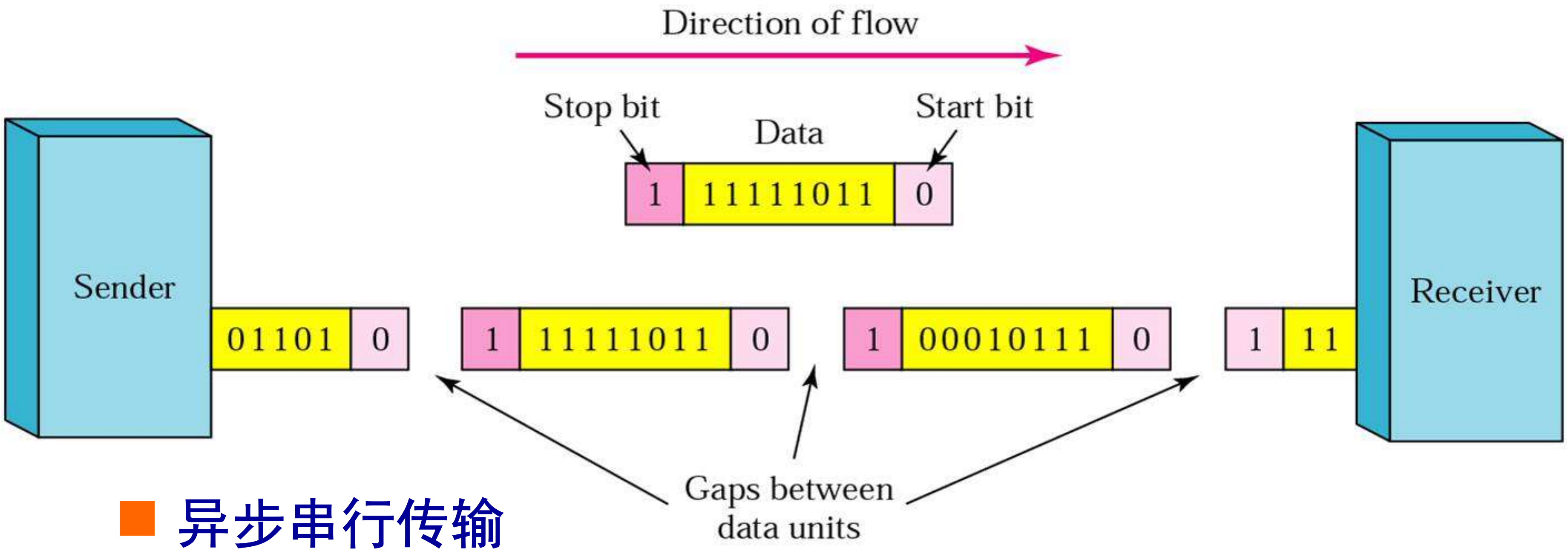
- Simplex（单工）
  - Transmission occurs in only one direction
  - eg. FM Radio broadcasting
- Half-duplex（半双工）
  - Transmission can be made in both direction, but at one time only one direction is available
  - eg. talkie and walkie （对讲机）
- Full-duplex（全双工）
  - Transmission can be made in both directions simultaneously
  - eg. Phone conversation

Serial transmission(串行传输) vs. Parallel transmission（并行传输）



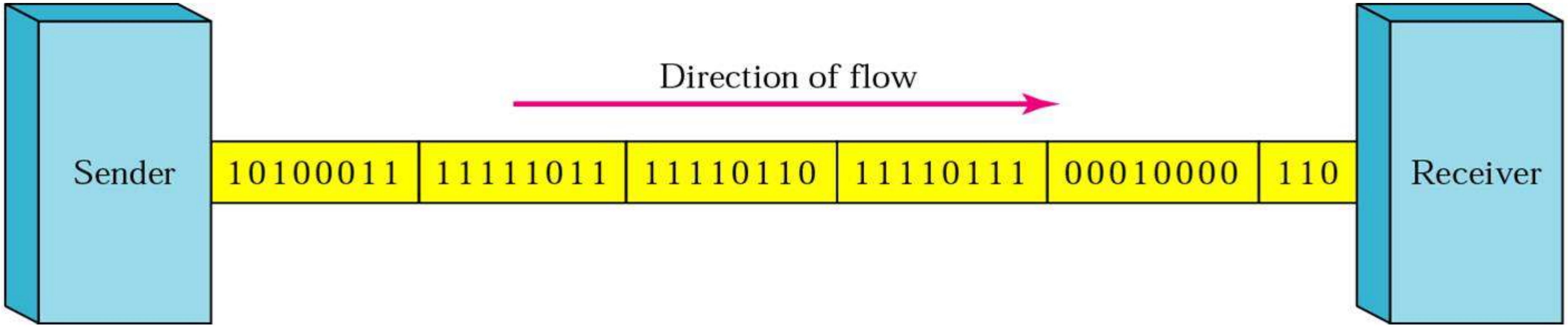
[Forouzan]

Asynchronous Transmission(异步传输) vs. Synchronous Transmission（同步传输）



- 异步串行传输
  - 独立时钟，无须同步
  - 以字符为单位进行传输
  - 发送两个字符之间的间隔是任意的
  - 接收方依靠字符中的起始位和停止位来同步

同步串行传输和异步串行传输



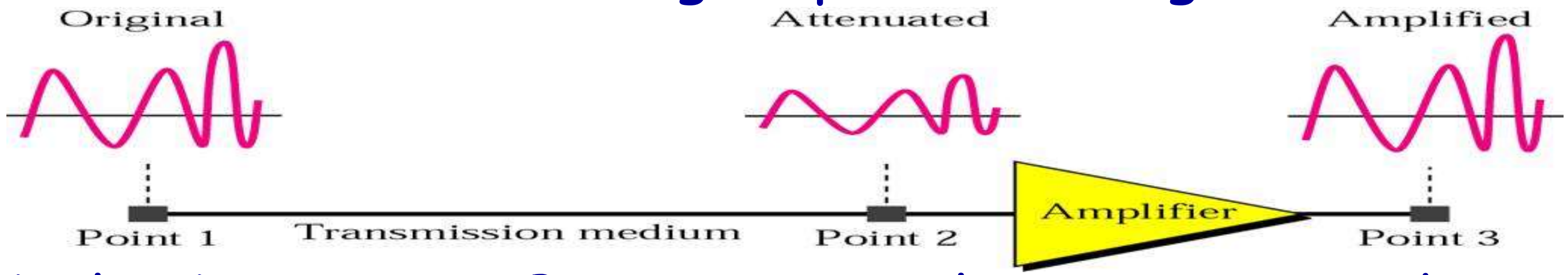
- 同步串行传输
  - 以时钟信号线对传输的数据线上的信号进行比特同步
  - 以数据块（帧或分组）为单位传输

Transmission Impairment(1)

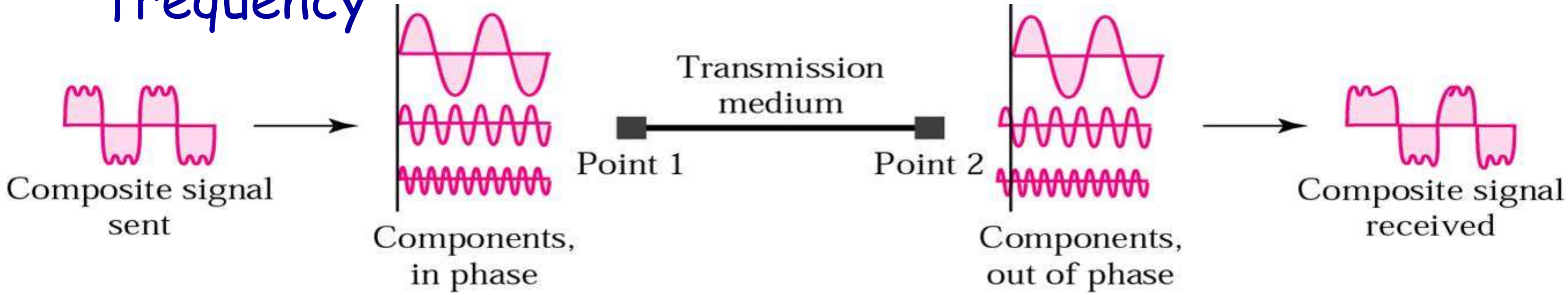
- Signal received may differ from signal transmitted
- Analog - degradation of signal quality
- Digital - bit errors
- Caused by
  - Attenuation(衰减) and attenuation distortion （失真）
  - Delay distortion(时延失真)
  - Noise(噪声)

Transmission Impairment(2)

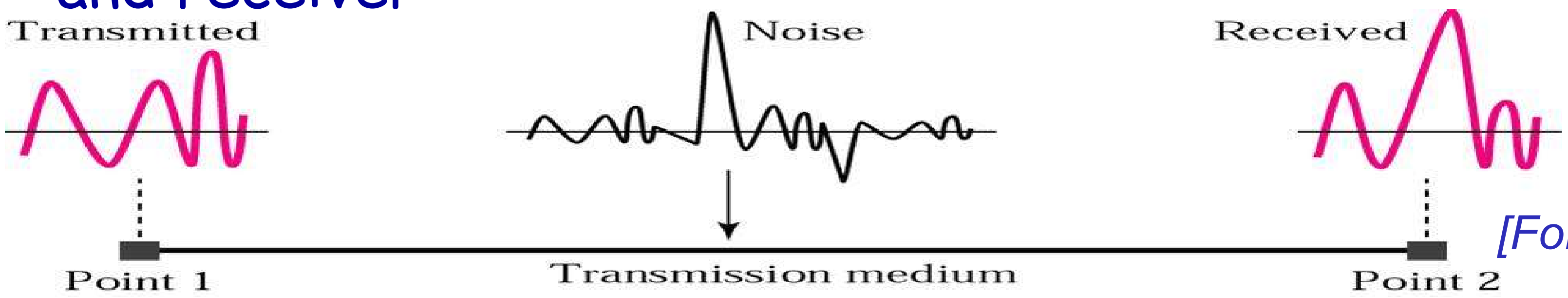
- Attenuation: loss of signal power through distance



- Delay Distortion: Propagation velocity varies with frequency



- Noise: Additional signals inserted between transmitter and receiver



[Forouzan]

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Bandwidth

- No transmission facility can transmit signal without **losing some power**
- All transmission facilities diminish different Fourier components by **different amounts**
- Range of frequencies transmitted without being strongly (0.5) attenuated is called **bandwidth**
  - Telephone wire's bandwidth: 1 MHz
  - UTP 5: 100MHz
- bandwidth is a **physical property** of the transmission medium, depends on
  - construction
  - thickness
  - length
- bandwidth：Hz, bps

Maximum Data Rate

Over a certain channel, what is the maximum speed to send data?

Capacity for noiseless channel(无噪声信道)

- Nyquist proved that: if an arbitrary signal has been run through a low-pass filter of bandwidth B, the filtered signal can be completely reconstructed by making only 2B samples per second
- Given binary signal, data rate supported by B Hz is 2B bps
- Nyquist Bit Rate(奈奎斯特公式，1924)

$$bit\ rate = 2 \times bandwidth \times \log_2 V$$

V is the number of signal levels used to represent data.

- Increasing the levels of a signal may reduce the reliability of the system.

Noisy Channel: Shannon Capacity （香农公式，1948）

- The theoretical highest data rate for a noisy channel
- Shannel's theorem

$$Capacity = bandwidth \times \log_2 (1 + \frac{signal\_power}{noise\_power})$$

- where capacity is in bits/second, bandwidth is in hertz, and signal and noise powers are measured in the same physical units, such as watts
- Usually, the SNR (Signal-to-Noise Ratio) is expressed on a log scale as: $S/N_{db} = 10 \log_{10} \underline{S/N}$
- The units of this log scale are called **decibels(dB)**

Example of Computing Channel Capacity

(1) A channel has B = 1MHz and SNR_{dB}=24dB,

1) what is the channel capacity limit?

$SNR_{dB} = 10 \log_{10} (SNR)$ $SNR = 251$

$C = B \log_2(1 + SNR) \approx 8$ Mbps

2) Assume we can achieve the theatrical C, how many signal levels are required?

$C = 2 B \log_2 V$

V = 16 levels

(2) Voice phone line 300Hz-3300Hz, S/N=30dB

$C = B\log_2(1+SNR) = 3000\log_2(1+1000)$ =30kbps

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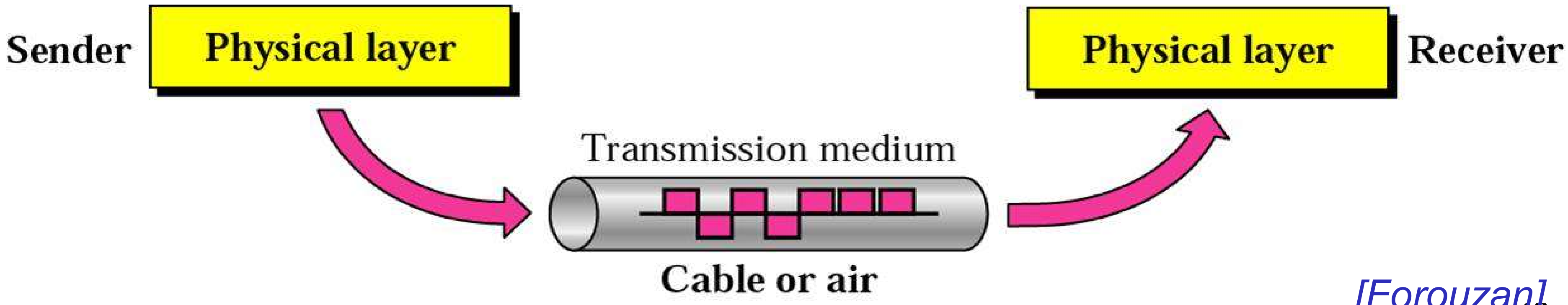
Transmission Media

- Magnetic Media
- Guided
  - Twisted Pair
  - Coaxial Cable
  - Fiber Optics
  - Power Lines
- Unguided (wireless transmission)
  - Radio
  - Microwave and Satellite



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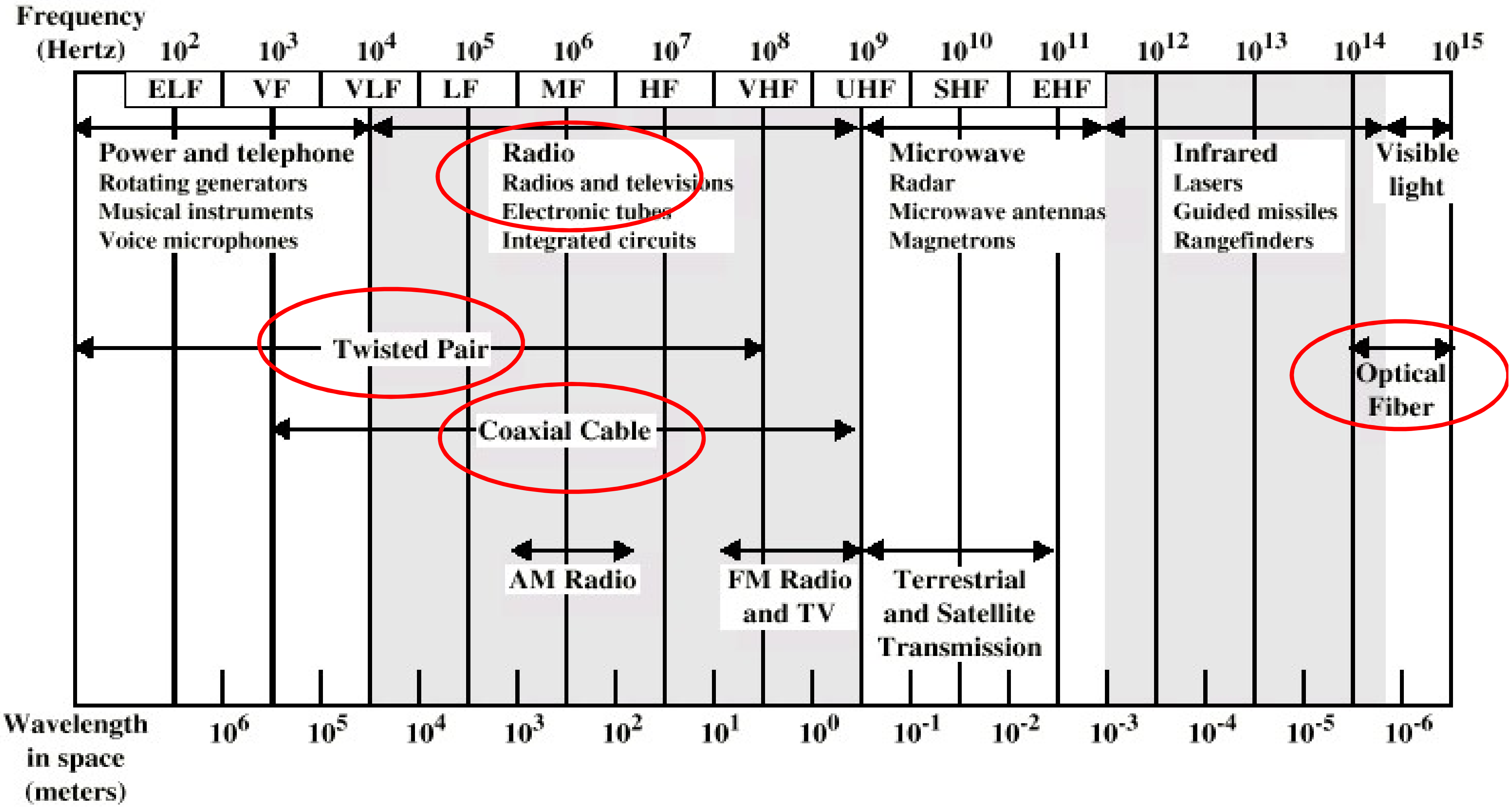
Magnetic Media

- Bandwidth (bps)
  - An industry standard Ultrium tape can hold 800 GB
  - A box can hold 1000 tapes: total capacity of 800 TB
  - Assumed that it takes 1 day to send the box by train from Beijing to Shenzhen
  - The effective bandwidth of this transmission is:

    800T x 8bits/86,400 sec=70Gbps.
- Cost
  - tape cost $4000 per box per usage
  - $1000 for shipping
  - $5000 for 800TB, so 1 GB for under 0.5 cents.

The Electromagnetic Spectrum

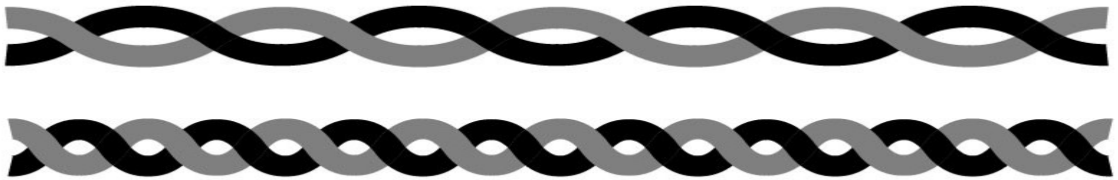


ELF = Extremely low frequency
VF = Voice frequency
VLF = Very low frequency
LF = Low frequency

MF = Medium frequency
HF = High frequency
VHF = Very high frequency

UHF = Ultrahigh frequency
SHF = Superhigh frequency
EHF = Extremely high frequency

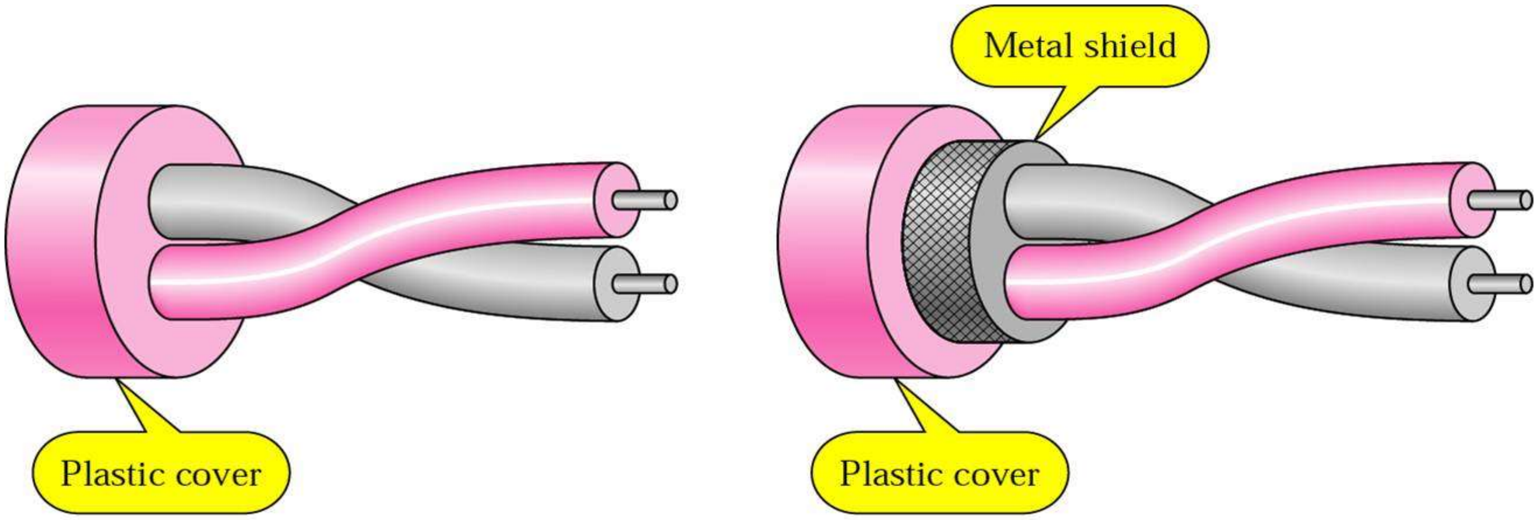
Twisted Pair(双绞线)



(b)

- Most common transmission media, Can be used for transmitting either analog or digital signals
- consists of two insulated copper wires, about 1 mm thick, the wires are twisted together in a helical form (螺旋状)
- When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively
- The bandwidth depends on
  - the thickness of the wire, the purity of the copper core
  - the distance traveled
  - the twists per meter
- Several megabits/sec can be achieved for a few kilometers

UTP and STP



a. UTP

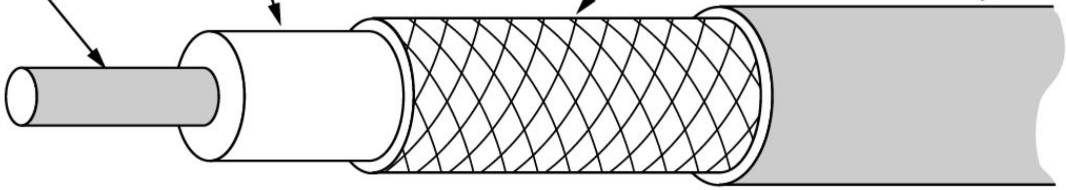
b. STP

Unshielded TP

Shielded TP

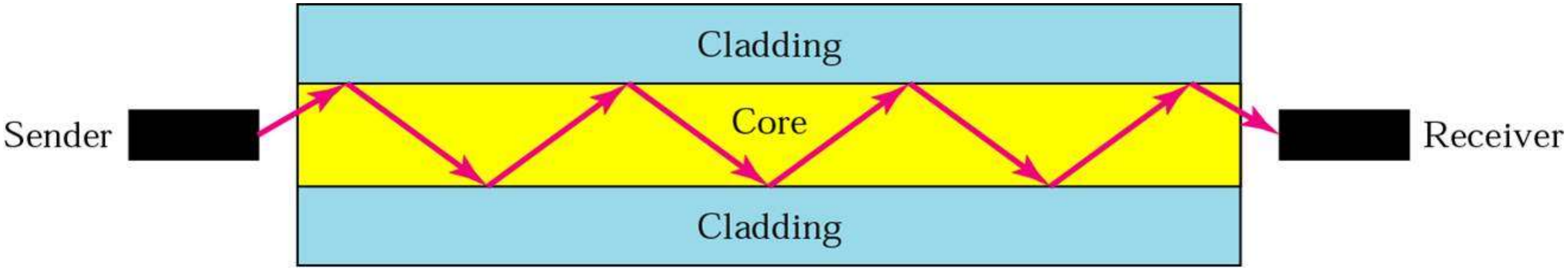
- Category 3 UTP: 16MHz
- Category 5 UTP: 100MHz
- Category 6 UTP: 250MHz
- Category 7 UTP: 600MHz

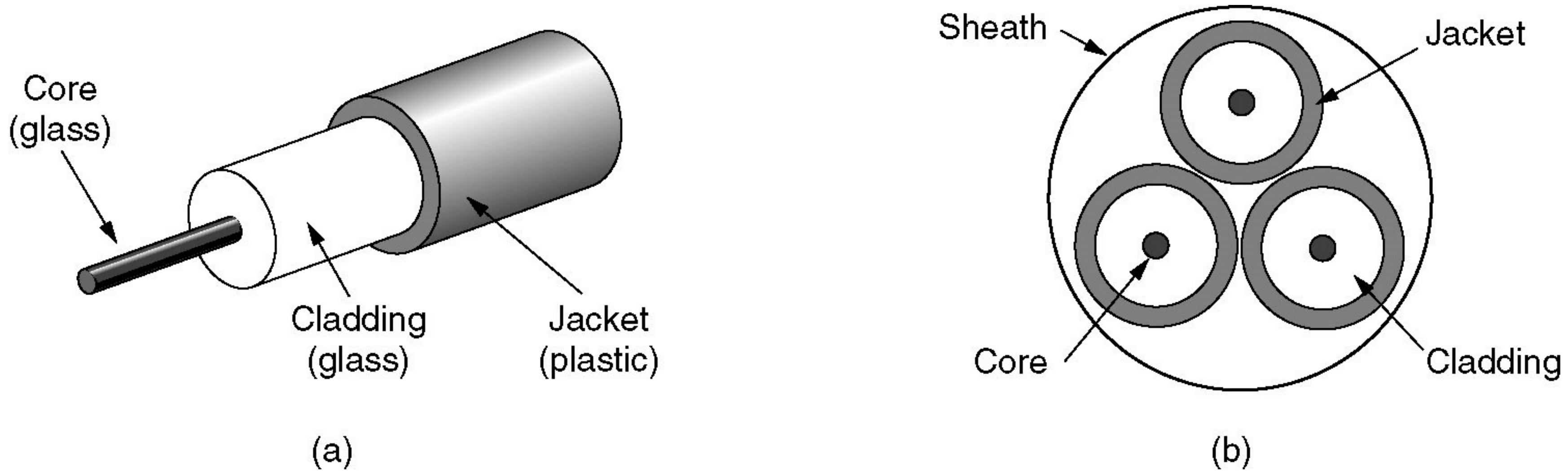
Coaxial Cable（同轴电缆）



- It has **better shielding** and **greater bandwidth** than UTP(span longer distances at higher speed)sulv
- Two kinds of coaxial cable are widely used
  - 50-ohm cable, is commonly used when it is intended for digital transmission
  - 75-ohm cable, is commonly used for analog transmission and cable television
- high bandwidth(1GHz)
- excellent noise immunity

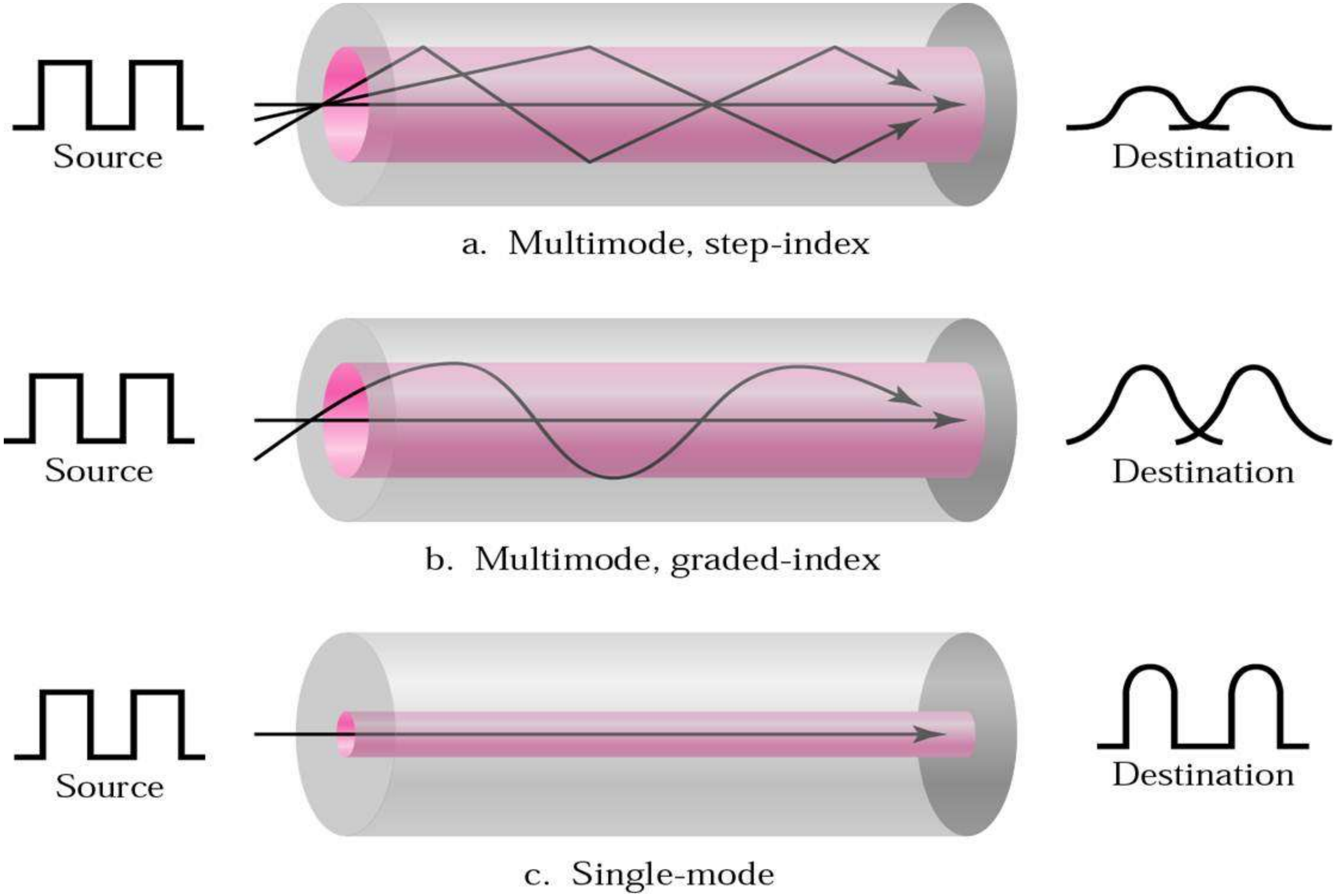
Fiber Cables





(a) Side view of a single fiber.
(b) End view of a sheath with three fibers.

Modes of Fiber



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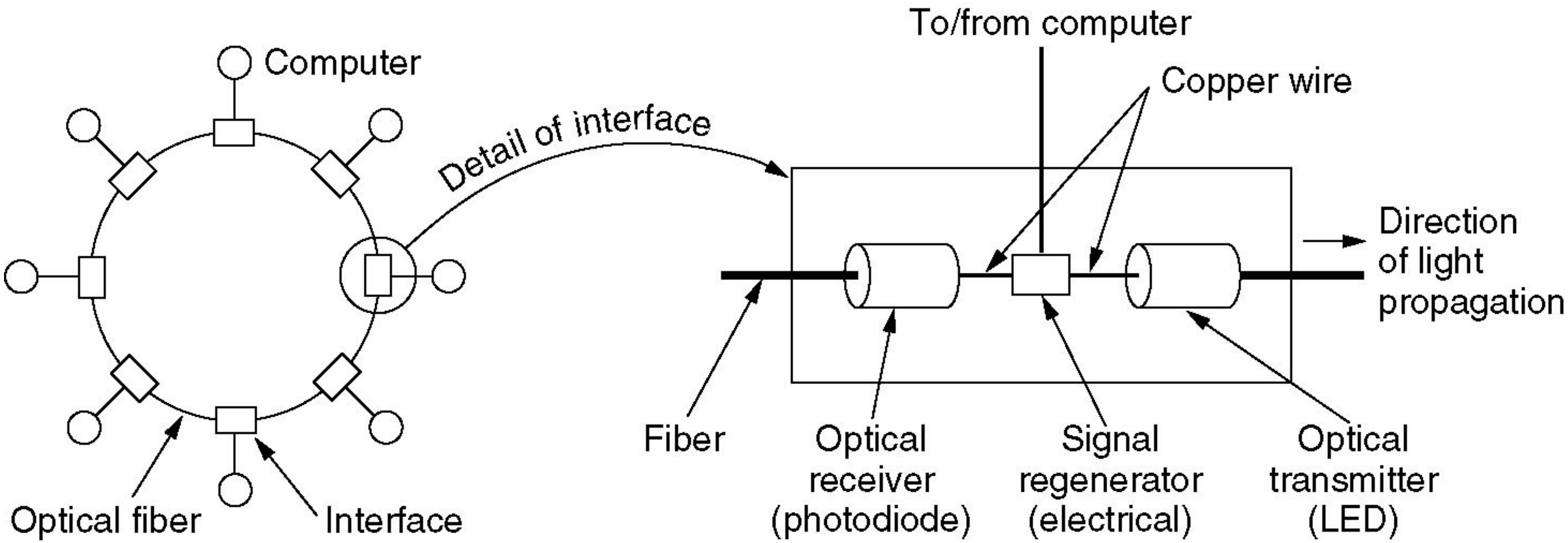
Fiber Optic Networks

A fiber optic ring with active repeaters.

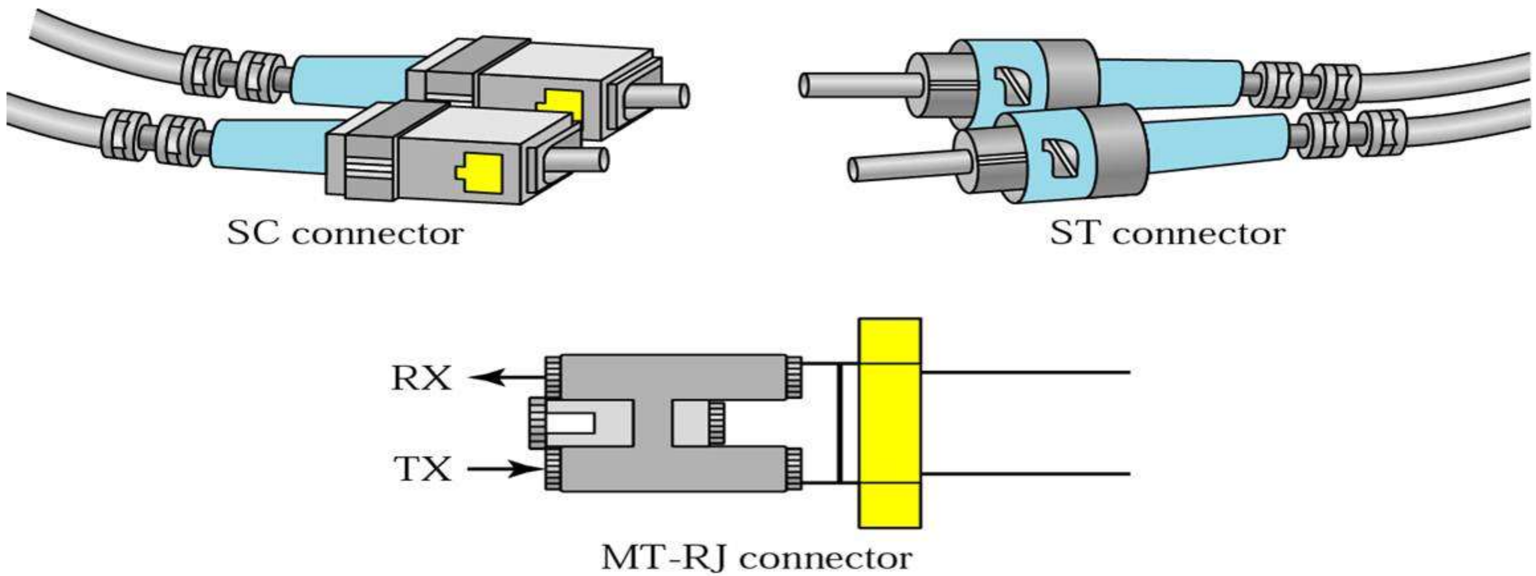


Father of Fiber Optic Communications---高锟

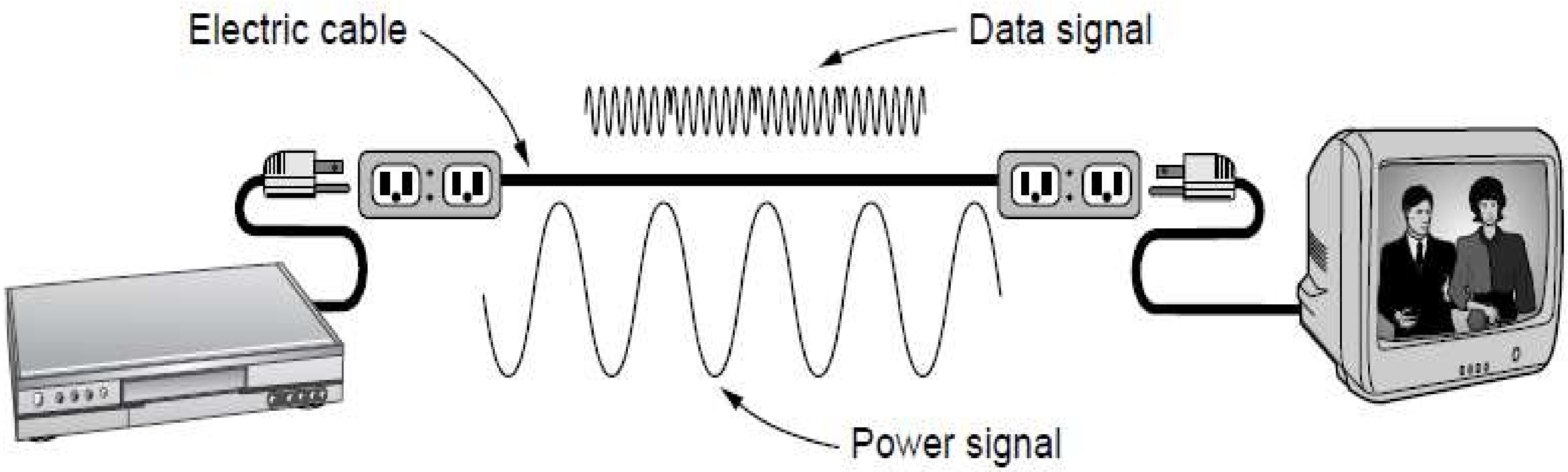
- 1964年提出在电话网络中以光代替电流，以玻璃代替导线
- 1965年，得到结论：玻璃光衰减基本限制在20dB/km以下，这是光通信的关键阈值，开启了寻找低损耗材料和合适纤维以达到这一标准的过程
- 1966年，发表《Dielectric-fibre surface waveguides foropticl frequenceies》，开创性地描述：长距离和高信息光通信所需要的绝缘性纤维的结构和材料特性。
- 2009年获得诺贝尔物理学奖

Advantages of Fiber

- High bandwidth
- Lightweight
- Security

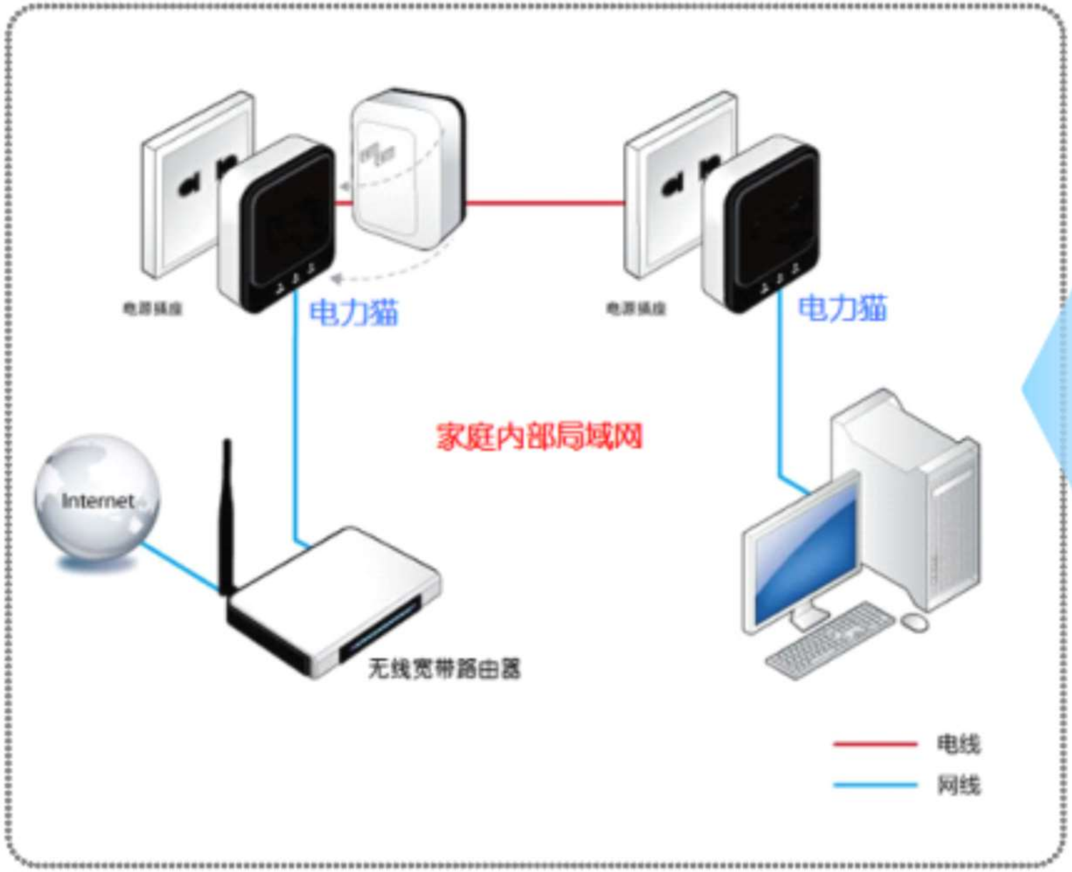


Power Line Communication



A network that uses household electrical wiring

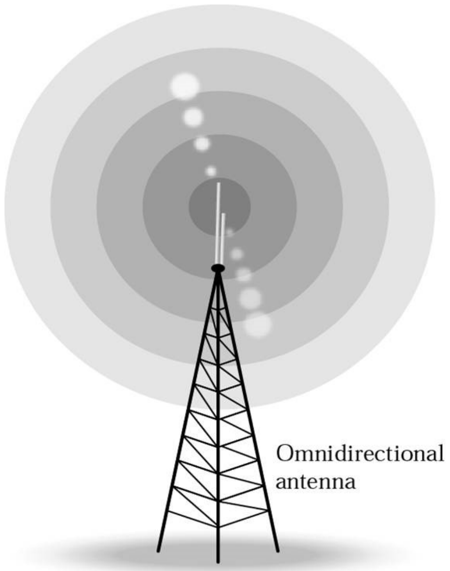
家庭应用：点对点联网


电源插座
电力猫
电源插座
电力猫
家庭内部局域网
Internet
无线宽带路由器
电线
网线

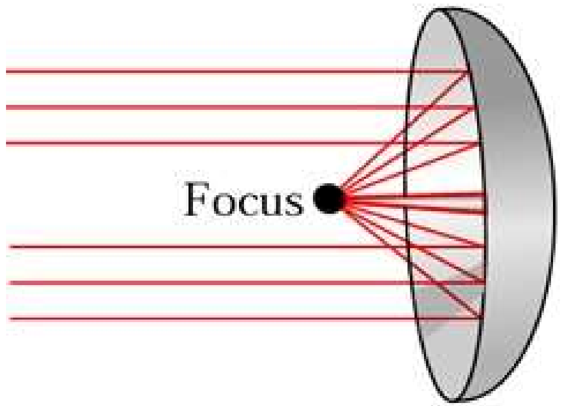

ZINWELL
PLQ-5100

Wireless Transmission

- Features
  - Comparatively higher bit error rate
  - Longer propagation delay
  - Omni-direction vs. Uni-direction



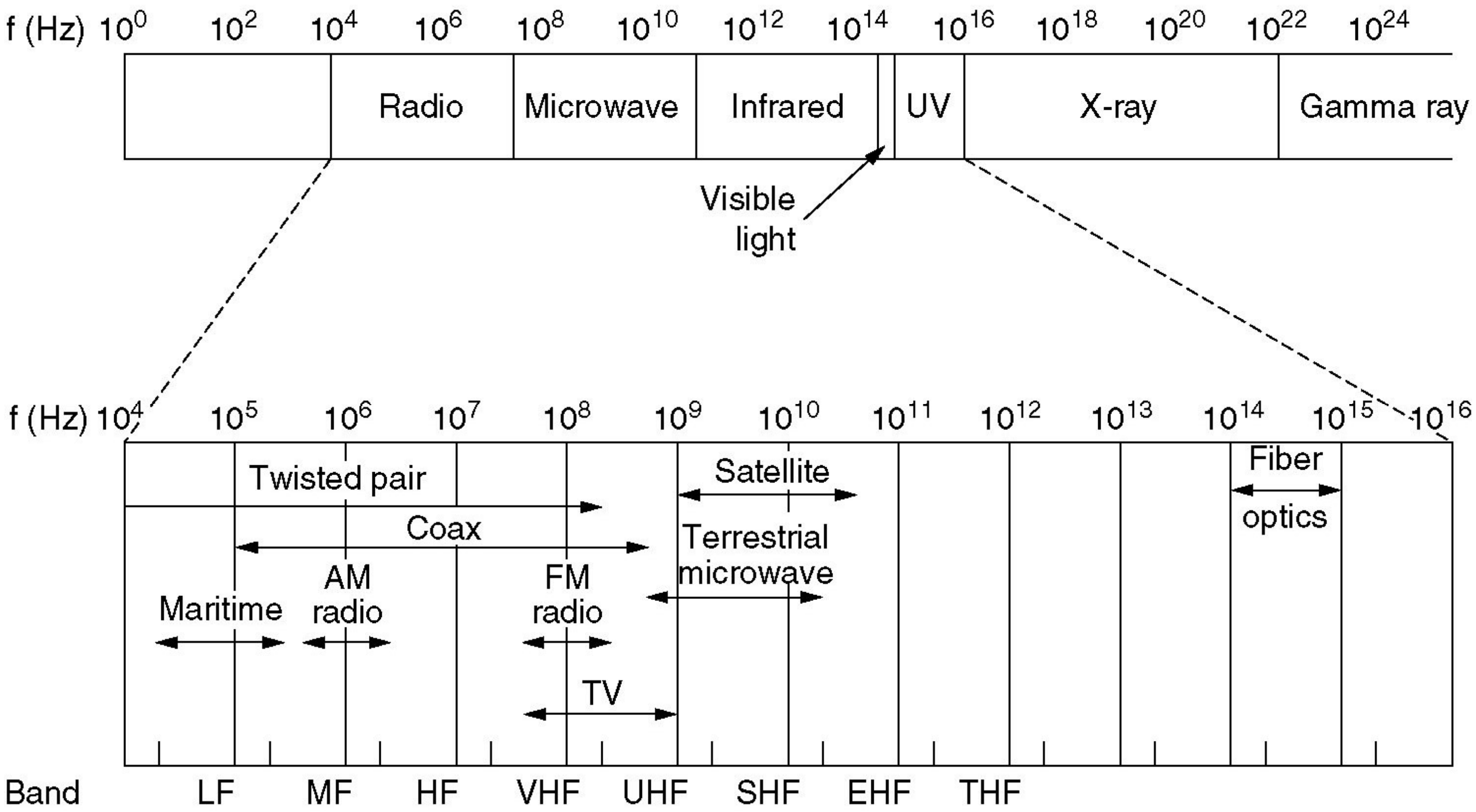
Omnidirectional antennas



Unidirectional antennas

[Forouzan]

Electromagnetic spectrum



Electromagnetic spectrum and use in communications

MF = Medium frequency
HF = High frequency
VHF = Very high frequency
UHF = Ultrahigh frequency
SHF = Superhigh frequency
EHF = Extremely high frequency

Transmission Frequency Band

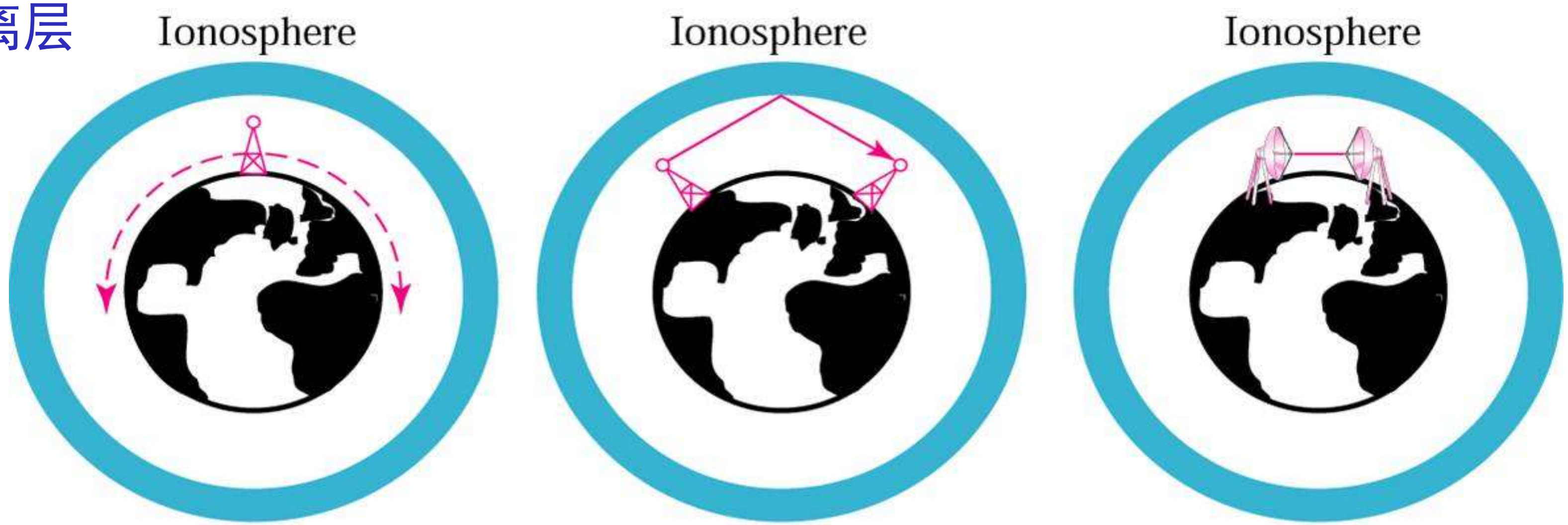
- Using narrow frequency band
  - Most transmission use narrow frequency band to get the best reception
- Spread Spectrum (bluetooth, WLAN, CDMA, ...)
  - FHSS: Frequency hopping spread spectrum
  - DSSS: Direct sequence spread spectrum
  - UWB：Ultra-WideBand

Category of Wireless Transmission

- Radio Transmission
- Microwave Transmission
- Communication Satellites
- Infrared and Millimeter Waves

Radio Propogation

电离层



Ground propagation
(below 2 MHz)

Sky propagation
(2 - 30 MHz)

Line-of-sight propagation
(above 30 MHz)

(a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth.

(b) In the HF band, they bounce off the ionosphere.

(c) In the VHF band and above, they use line-of-sight（视距）transmission, not affected by ionosphere.

[Forouzan]

Bands and Applications

Band	Range	Propagation	Application
VLF	3–30 KHz	Ground	Long-range radio navigation
LF	30–300 KHz	Ground	Radio beacons and navigational locators
MF	300 KHz–3 MHz	Sky	AM radio
HF	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF	3–30 GHz	Line-of-sight	Satellite communication
EHF	30–300 GHz	Line-of-sight	Long-range radio navigation

[Forouzan]

Microwave Transmission

(100M~10GHz)

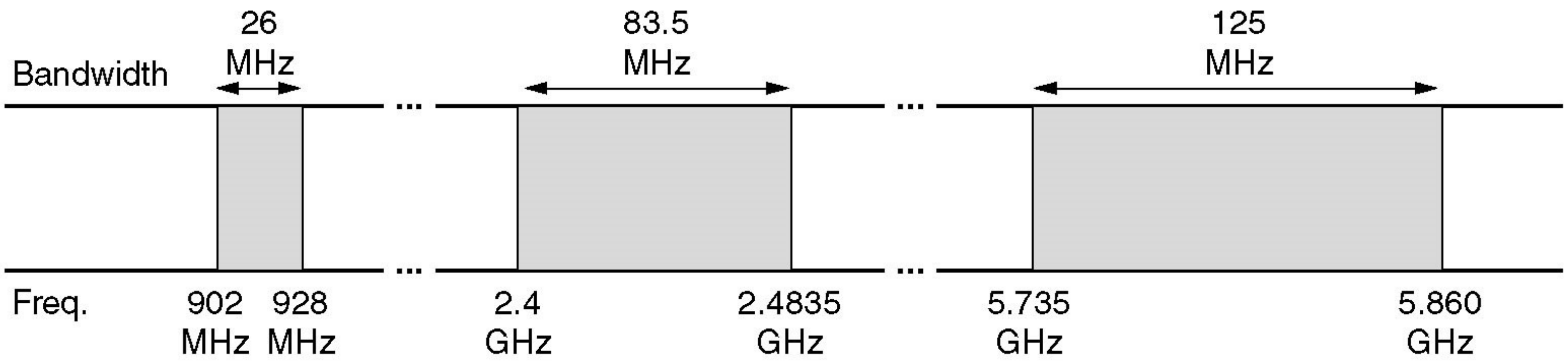
- Before fiber optic, microwaves formed the heart of the long-distance telephone transmission system
- waves travel in nearly straight lines, the transmitting & receiving antennas must be accurately aligned
- Distance between repeaters: 80km
- Multipath fading is often a serious problem (weather and frequency dependent)

(At about 4GHz, absorbed by rain)

Politics of Electromagnetic Spectrum

- National governments allocate spectrum
  - Worldwide:ITU-R; USA: FCC；China：Radio administration of the Ministry of industry and information technology (National Radio Office)
- ISM (Industrial, Scientific, Medical) bands
  - some frequency bands for **unlicensed usage**
  - FCC mandates that all devices in ISM bands use **spread spectrum techniques** to minimize interference between uncoordinated devices
  - **Power is under 1 watt**

（限制频率范围，调制技术，信号强度）



ISM bands in USA

Infrared Transmission

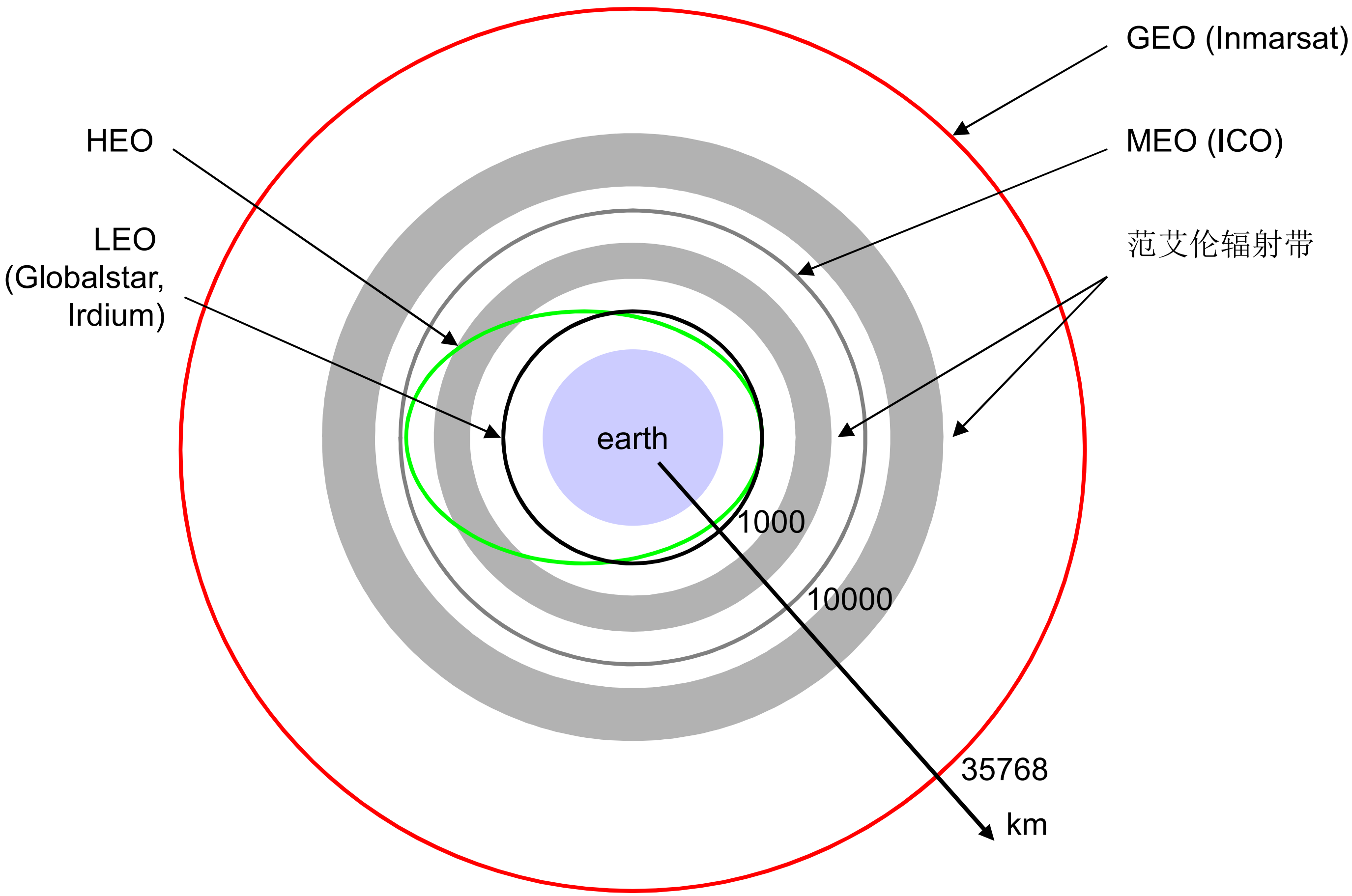
- widely used for short-range communication
- relatively directional, cheap
- do not pass through solid objects
- no government license is needed to operate an infrared system

Communication Satellites

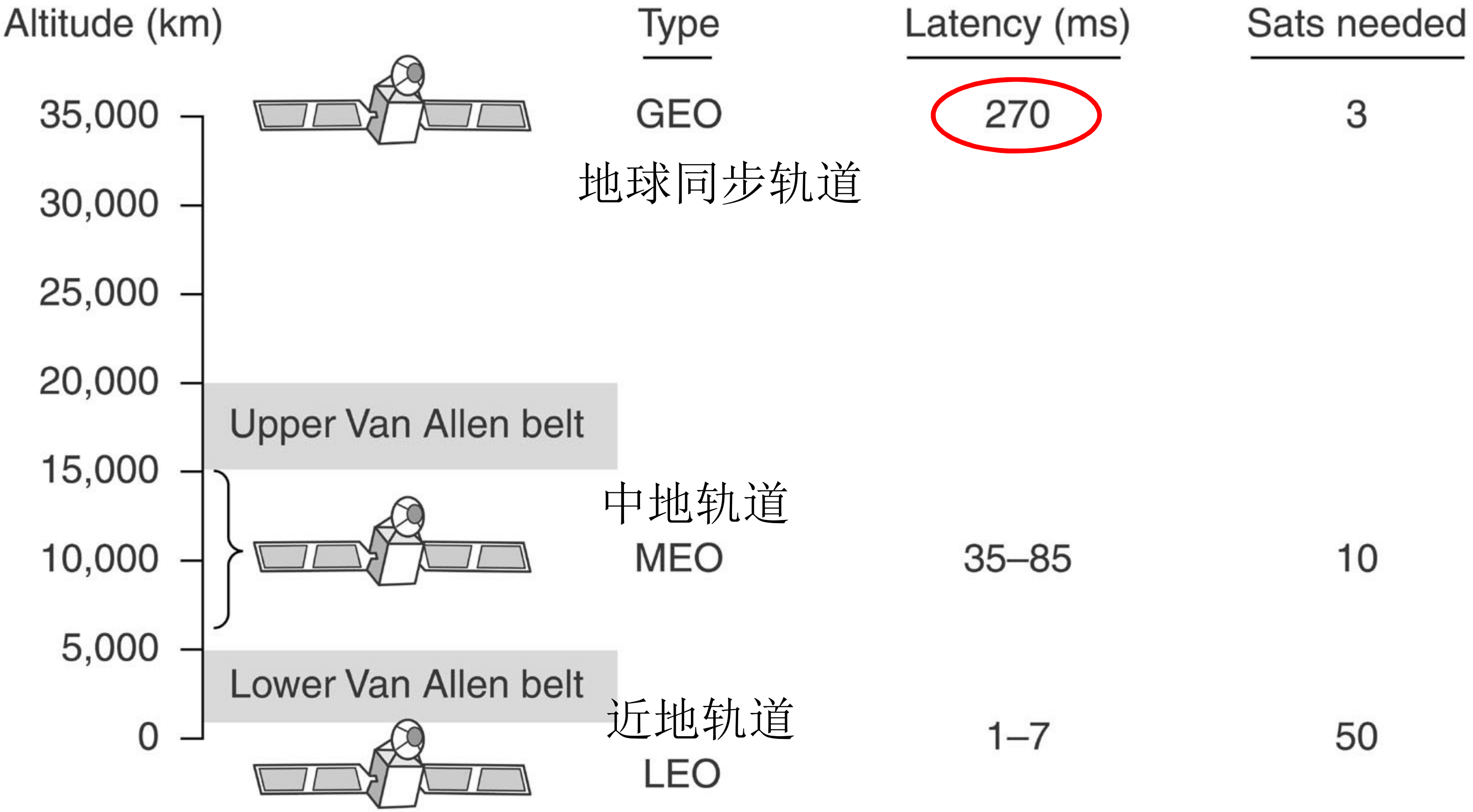
- Act as a big microwave repeater in the sky
  - Contains multiple antennas and multiple transponders
- Transponder
  - listen to some portion of the spectrum
  - amplifies the incoming signals
  - rebroadcasts it at another frequency (downward beams)—Bent pipe
- Downward beams
  - Broad: a substantial fraction of the earth’s surface
  - Narrow: spot beams cover an area hundreds of kms in diameter (bent pipe)

Communication satellites and some properties

- GEO
- MEO
- LEO



Communication satellites and some of their properties



Features of Satellite Communication

- The long round-trip distance introduces a substantial delay for GEO satellites.
  - Long propagation delay
- They are inherently broadcast media.
- The cost of transmitting a message is independent of the distance traversed.

The Principal Satellite Bands

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

A modern satellite has around 40 transponders, most often with a 36-MHz bandwidth

Outline

- Position and Functions of Physical Layer
- Basic concepts on data communications
- Theoretical basis for data communication
- Transmission medium
- *Digital modulation and encoding*
- Multiplexing
- Physical network: PSTN
- Physical network: Cable TV system
- Circuit switching vs. packet switching
- Physical layer protocol: 10BaseT

Signal Encoding Techniques

- Line encoding: Digital data, digital signal
- Modulation: Analog data, digital signal
- PCM: Digital data, analog signal
- Telephone: Analog data, analog signal

Line encoding: baseband transmission

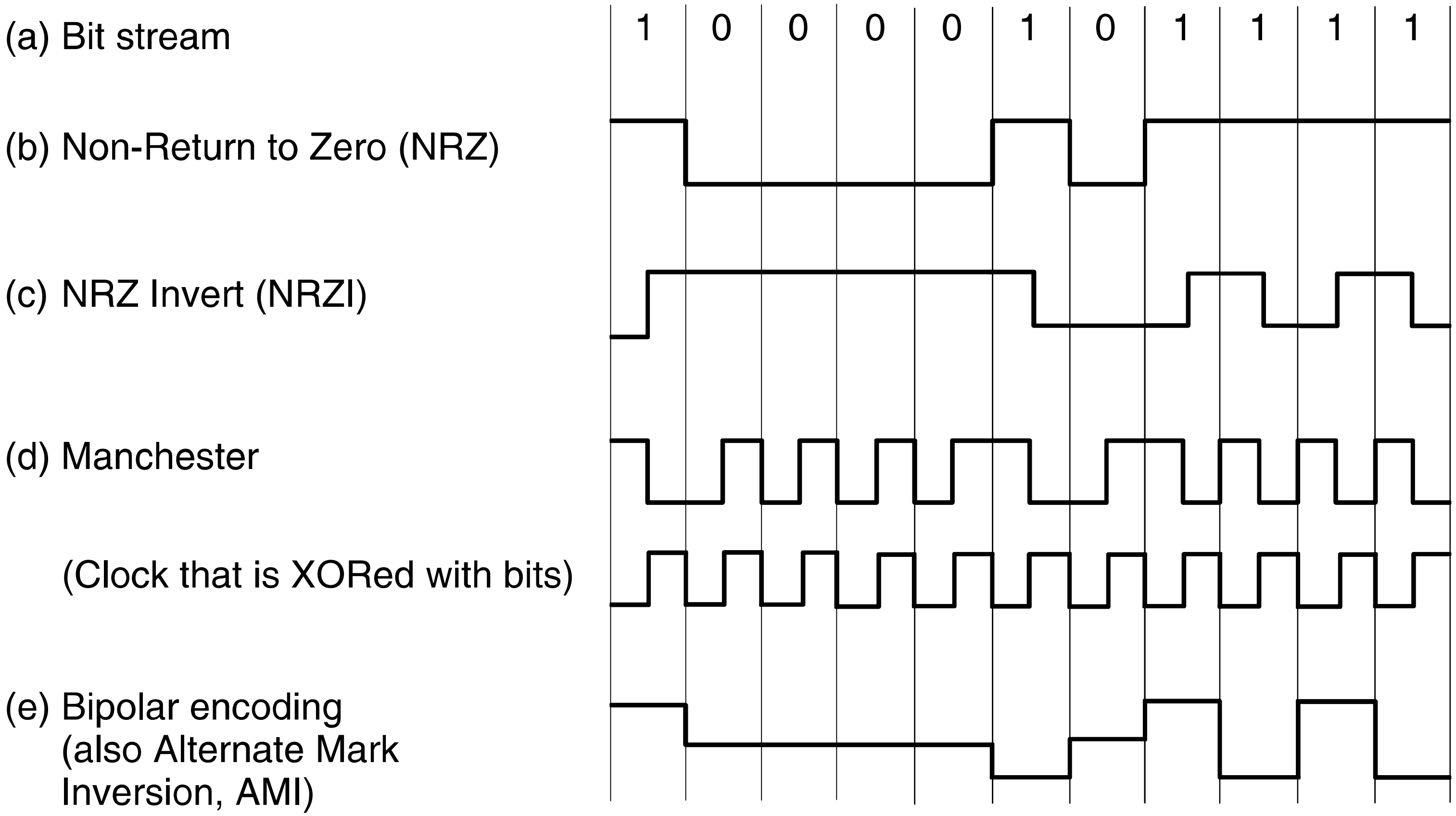
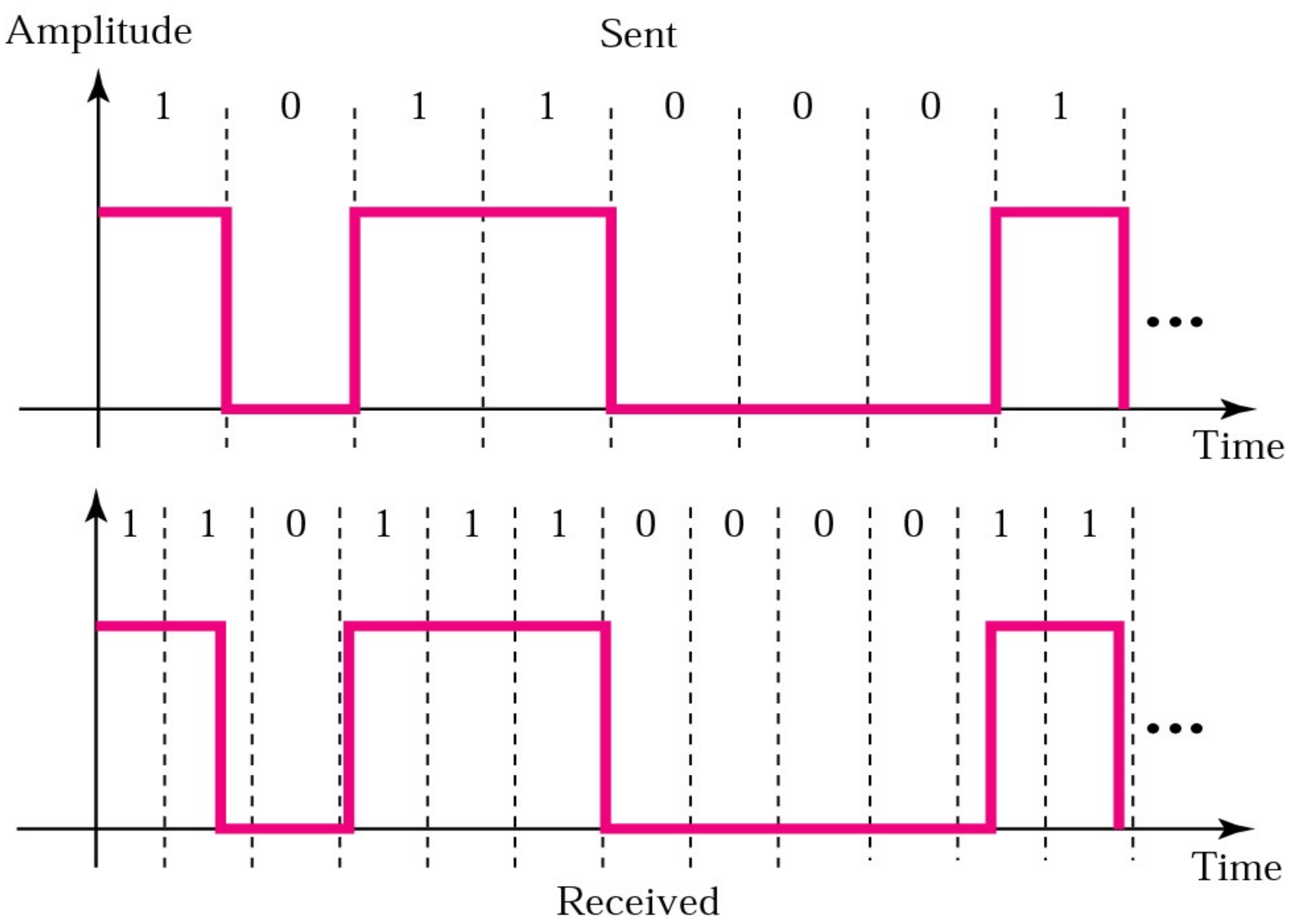


Figure 2-20. Line codes: (a) Bits, (b) NRZ, (c) NRZI, (d) Manchester, (e) Bipolar or AMI.

Loss of Synchronization

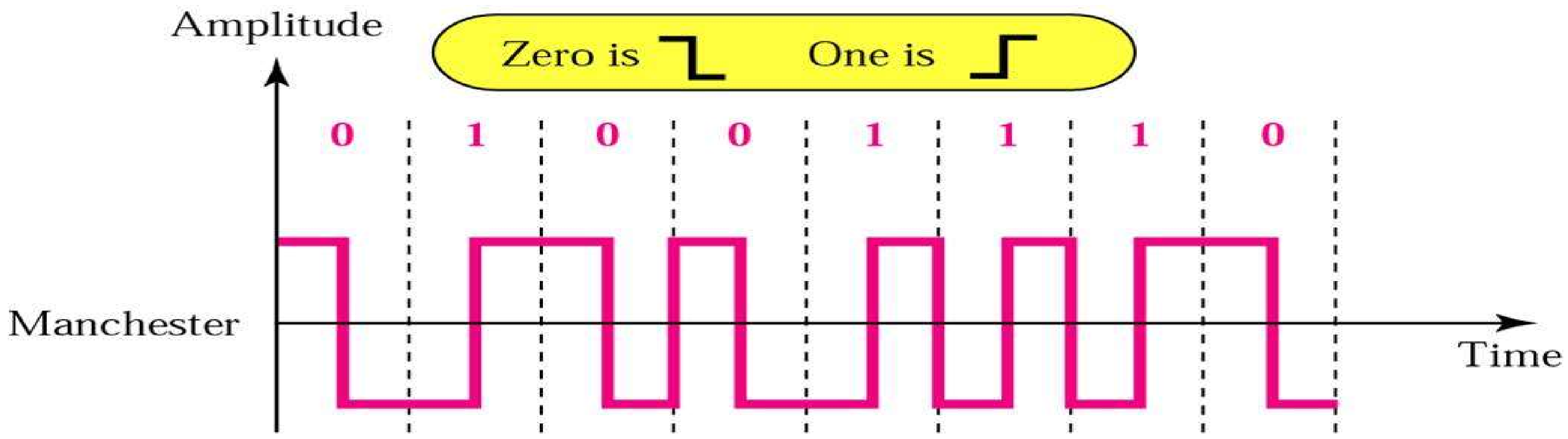


Baseband transmission

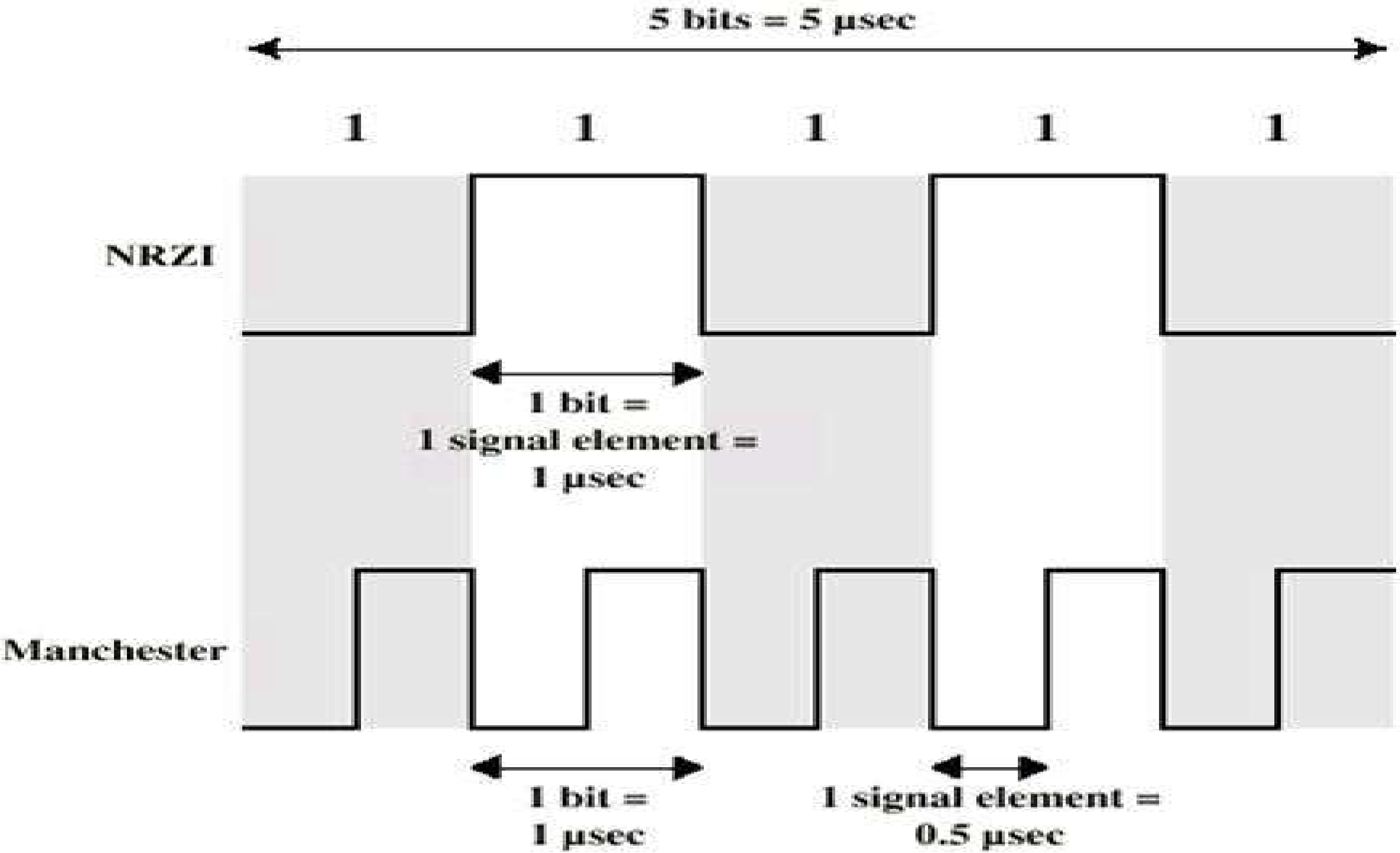
- Bandwidth Efficiency
  - NRZ need a bandwidth of at least B/2 Hz when the bit rate is B bits/sec (Nyquist rate: $C=2H\log_2 V$)
  - bit rate=baud rate * the number of bits per symbol
  - the number of signal levels does not need to be a power of two
- Clock Recovery
  - **Manchester/**NRZI
  - 4B/5B never be a run of more than three consecutive 0s
- Balanced Signals
  - Signals that have as much positive voltage as negative voltageliangge
  - Bipolar encoding: use two voltage levels to represent a logical 1.(AMI)
  - Using mapping to achieve balance. 8B/10B

Manchester encoding

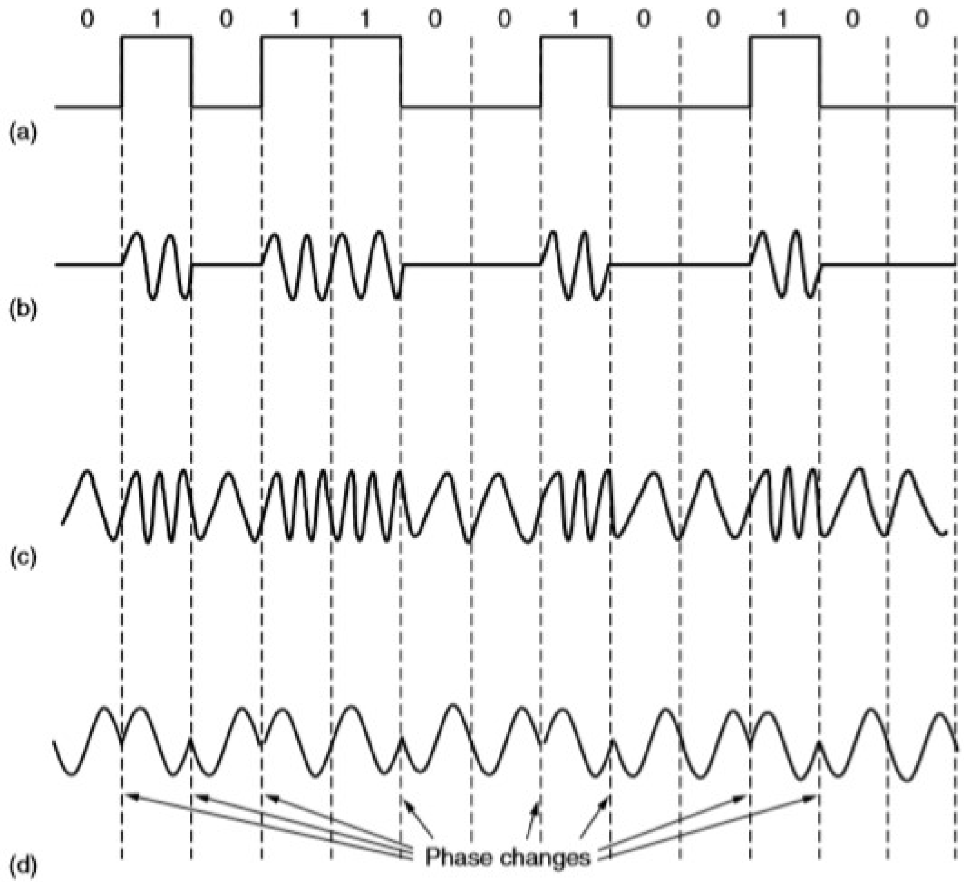
- Manchester
  - Transition in middle of each bit period
  - Transition serves as clock and data
  - Low to high represents 1,High to low represents 0
  - Used by IEEE 802.3 LAN



NRZ vs. Manchester



Modulations(调制)



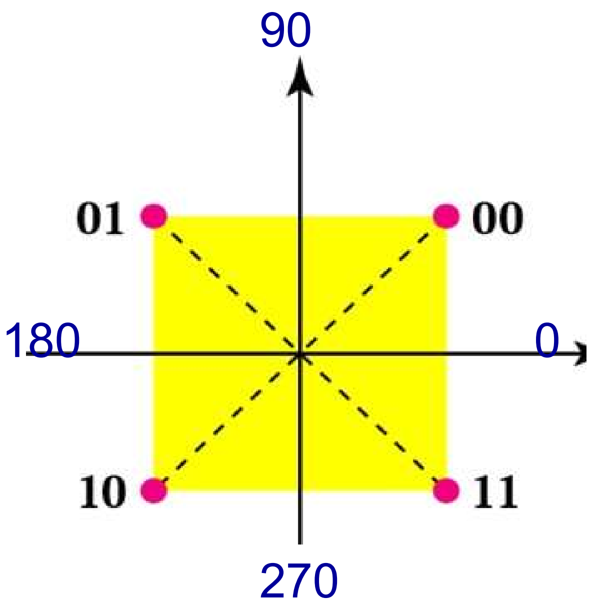
(a) A binary signal
(b) Amplitude shift keying
(c) Frequency shift keying
(d) Phase shift keying

Combination of modulation techniques

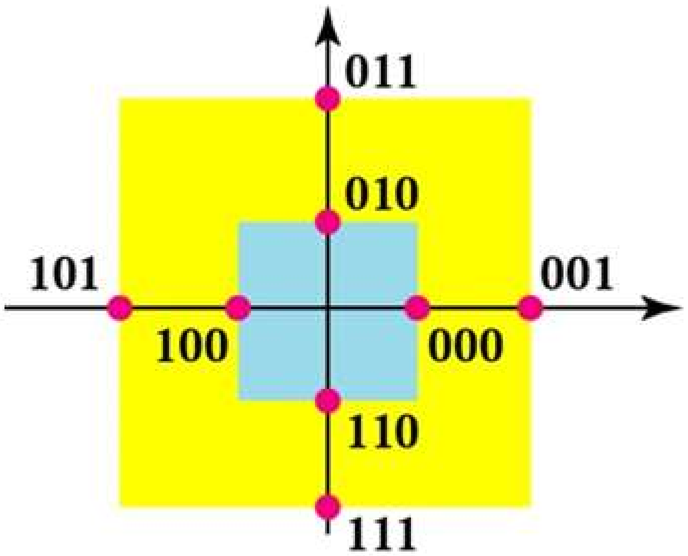
- Modems use a combination of modulation techniques to transmit multiple bits per baud
  - QPSK: Quadrature Phase Shift Keying-正交相移键控
  - QAM: Quadrature Amplitude Modulation-正交幅度调制
- Constellation Diagram
  - The phase of dot is indicated by the angle of a line from it to the origin makes with the positive X-axis.
  - The amplitude of dot is the distance from the origin.

QPSK & QAM

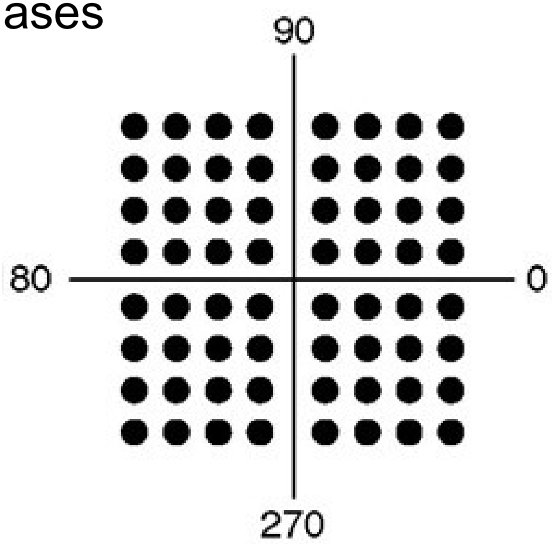
QPSK(4-PSK)
1 amplitude
ases



QPSK(4-PSK)
1 amplitude
4 phases



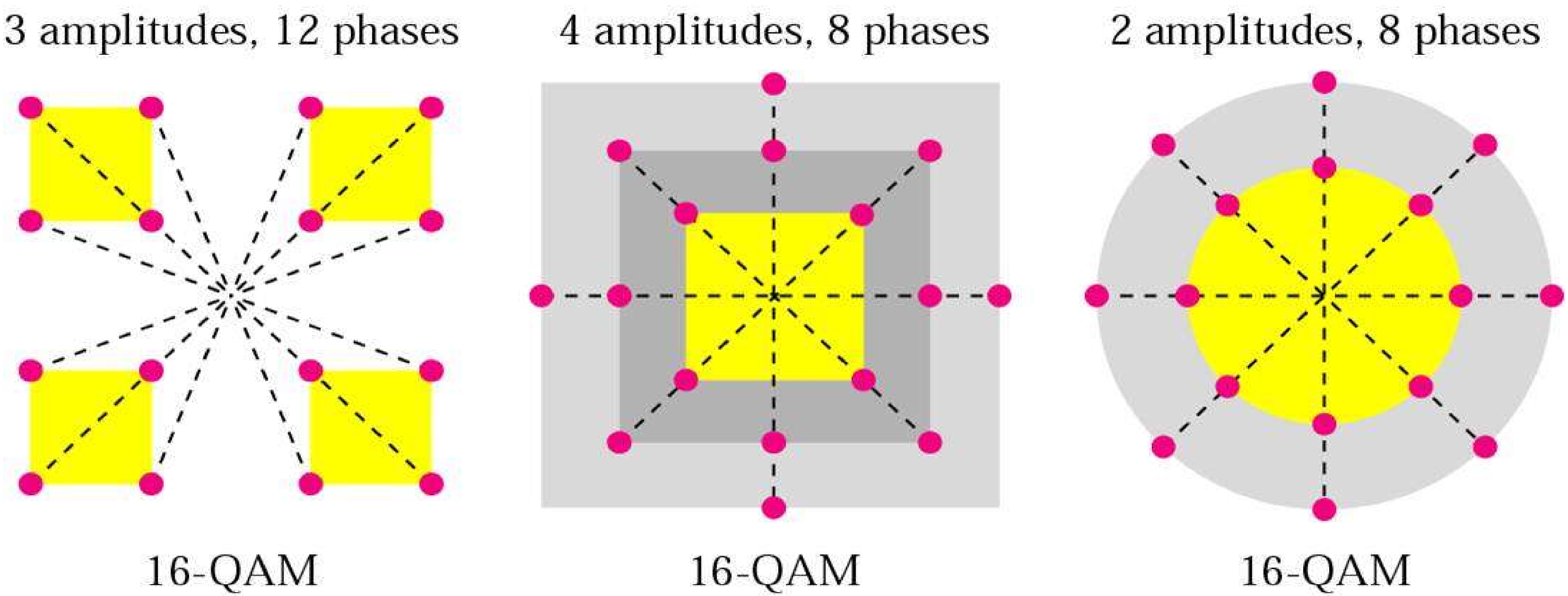
QAM-16



QAM-64

[Forouzan]

QAM-16 constellations



多种实现方式

[Forouzan]

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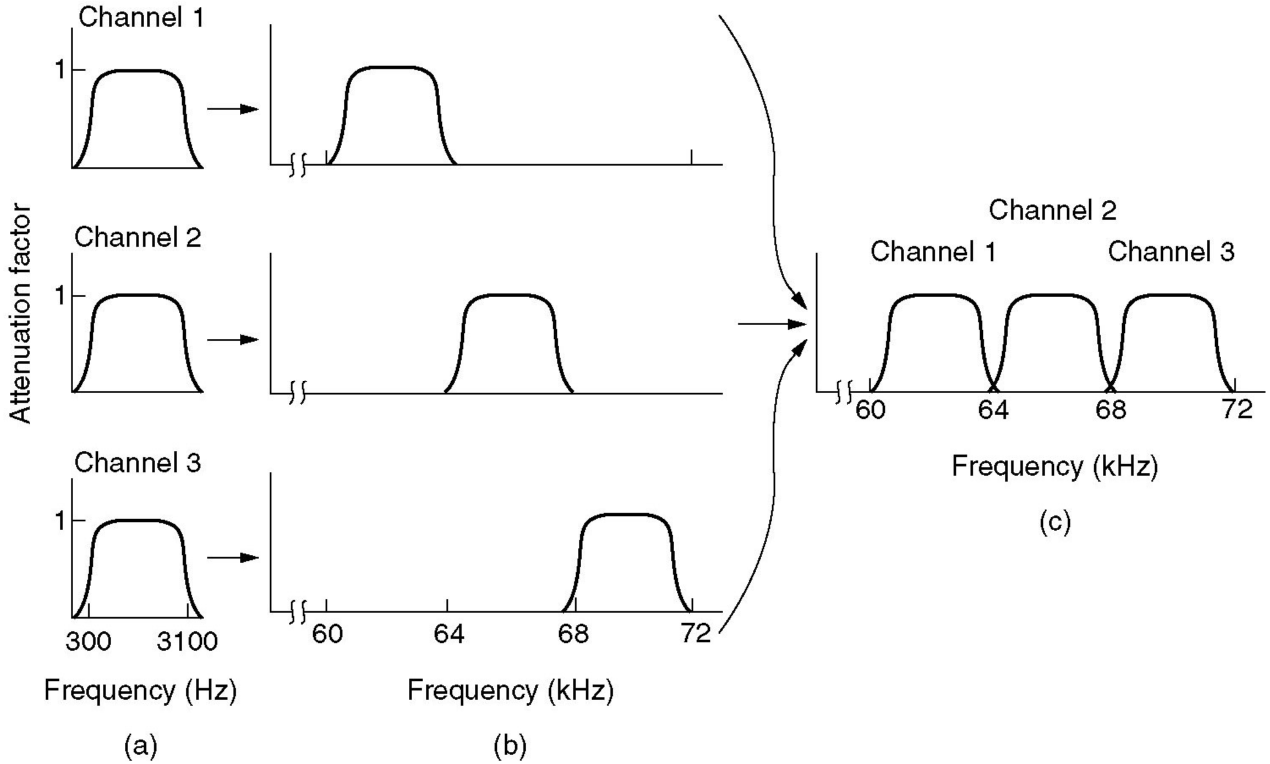
Multiplexing（复用）

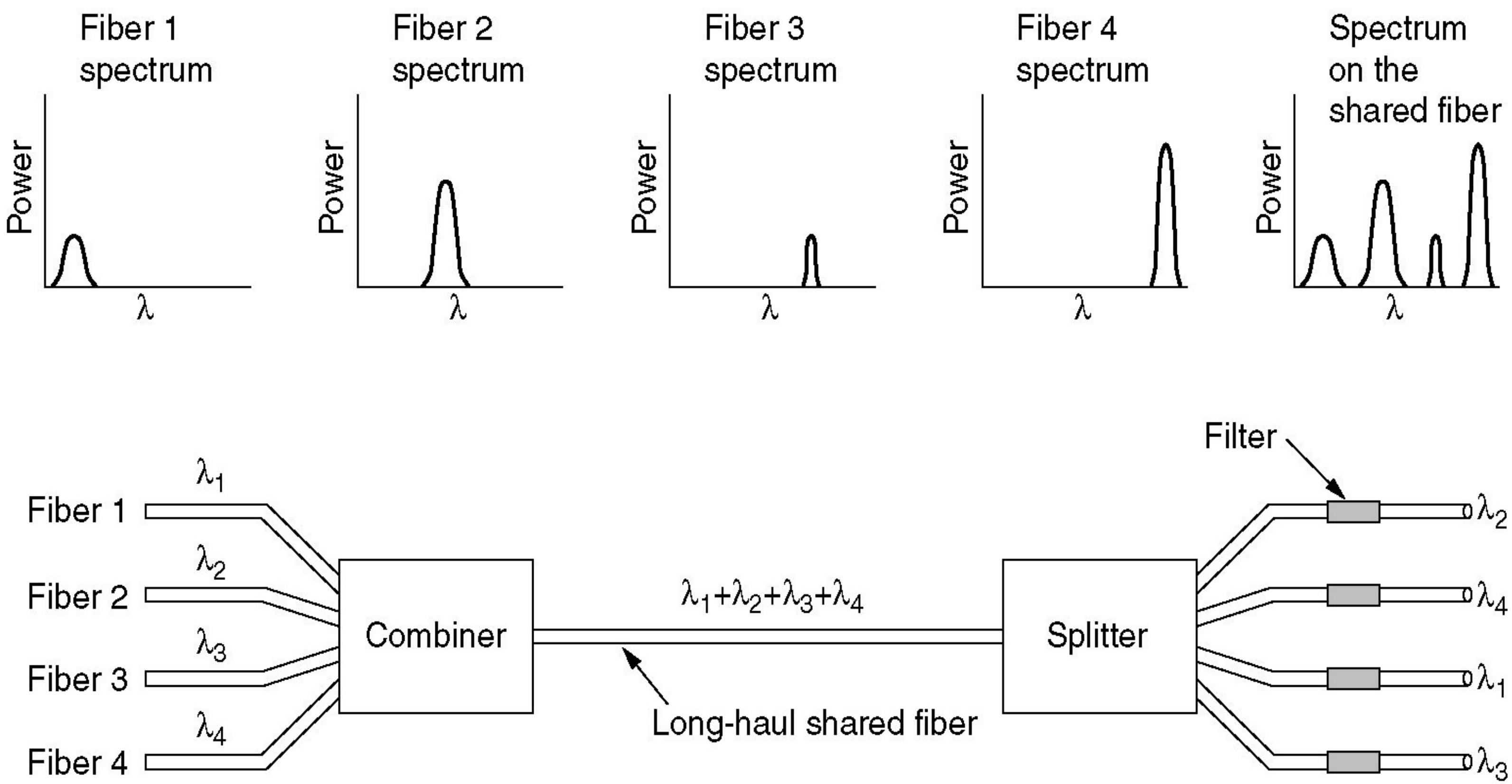
- “refer to a process where multiple analog message signals or digital data streams are combined into one signal” *(Wikipedia)*
  - TDM（时分复用）
  - FDM（频分复用）
  - WDM（波分复用）
  - CDM（码分复用）

Frequency Division Multiplexing



(a) The original bandwidths.
(b) The bandwidths raised in frequency.
(b) The multiplexed channel.

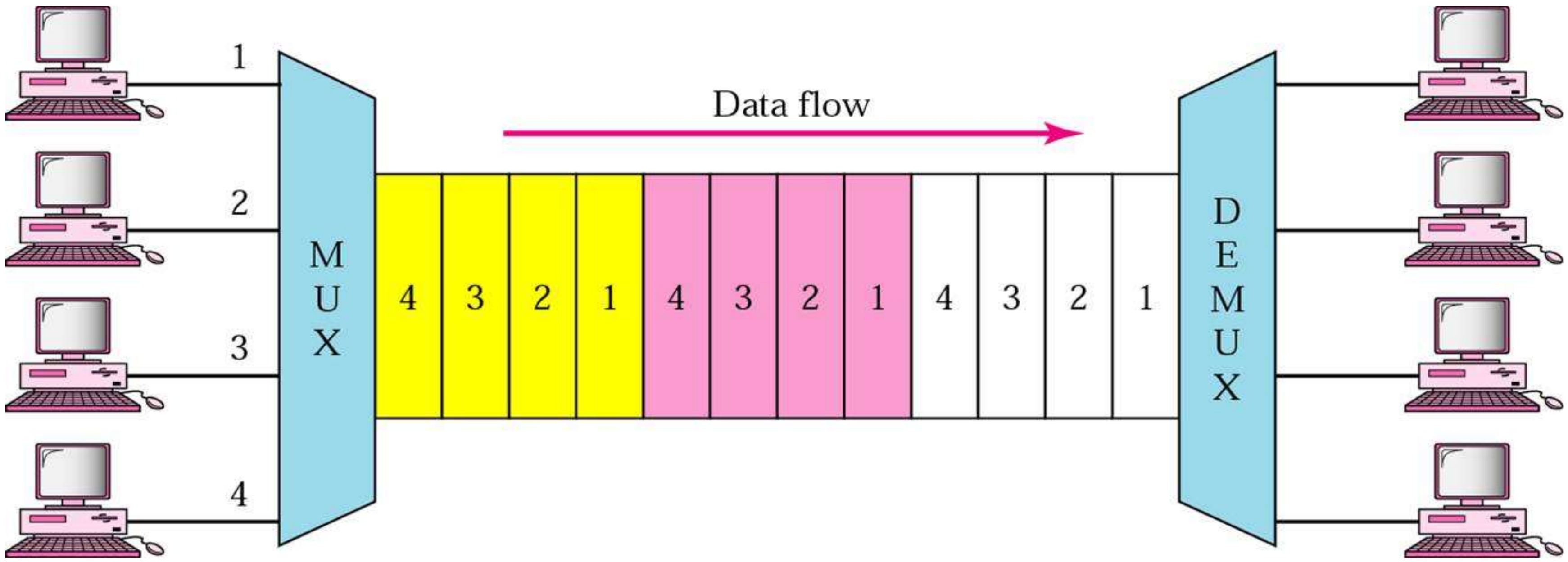
Wavelength Division_Multiplexing



WDM (Wavelength Division Multiplexing).
DWDM (Dense WDM).

Time Division Multiplexing

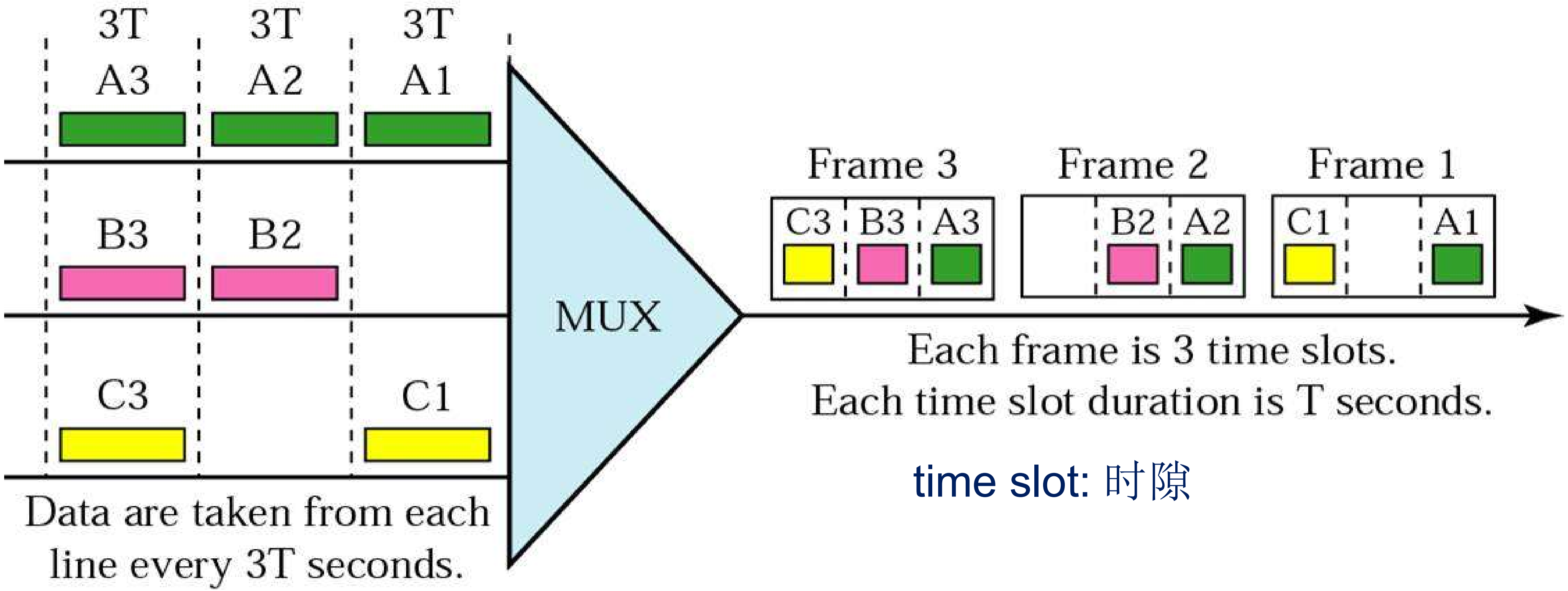
- Signals are transferred apparently simultaneously as sub-channels in one communication channel, but are physically taking turns on the channel.
- The time domain is divided into several recurrent timeslots of fixed length, one for each sub-channel.



[Forouzan]

TDM Frames（复用帧）

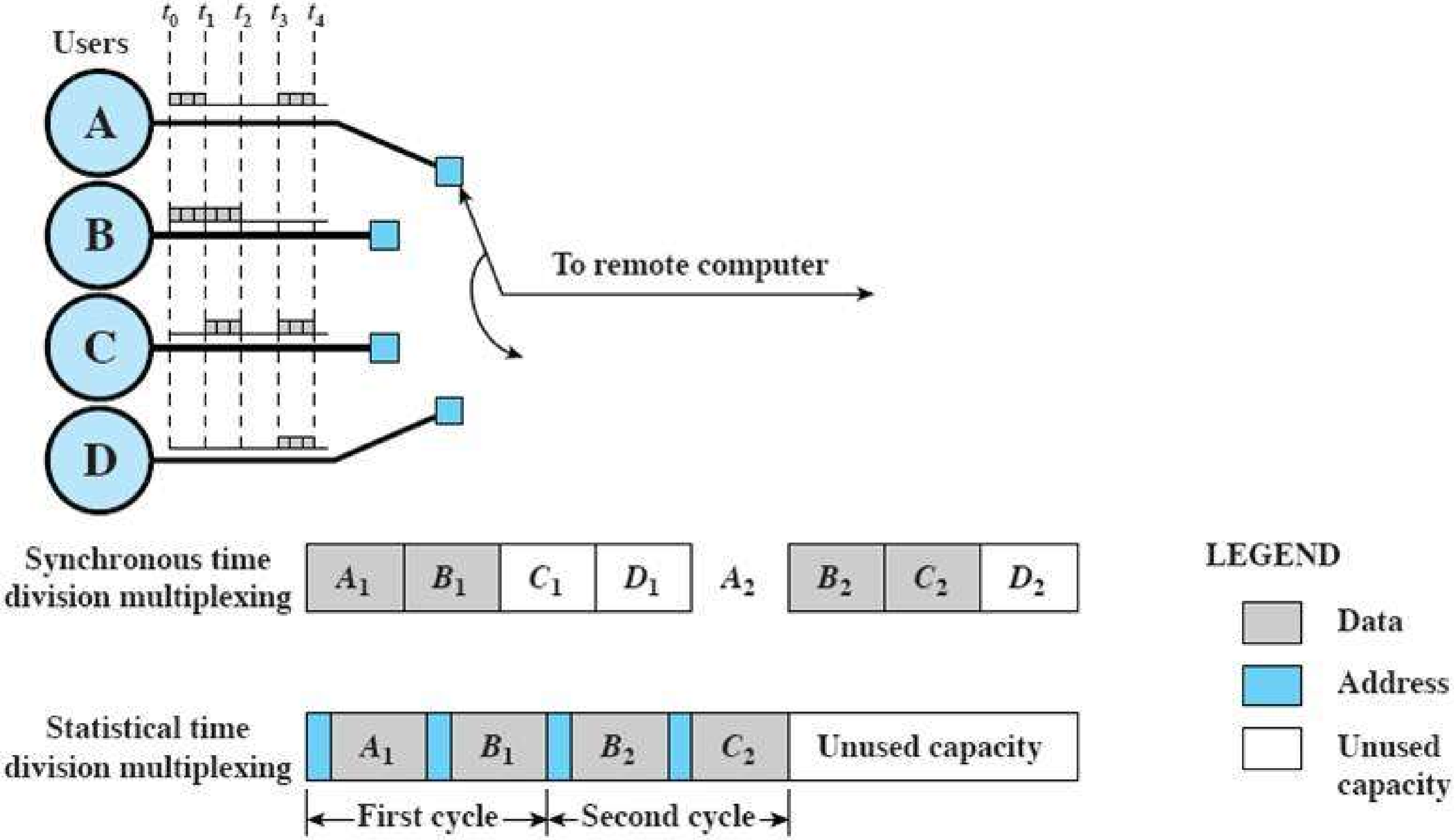
- May cause waste of resources



time slot: 时隙

[Forouzan]

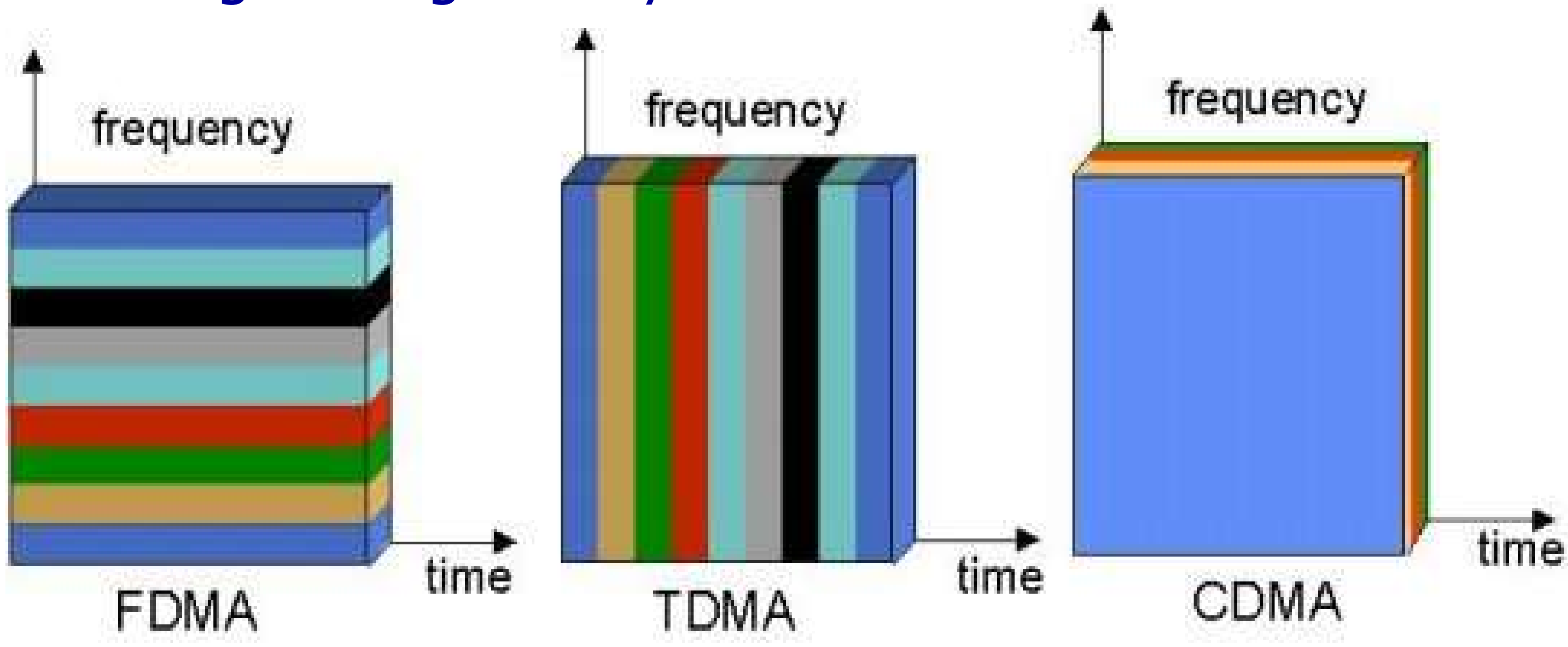
Synchronous TDM vs. Statistical TDM



[Stallings]

CDM（码分复用）

- A form of spread spectrum communication
- Code Division Multiple Access:Each station can transmit over the entire frequency spectrum all the time.
- Multiple simultaneous transmissions are separated using coding theory.



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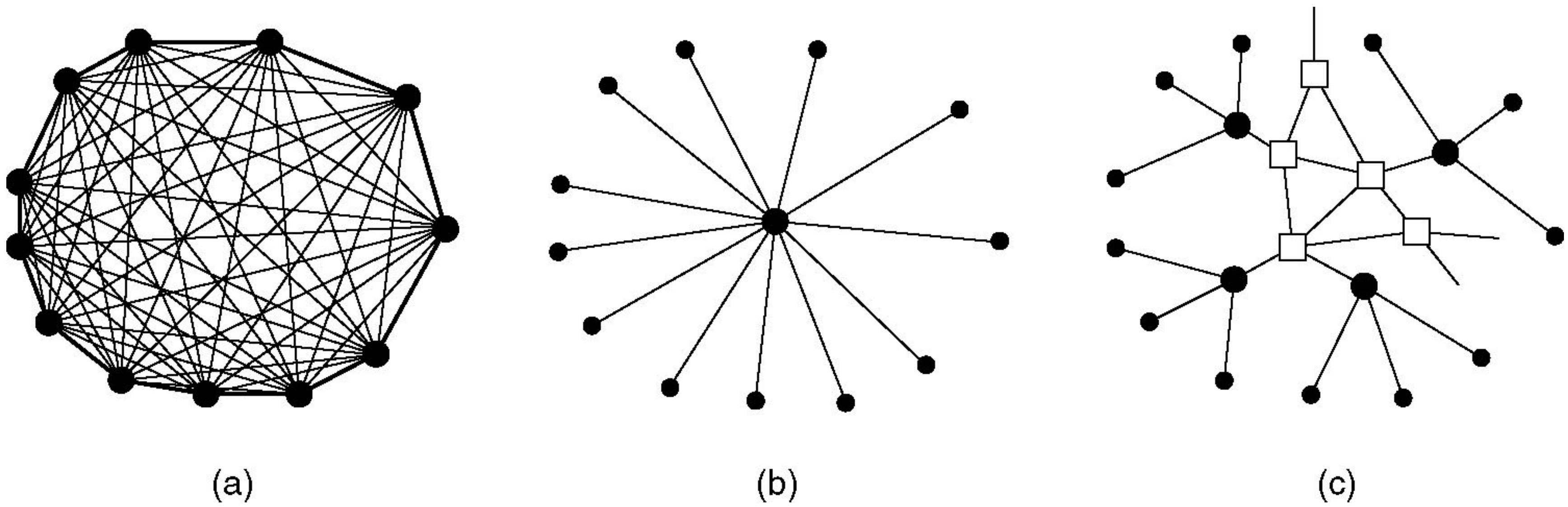
Structure of the Telephone System



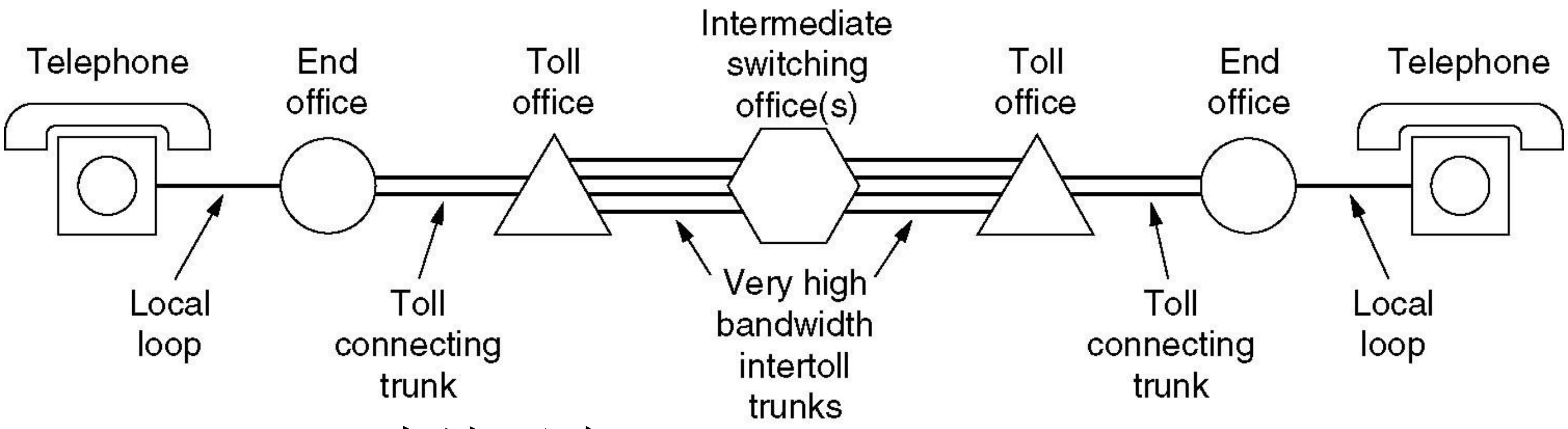
(a) Fully-interconnected network.

(b) Centralized switch.

(c) Two-level hierarchy.

Structure of Telephone System

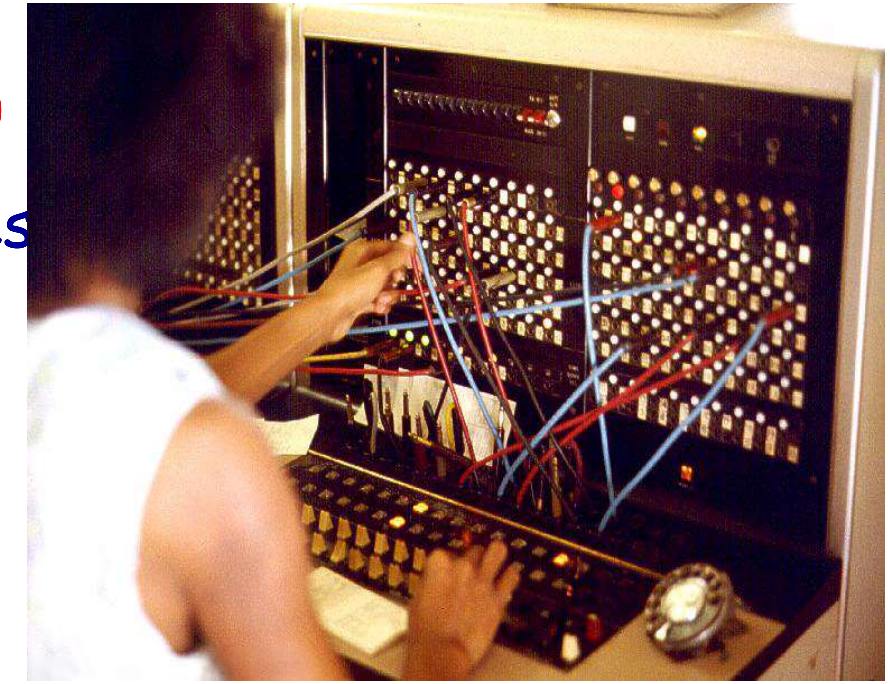
- A typical circuit route for a medium-distance call

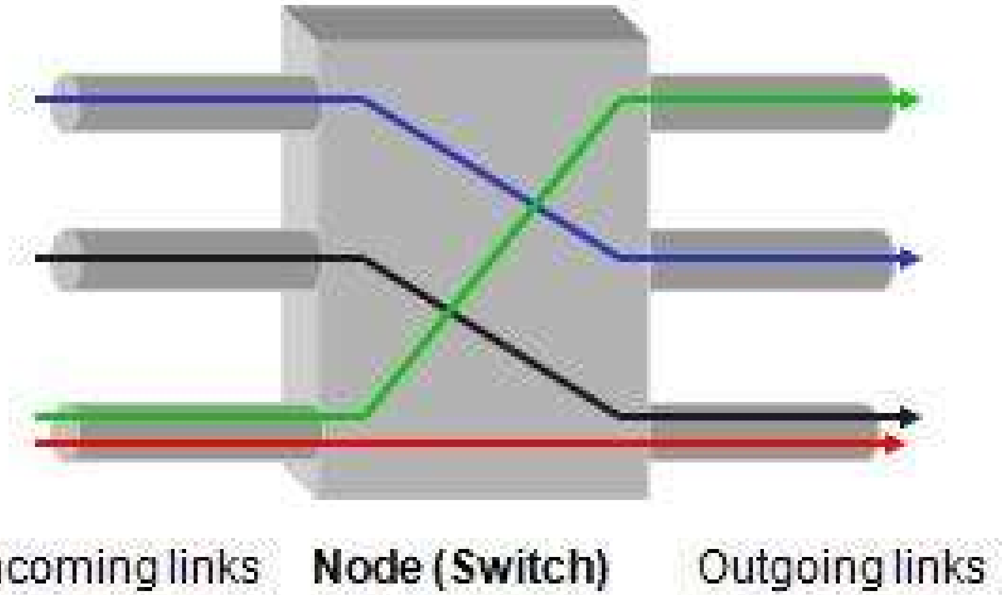


- Local loops 本地环路
  - Analog twisted pairs going to houses and businesses
- Trunks 中继
  - Digital fiber optics connecting the switching offices
- Switching offices 交换局(End office端局、Tandem office汇接局、Toll office长途局)
  - Where calls are moved from one trunk to another

What is Switching?(交换)

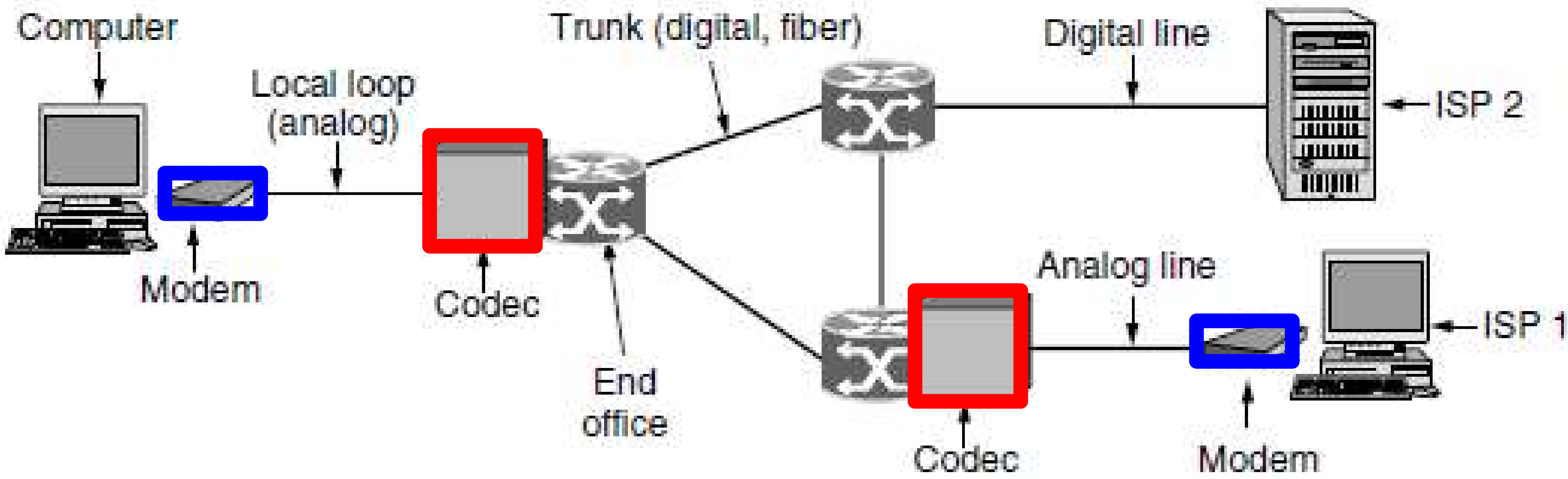
- In telephone system, switches interconnect telephone subscriber lines or virtual circuits of digital systems to establish telephone calls between subscribers.





Digital and Analog Transmission in PSTN

The use of both analog and digital transmissions for a computer to computer call. Conversion is done by the Modems and Codecs.



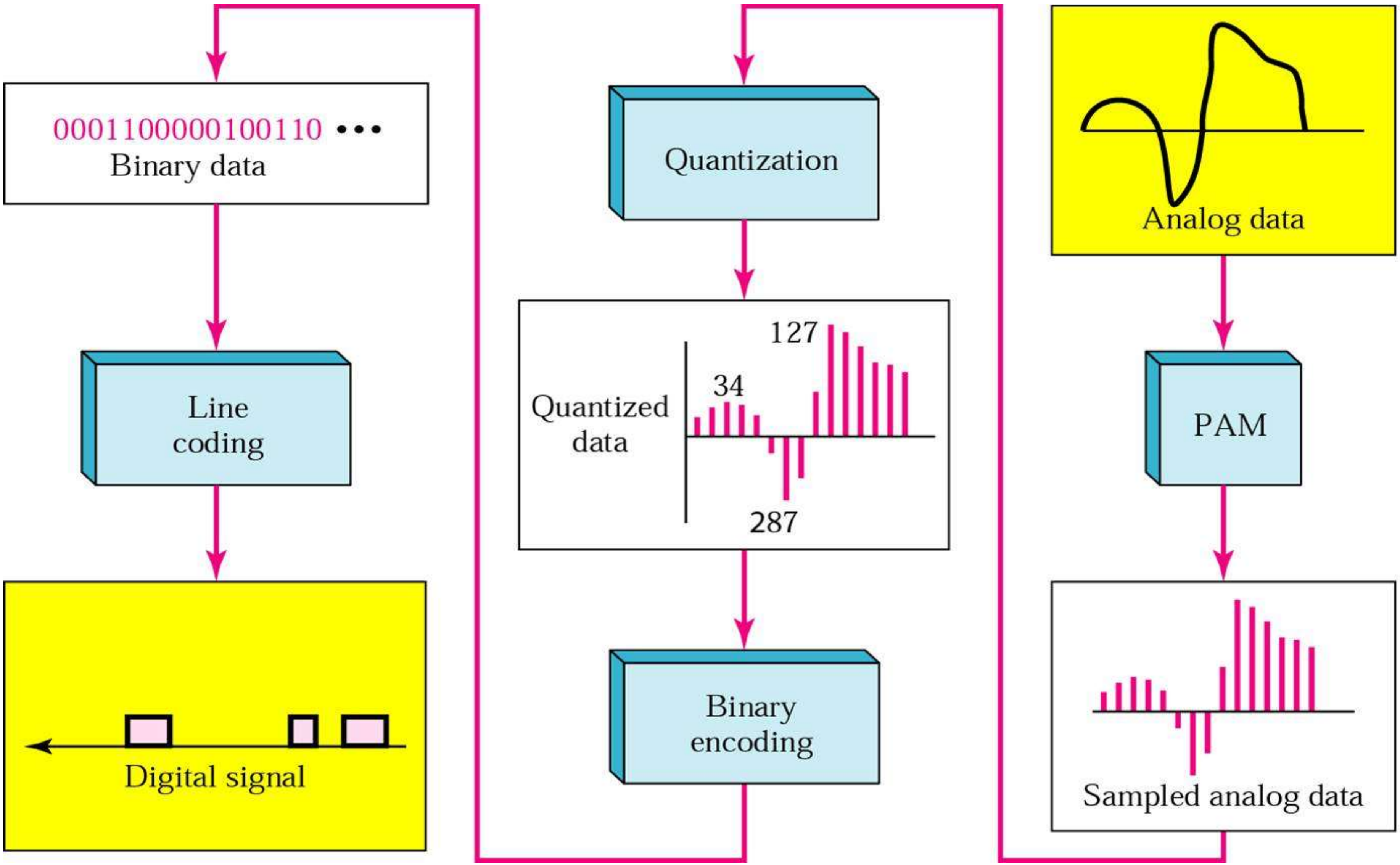
Codec(编码解码器):

- 8000 samples per second，1 sample = 8 bits (非线性编码)

Digitized Analog Signals

- 数字化语音信号PCM：Pulse Code Modulation（脉冲编码调制）
- Device: Codec编码解码器(digitized analog signals)
- Procedures
  - Sampling（采样）
    - According to Nyquist theorem, the sampling rate must be at least 2 times the highest frequency.
  - Quantizing（量化）
    - Dividing signal strength into levels linearly or logarithmically
    - A Law algorithm：Used in European and China
    - Mu Law(μ-Law) algorithm:Used in North America and Japan
  - Encoding（编码）

From analog signal to PCM digital code

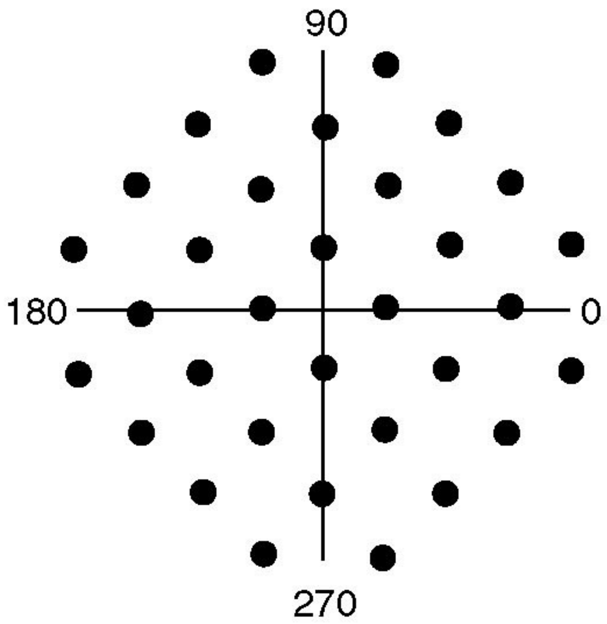


[Forouzan]

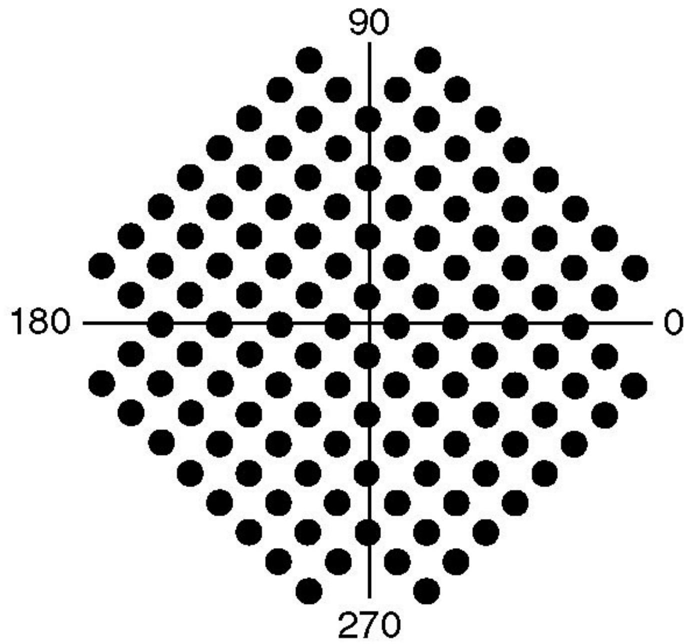
Modems

- Modems use a combination of modulation techniques to transmit multiple bits per baud （multiple amplitudes and multiple phase shifts）
  - QPSK: Quadrature Phase Shift Keying-相移键控
  - QAM： Quadrature Amplitude Modulation-正交幅度调制
- **Modems sample 2400 times/sec** and focus on getting more bits per sample
  - V.32: 2400 baud * (5-1) bits = 9.6Kbps
  - V.32bis: 2400 baud * (7-1) bits = 14.4Kbps
  - V.34: 2400 baud * 12 bits = 28.8Kbps
  - V.34bis: 2400 baud * 14 bits = 33.6Kbps
- V.90(56kbps)

V.32 & V.32bis



(a)

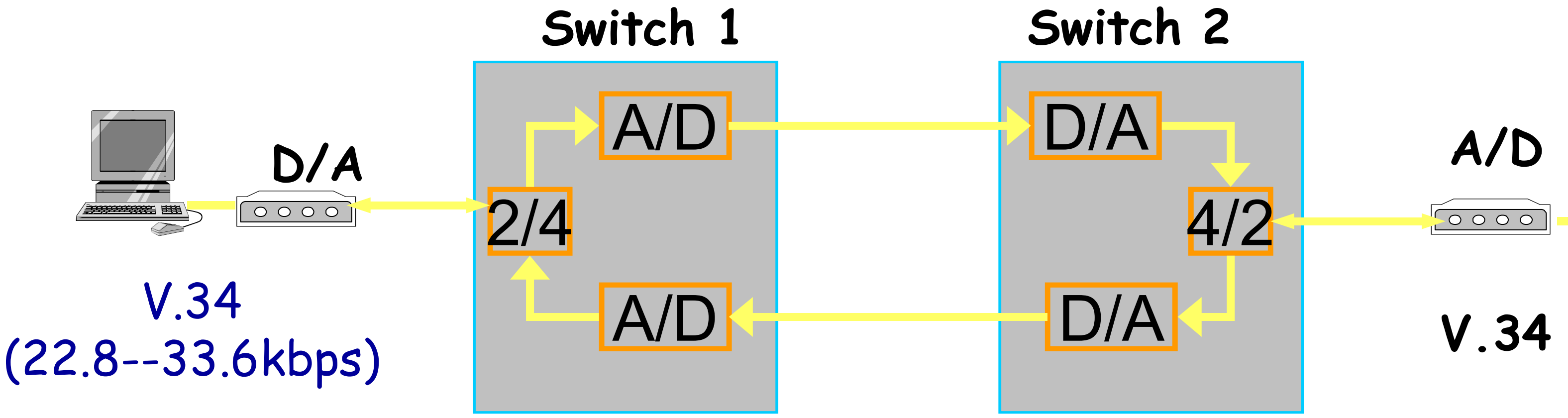


(b)

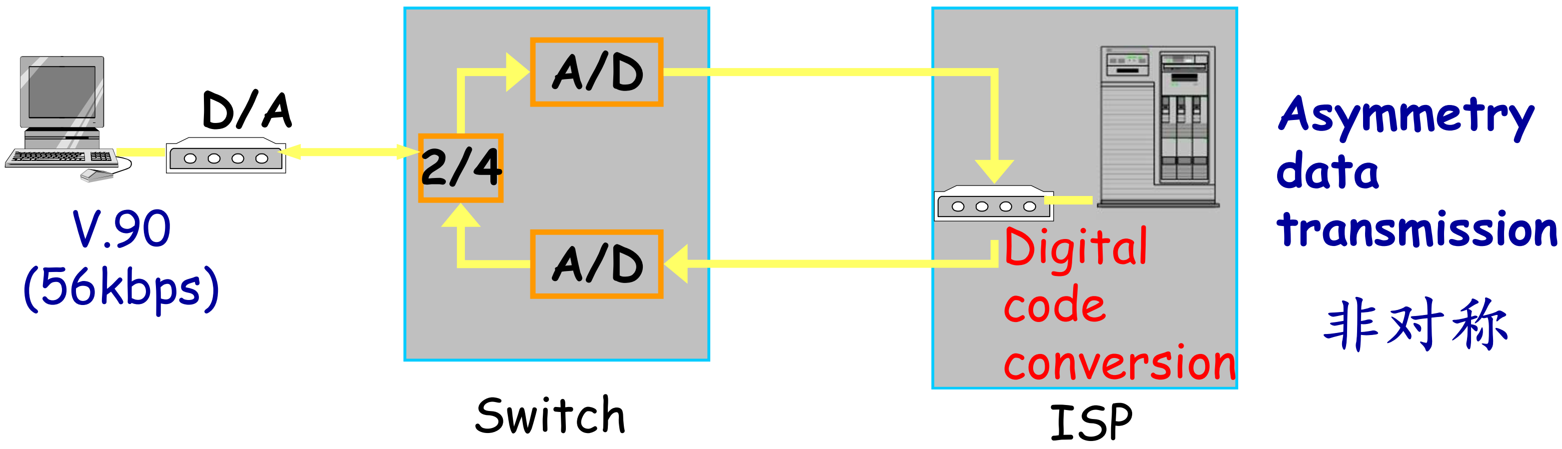
(a) V.32 for 9600 bps.

(b) V32 bis for 14,400 bps.

Maximum Data Rate of Dial-up

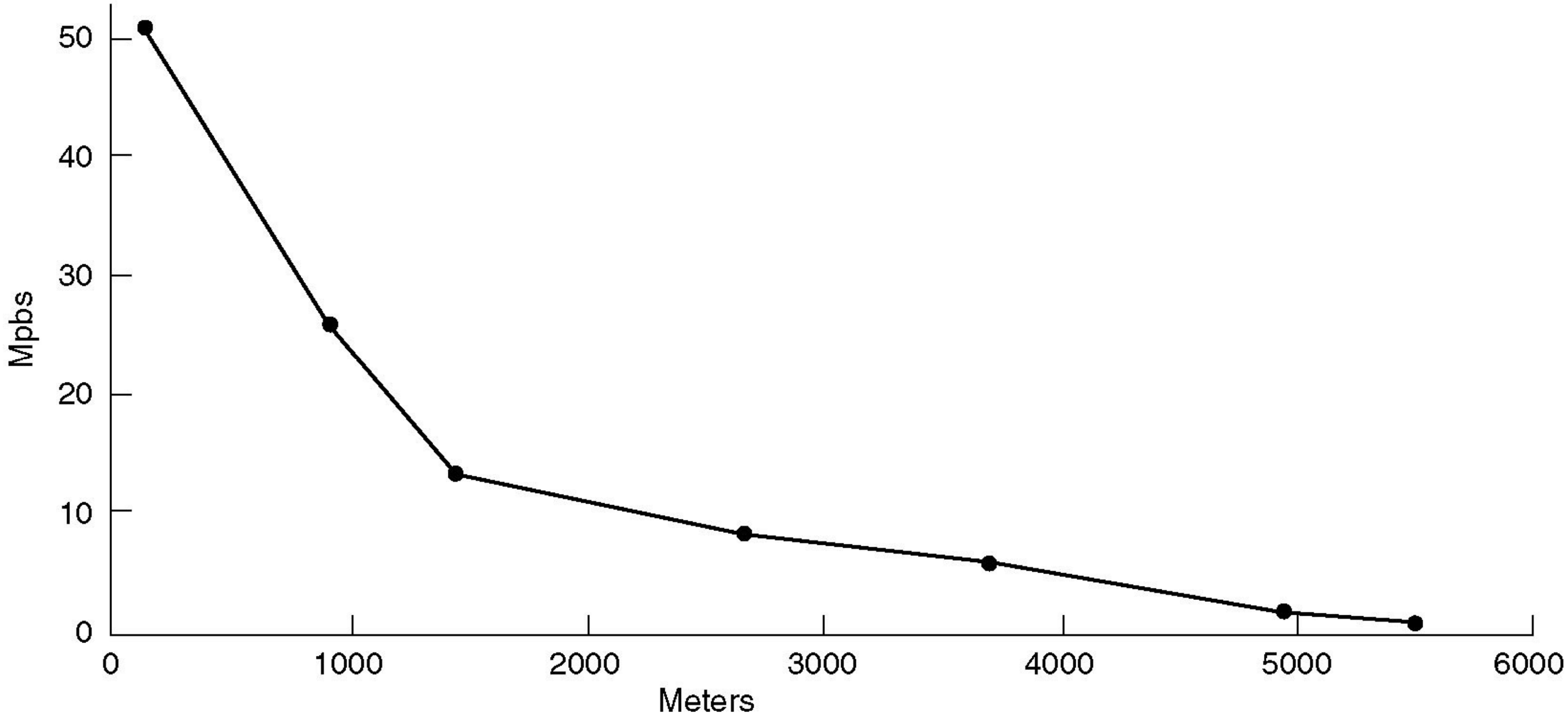


Noise comes from local loops. Shannon limit of telephone system is about 35kbps



DSL: Digital Subscriber Lines (数字用户线)

Bandwidth versus distances over UTP3 for DSL

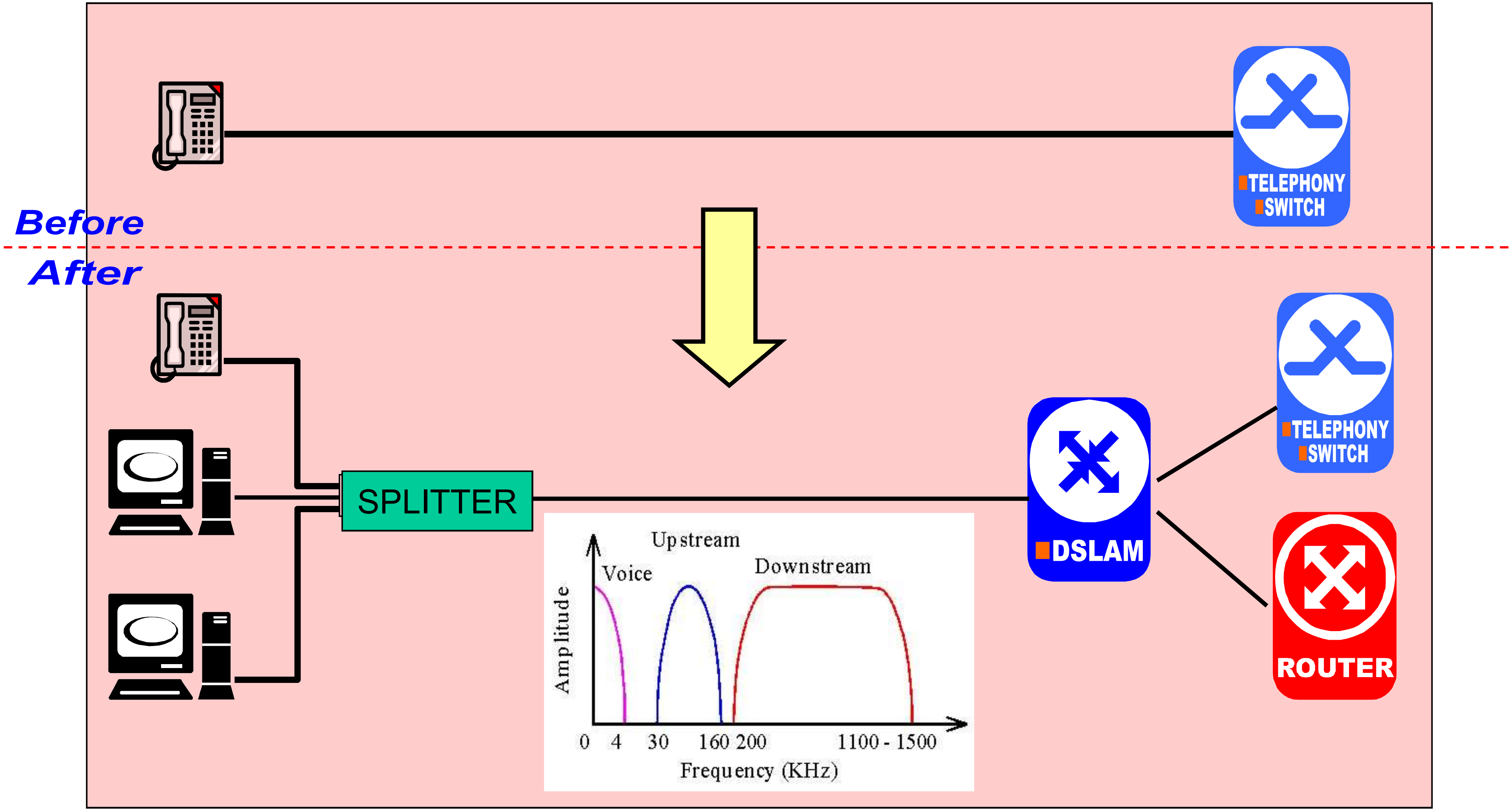


xDSL Services

- Goal of xDSL services
  - The services must work over the existing **UTP3** local loops
  - They must not affect customers' existing telephones and fax machine
  - They must be much **faster than 56kbps**
  - They should be always on, with just a monthly charge but no per-minute charge

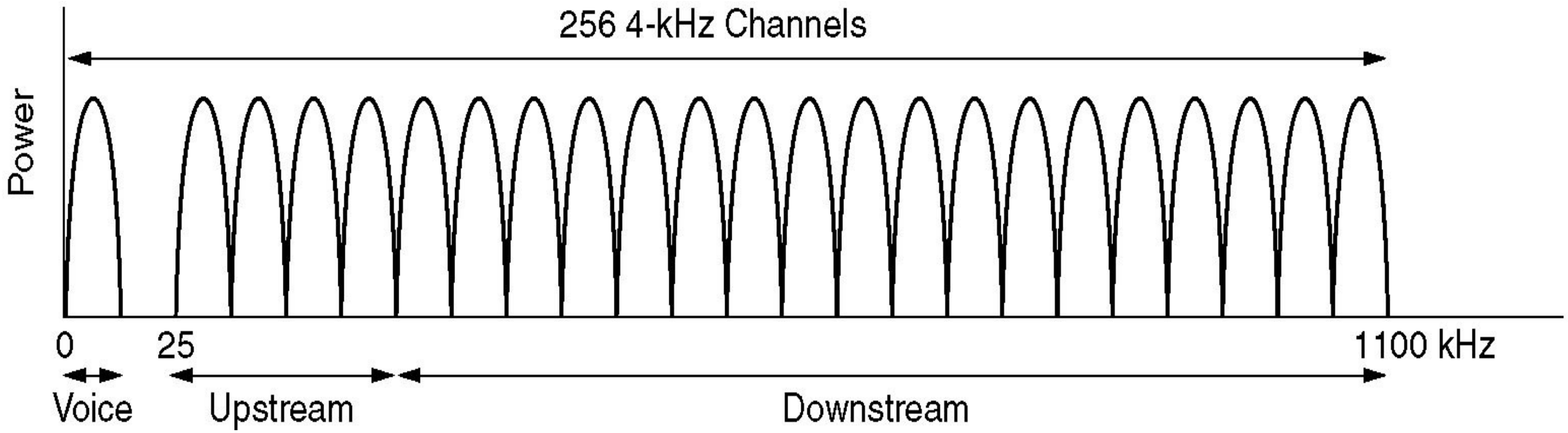
- ADSL: Asymmetric DSL(非对称数字用户线)

ADSL接入



Before
After
TELEPHONY
SWITCH
SPLITTER
DSLAM
TELEPHONY
SWITCH
ROUTER
Upstream
Voice
Downstream
Amplitude
0 4 30 160 200 1100 - 1500
Frequency (KHz)

ADSL: Discrete Multitone Modulation



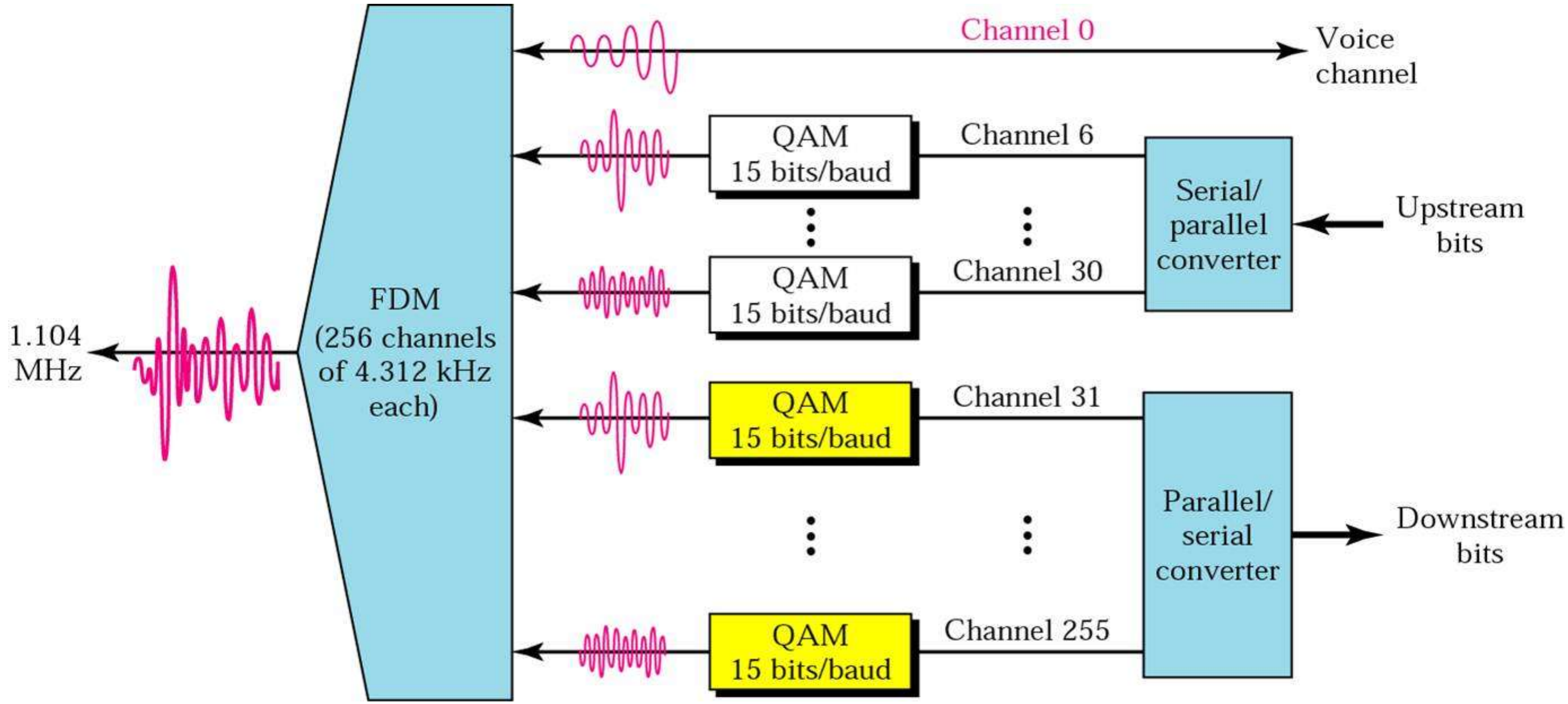
- Divide 1.1 MHz into 256 channels of 4312.5 Hz
- Channel 0 for POTS(电话)
- 1-5 not used, to keep voice signal and data signals from interfering with each other
- 250 channels
  - Upstream control (1)
  - Downstream control (1)
  - User data (248)

FDM(频分复用)

ADSL Modulation Scheme

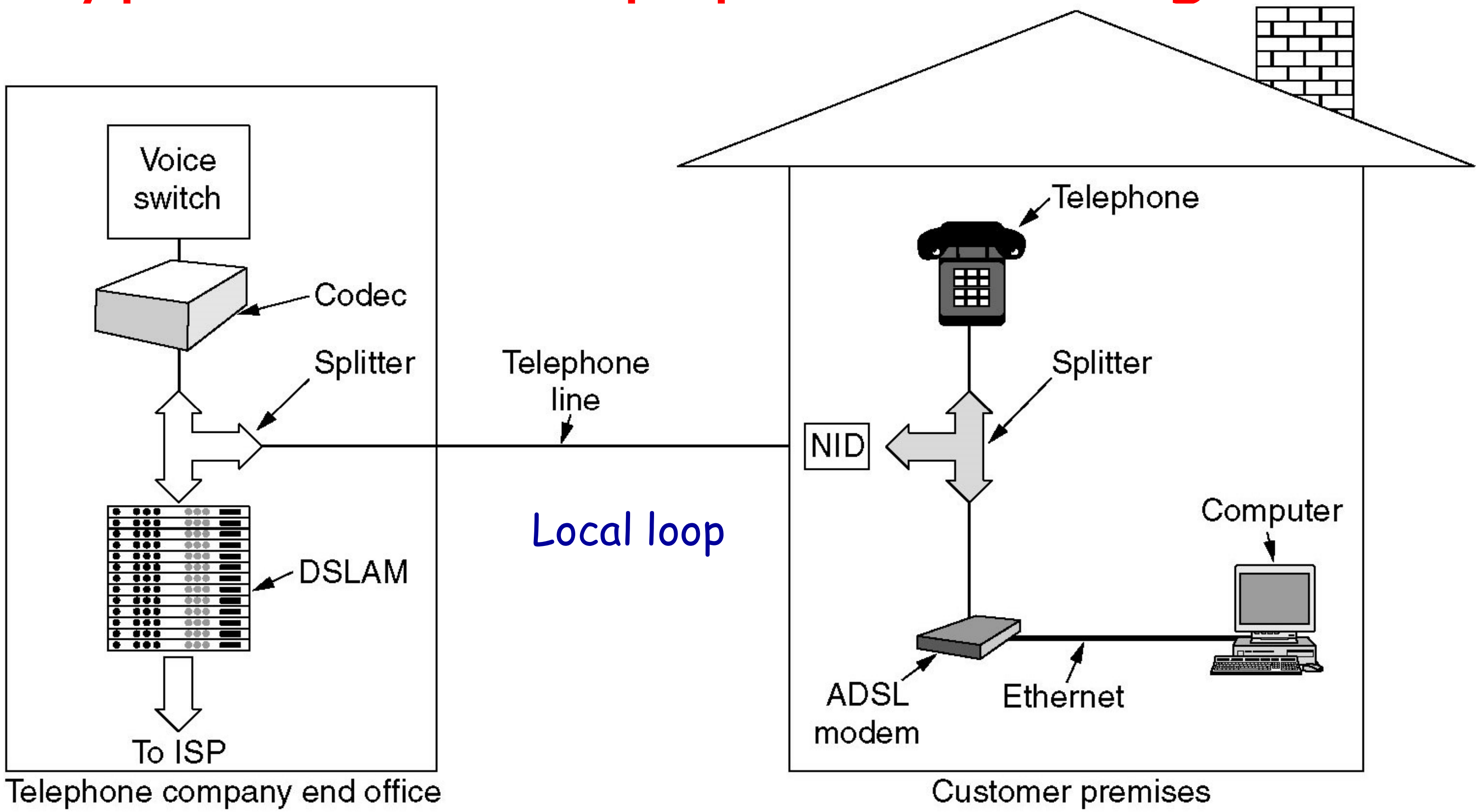
- Within each channel, a modulation scheme similar to V.34 is used
  - Sampling rate is **4000 baud**
  - The line quality in each channel is monitored and data rate adjusted continuously
  - The actual data are sent with QAM(正交幅度调制), with up to 15 bits/baud
- Typical data rate
  - Standard service: 1M bps(downstream), 256 kbps(upstream)
  - Improved service: 4Mbps, 1Mbps
  - Premium service: 8 Mbps, 2M kbps

DMT: Discrete Multitone Modulation (离散多音调制)



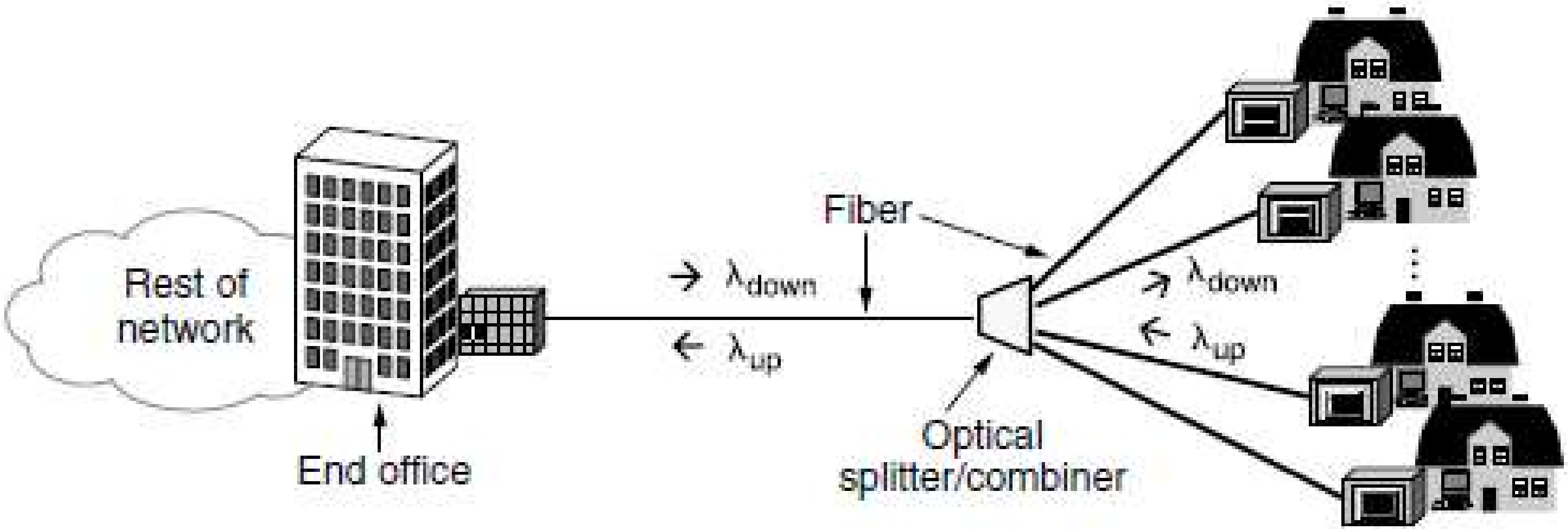
QAM： Quadrature Amplitude Modulation-正交幅度调制

A typical ADSL equipment configuration



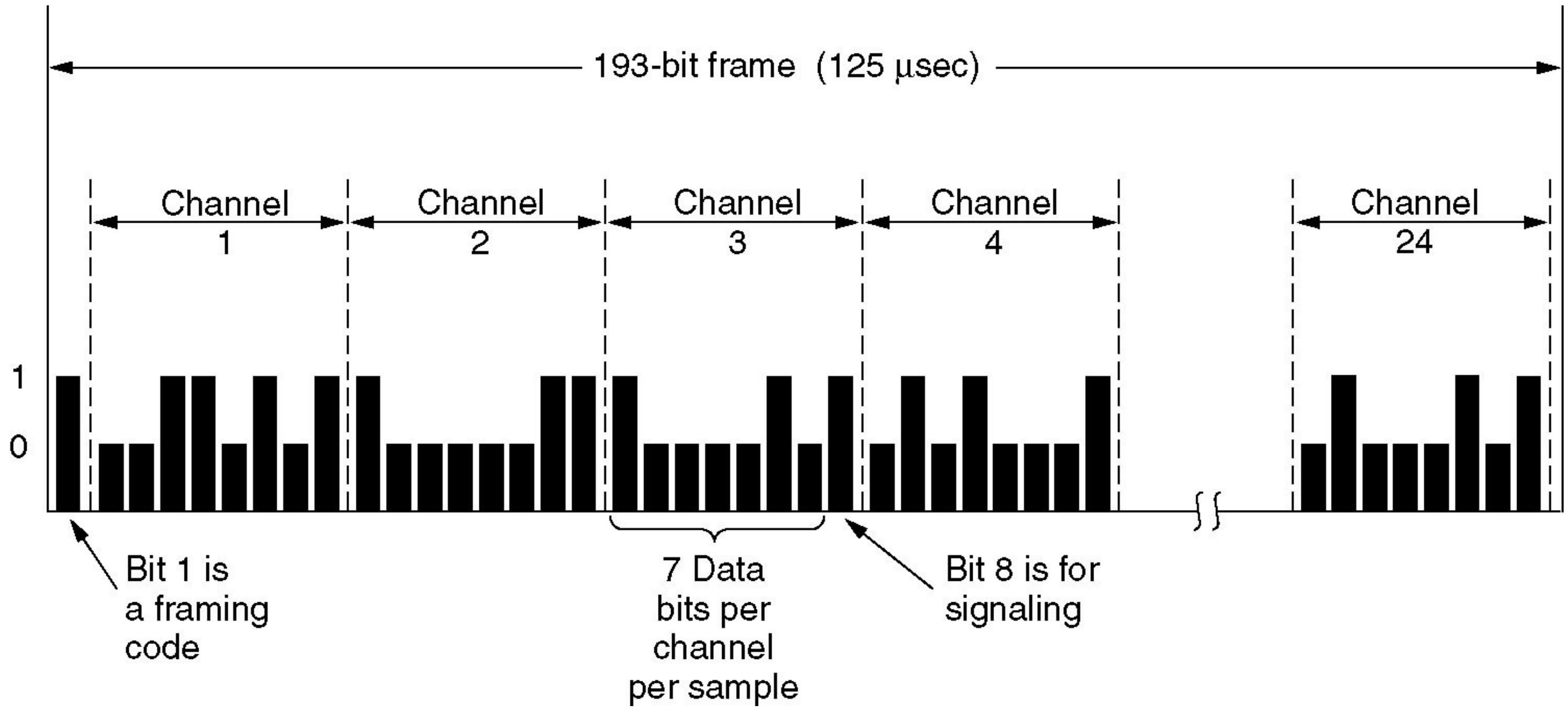
- Spliter:
- DSLAM: Digital Subscriber Line Access Multiplexer
- NID: Network Interface Device

FTTH（光纤入户）



PON(Passive Optical Network) 无源光网络

Time Division Multiplexing in PSTN(1)

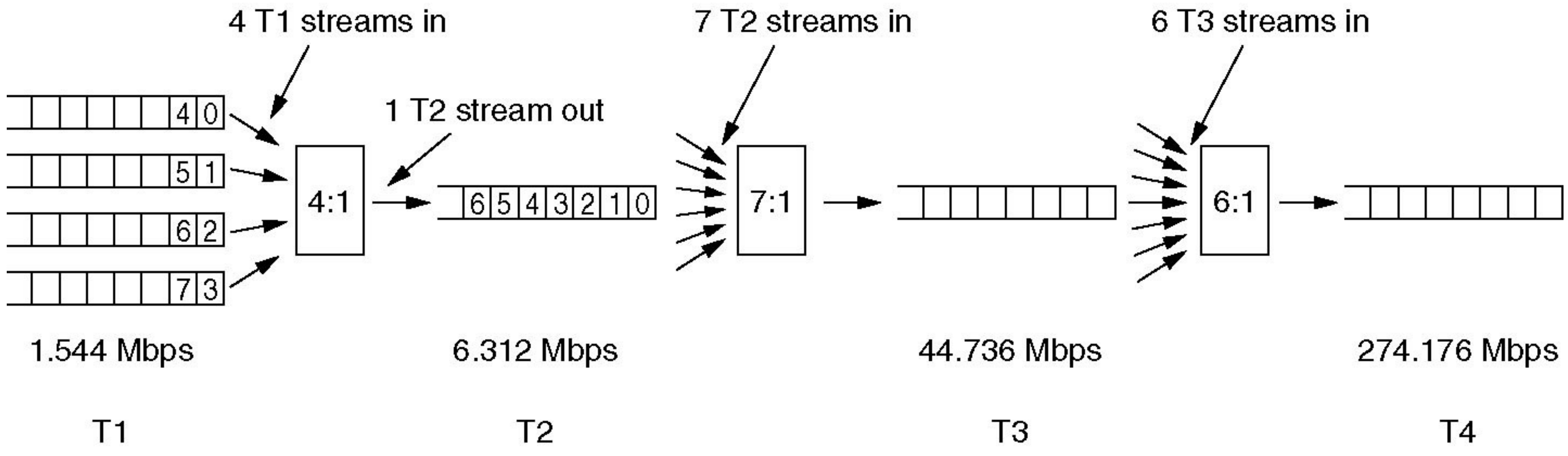


T1 ---1.544 Mbps

E1 ---2.048Mbps

Time Division Multiplexing (2)

Multiplexing T1 streams into higher carriers.



Summary

- 调制技术：改变基带信号的频率
  - 典型技术：ASK,FSK,PSK,QPSK,QAM
  - 典型设备：Modem(2400baud), ADSL(4000baud)
- 复用技术：FDM,TDM,CDM,WDM
- 三个设备：
  - 数字发送器---线路编码技术，数字基带信号，抗干扰、带宽效率
  - 调制解调器---调制解调技术，信号频率变换，通带信号，长距离传输、有效利用带宽
  - 编码解码器---采样、量化、编码技术，将模拟信号数字化，将模拟信源的输出变为数字数据

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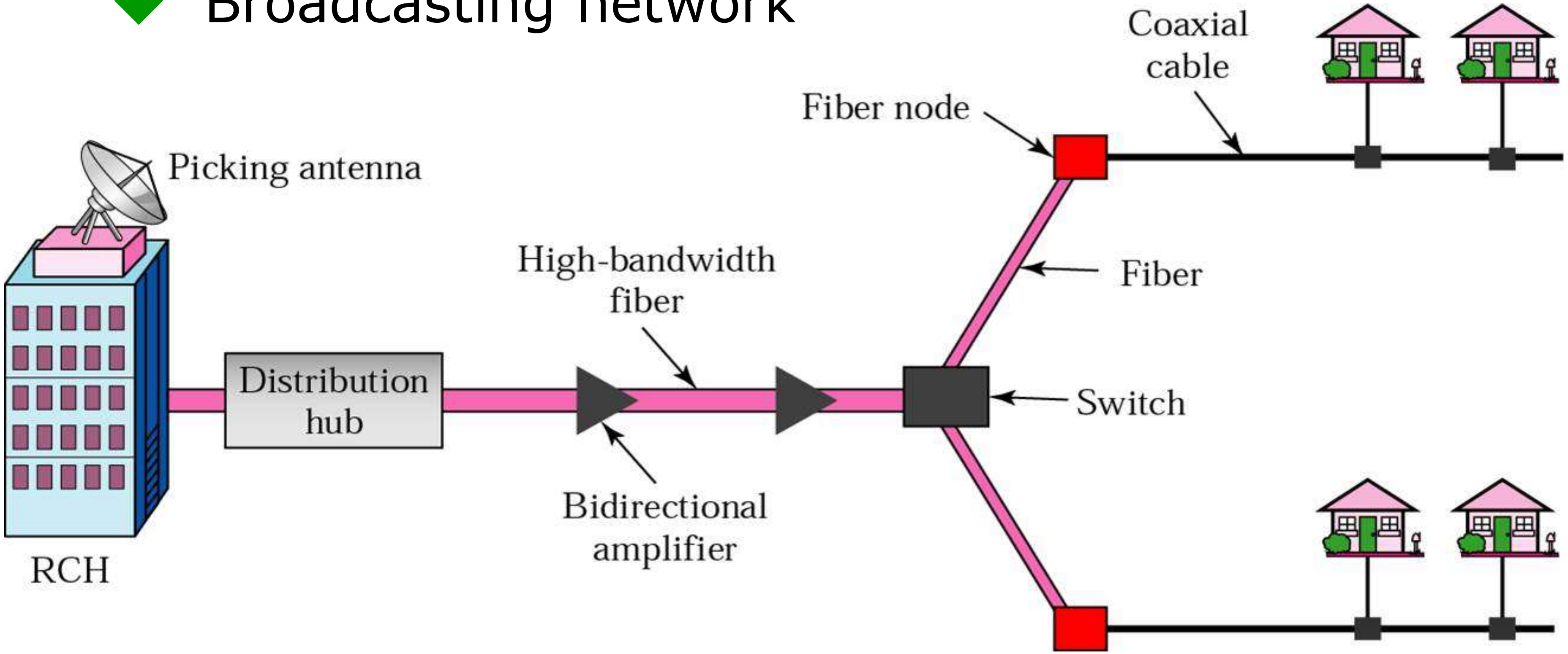
Structure of Cable TV System

- HFC: Hybrid Fiber-Coaxial(混合光纤同轴)
  - Tree topology
  - Broadcasting network

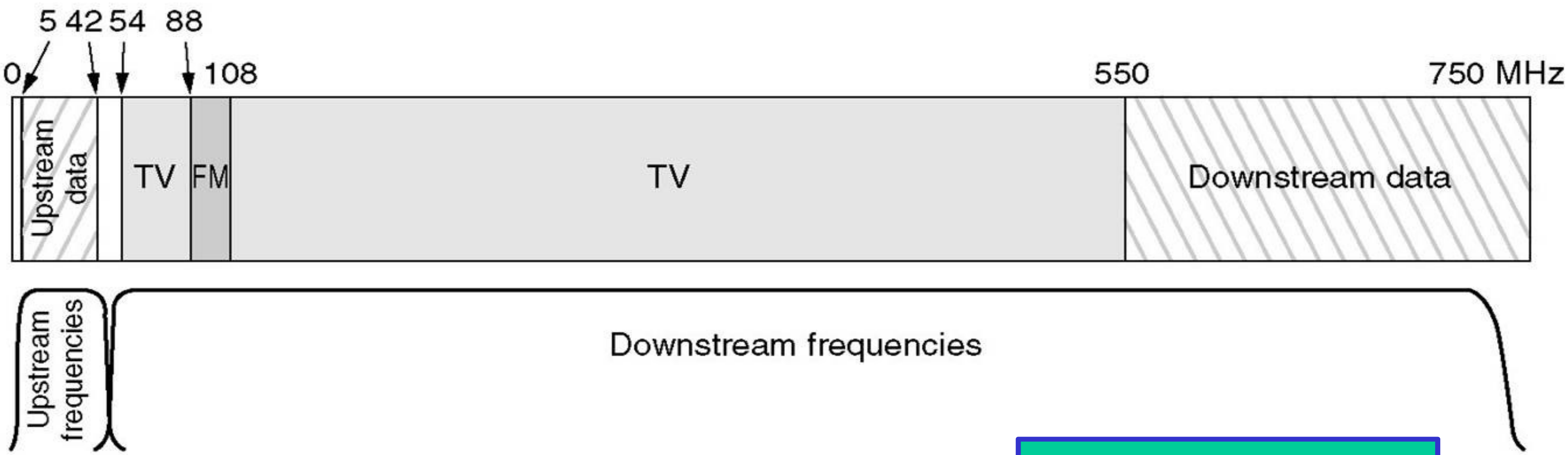


- RCH: regional cable head

[Forouzan]

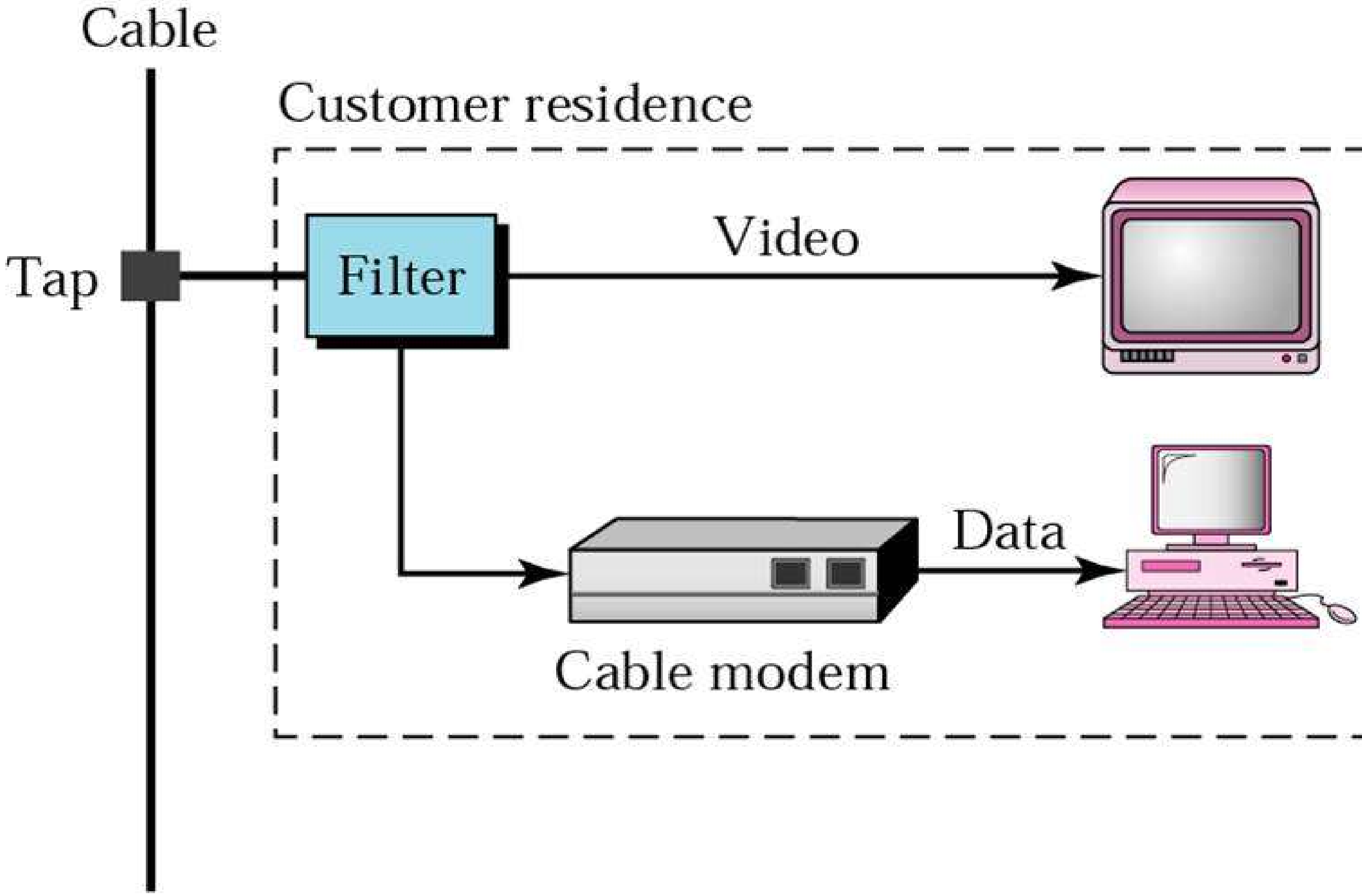
Spectrum Allocation in Typical CATV System

- **Asymmetric Channel**
  - Downstream data modulation: 64-QAM, with theoretical data rate of 30 Mbps
  - Upstream data modulation: QPSK(正交相移键控), with theoretical data rate of 12 Mbps.



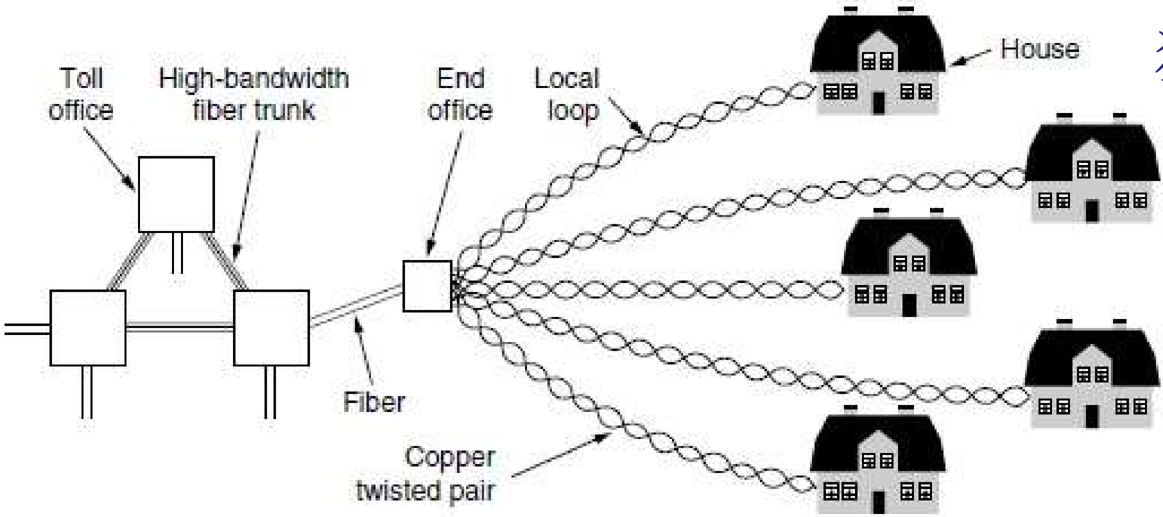
FDM(频分复用)

Cable Modem

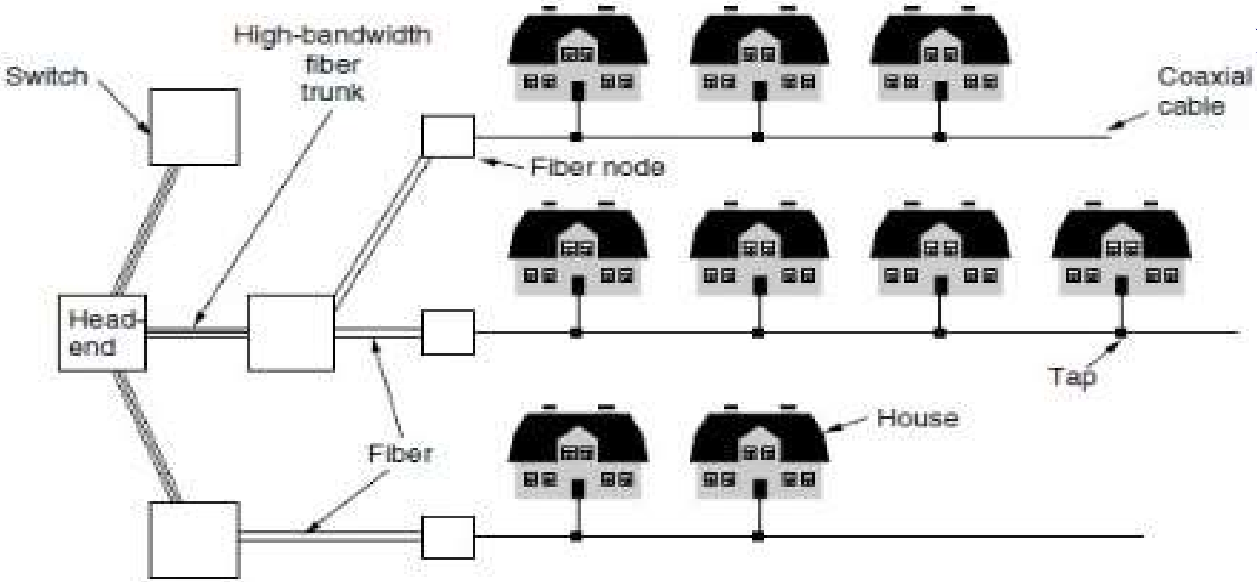


Cable
Customer residence
Tap
Filter
Video
Data
Cable modem

ADSL vs. Cable



独占DSL



共享电缆

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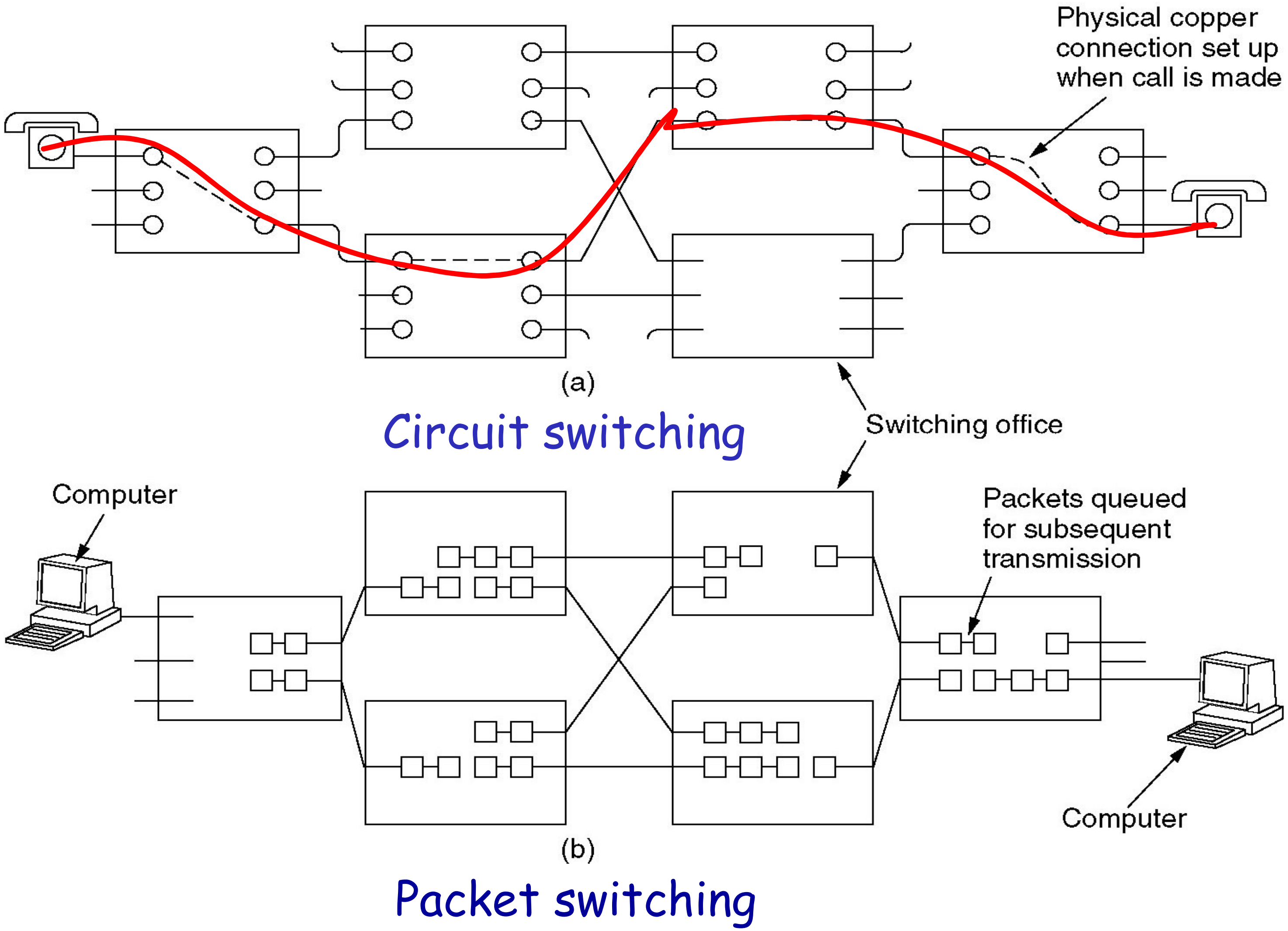
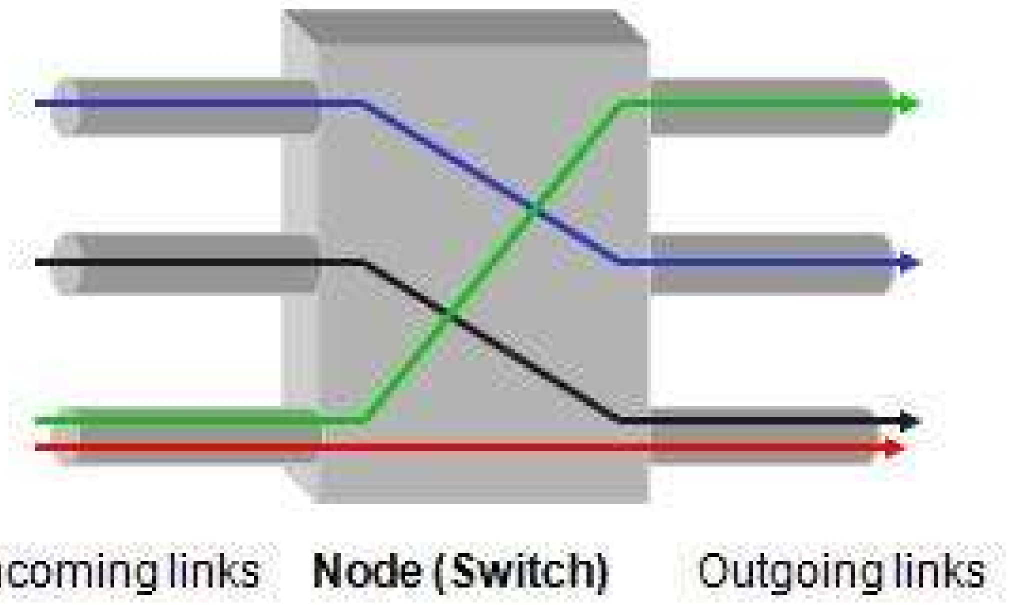
Circuit Switching vs. Packet Switching

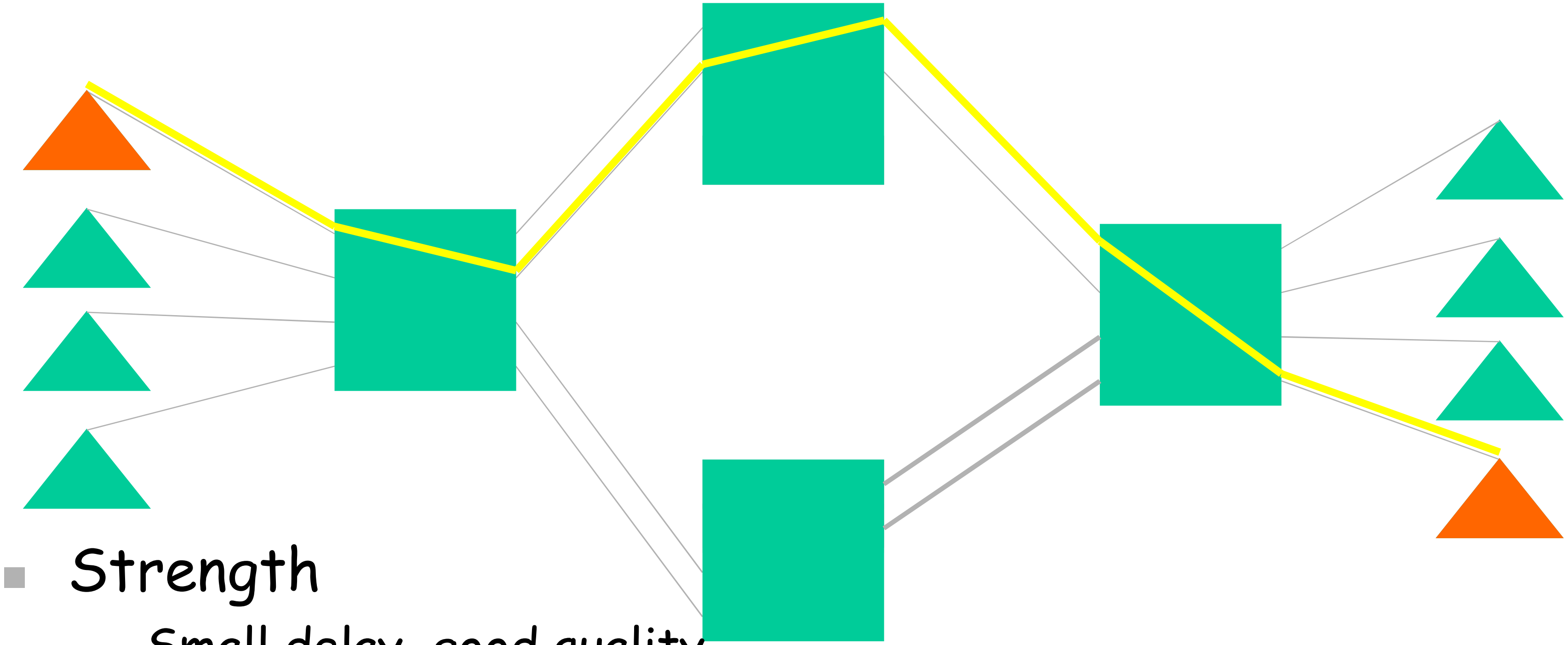


Circuit switching（电路交换）

- Example: Telephone network
- Dedicated communication path between 2 stations
- Resources allocated in advance
- 3 phases: circuit establishment, data transfer and circuit termination



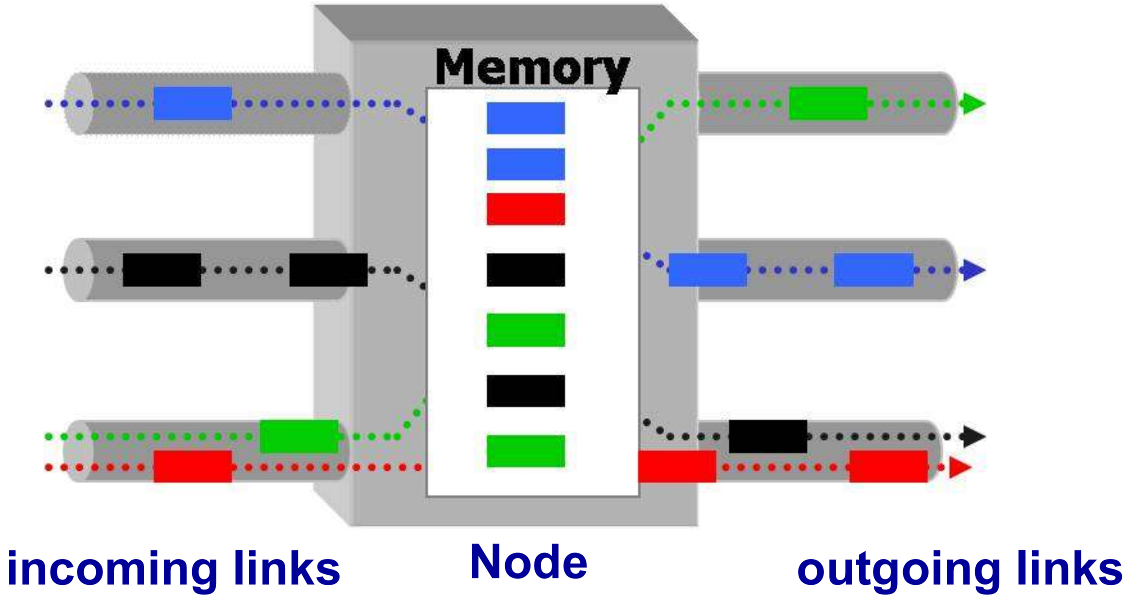
Demo of Circuit Switching



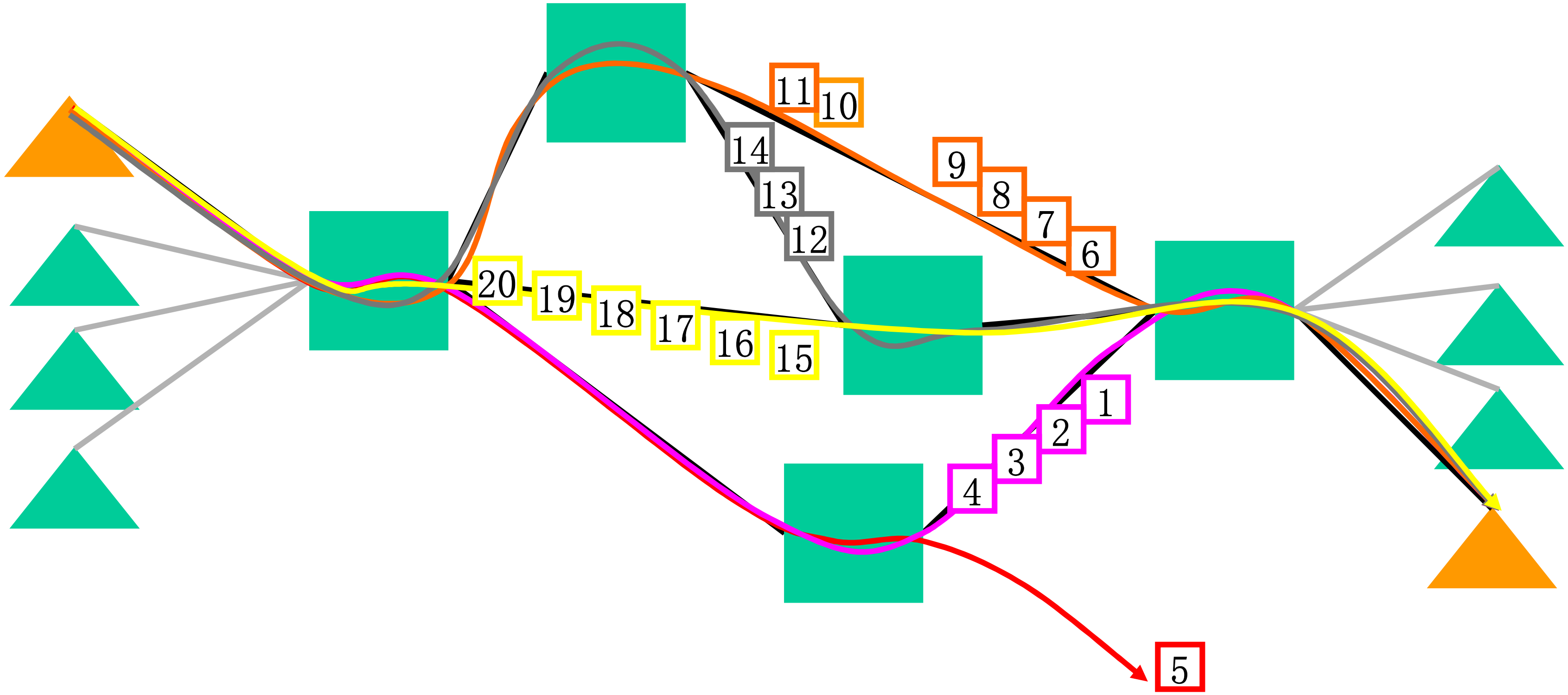
- Strength
  - Small delay, good quality
  - Simple control in switches
- Weakness
  - Time consumed in call set up
  - Fixed bandwidth, not suitable for various rate services
  - Channel waste when both parties keep silent

Packet Switching（分组交换）

- Example：Post office
- Data sent as formatted bit-sequences (Packets)
- Each packet is independently routed
- At each node the entire packet is received, stored briefly, and then forwarded to the next node (Store-and-Forward，存储-转发)
- Dynamic resource allocation



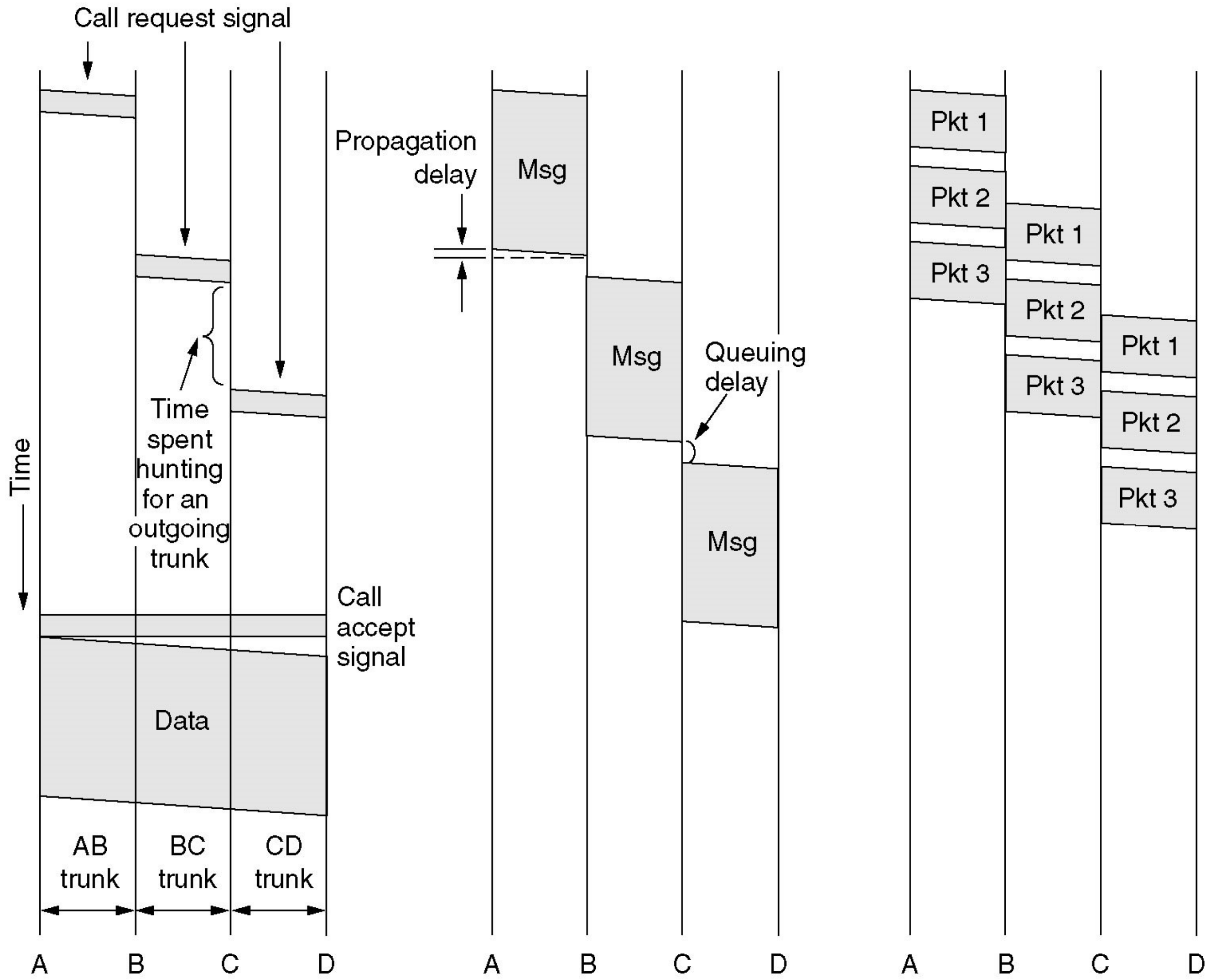
Demo of Packet Switching



Basic Operation of Packet Switching

- Data transmitted in small packets
  - Typically 1000 octets (bytes)
  - Longer messages split into series of packets
  - Each packet contains a portion of user data plus some control information
- Control information
  - Routing (addressing) information

Timing of CS, MS and PS



(a) Circuit switching (b) Message switching (c) Packet switching

报文交换

Circuit switched and Packet-switched: Comparison

Item	Circuit-switched	Packet-switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
When can congestion occur	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Transparency	Yes	No
Charging	Per minute	Per packet

Outline

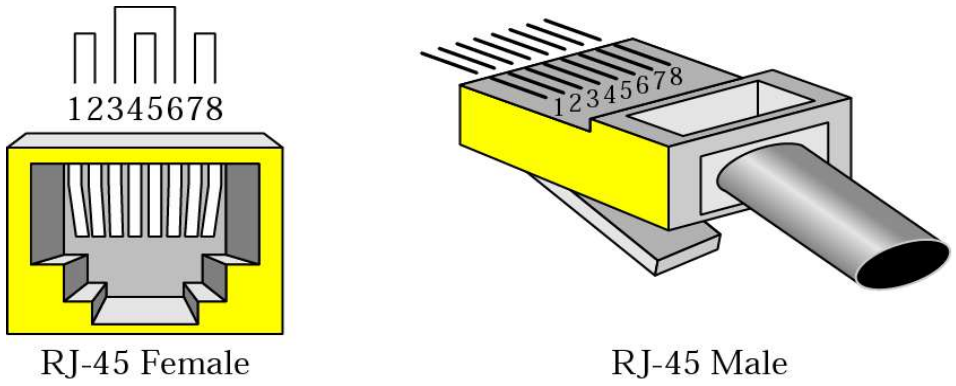
- Position and Functions of Physical Layer
- Basic concepts on data communications
- Theoretical basis for data communication
- Transmission medium
- Digital modulation and encoding
- Multiplexing
- Physical network: PSTN
- Physical network: Cable TV system
- Circuit switching vs. packet switching
- *Physical layer protocol: 10BaseT*

Physical Interface Features

- Mechanical features(机械特性):指明接口所用接线器的形状和尺寸、引线数目和排列、固定和锁定装置等
- Electrical features(电气特性):指明在接口电缆的各条线上出现的电压的范围
  - Encoding, Data rate, cable length
- Functional features(功能特性):指明某条线上出现的某一电平的电压表示何种意义
- Procedure features(过程特性)：指明对于不同功能的各种可能事件的出现顺序

Physical Interface: Ethernet

- IEEE802.3，10BaseT
  - Data rate: 10Mbps, Twist pair as transmission media
  - Mechanical features
    - RJ-45 Interface
  - Electrical features
    - Manchester Encoding
    - 2.5v，-2.5v
  - Functional features
    - 1 pair for sending(pin 1&2)，1 pair for receiving(pin 3&6)
    - Full duplex communication



Key Points of Physical Layer

- Concepts
  - Data vs. Signal
  - Analog vs. Digital
  - Bandwidth vs. Channel capacity
  - Asynchronous vs. Synchronous
  - Baud rate vs. Bit rate
  - Transmission delay vs. Propagation delay
- Nyquist theorem and Shannon theorem
- Transmission media
- Principles of modulation (QPSK & QAM)
- Line encoding (Manchester)
- Idea of ADSL and PCM
- Multiplexing: FDM, WDM, TDM and CDMA
- Circuit Switching vs. Packet Switching

Terms

缩写	英文全称	中文
	analog signal	模拟信号
	Asynchronous Transmission	异步传输
ADSL	Asymmetric Digital Subscriber Line	非对称数字用户线
	bandwidth	带宽
	base station	基站
	baud	波特
BER	Bit Error Rate	比特误码率
	bit rate	数据率(比特率)
	bluetooth	蓝牙
	broadband	宽带
	cable headend	线缆头端
CATV	Community Antenna Television	社区天线电视(有线电视)
	cellular networks	蜂窝网络
	channel	信道
	channel capacity	信道容量

缩写	英文全称	中文
	circuit	电路
CDMA	Code Division Multiple Access	码分多址
CODEC	Coder-Decoder	编码解码器
	chip sequence	码片序列
CS	Circuit Switching	电路交换
	Coaxial Cable	同轴电缆
	cut-through switching	直通交换
	data link layer	数据链路层
	dial-up	拨号
	digital signal	数字信号
DCE	Data Communications Equipment	数据通信设备
	Differential Manchester	差分曼彻斯特
DMT	Discrete MultiTone	离散多音(调制)
DSL	Digital Subscriber Line	数字用户环线
DSLAM	Digital Subscriber Line Access Multiplexer	数字用户环线接入复用器

缩写	英文全称	中文
DTE	Data Terminal Equipment	数据终端设备
	error detection	差错检测
	Ethernet	以太网
FDM	Frequency Division Multiplexing	频分复用
	fiber	光纤
	frame	帧
FTTH	Fiber to the Home	光纤到户
	full duplex	全双工
GEO	Geosynchronous orbit	地球同步轨道
GPS	Global Positioning System	全球定位系统
GSM	Global System for Mobile communications	全球移动通信系统
	half duplex	半双工
	harmonics	谐波
	HUB	集线器

缩写	英文全称	中文
ISM	Industrial, Scientific, Medical	工业、科学、医药（频段）
ISP	Internet Service Provider	互联网服务提供商
LAN	Local Area Network	局域网
	Line coding	线路编码
	Magnetic Media	磁介质
MAN	Metropolitan Area Network	城域网
	Manchester Encoding	曼彻斯特编码
	Message Switching	报文交换
MODEM	Modulator-Demodulator	调制解调器
	microwave	微波
NFC	Near Field Communication	近场通信
NRZ	Nonreturn-to-Zero	不归零
	Nyquist theorem	奈奎斯特定理

缩写	英文全称	中文
	Packet Switching	分组交换
PAD	Personal Digital Assistant	个人数字助手(平板电脑)
PAN	Personal Area Network	个域网
PCM	Pulse Code Modulation	脉冲编码调制
	periodic signal	周期性信号
	physical layer	物理层
	physical medium	物理媒体/介质
PLC	Power Line Communication	电力线通信
	Port	端口
	parallel transmission	并行传输
PSK	Phase Shift Keying	相移键控
	power-line networks	电力线网络
	propagation delay	传播时延
PSTN	Public Switched Telephone Network	公众交换电话网络

缩写	英文全称	中文
	Radio	无线电
QAM	Quadrature Amplitude Modulation	正交幅度调制
QPSK	Quadrature Phase Shift Keying	正交相移键控
RFID	Radio Frequency Identification	射频识别
	satellite	卫星
	serial transmission	串行传输
	Shannon theorem	香农定理
SMS	Short Message Service	短消息业务
	sensor networks	传感网
	signal	信号
	simplex	单工
	statistical multiplexing	统计复用
STP	Shielded Twisted Pair	屏蔽双绞线
	store-and-forward switching	存储转发交换
	subnet	子网

缩写	英文全称	中文
	switch	交换机
	switched Ethernet	交换式以太网
	Synchronous Transmission	同步传输
TDM	Time Division Multiplexing	时分复用
	Transmission Media	传输媒体(介质)
	transmission delay	发送时延
	transport layer	传输层
UTP	Unshielded Twisted Pair	非屏蔽双绞线
VoIP	Voice over IP	IP话音
VPN	Virtual Private Networks	虚拟专用网
WAN	Wide Area Network	广域网
WCDMA	Wideband Code Division Multiple Access	宽带码分多址接入
WDM	Wave Division Multiplexing	波分复用
WLAN	Wireless Local Area Network	无线局域网

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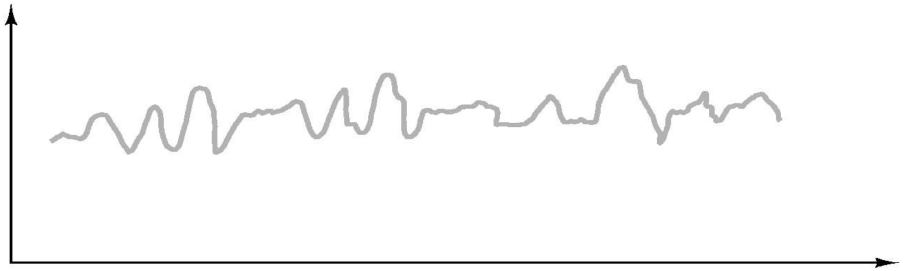
  - “计算机网络”, 谢希仁，第五版，电子工业出版社，2008年，which are marked with *[谢]*

Self-study after class

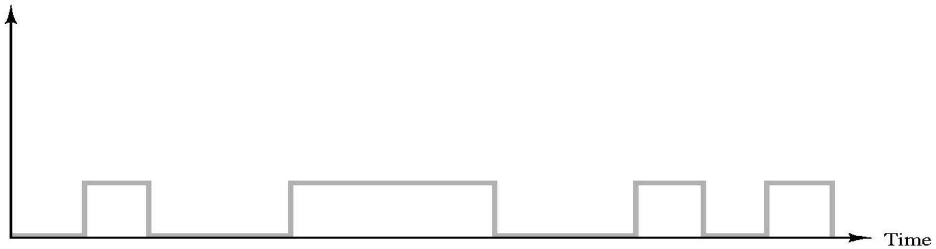
- More for signals
- Fourier Transform to understand bandwidth-limited signals
- Example of Satellites
- More line encoding schemes
- CDMA
- Cable Modem
- Physical layer protocol: RS-232

Signals: Analog vs. Digital

- Signal(信号): 数据的电编码或电磁编码，数据以信号的形式传播。
  - a function s(t) that varies with time (t stands for time)
- Analog: varies continuously（模拟信号）
  - Example: voltage representing audio (analog phone call)

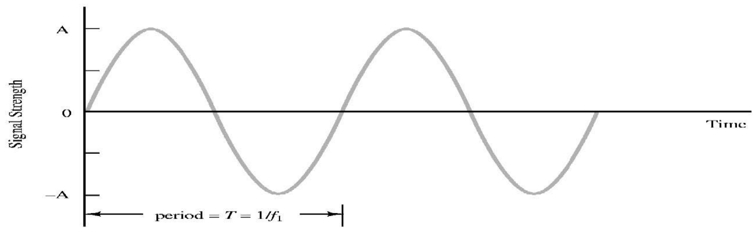


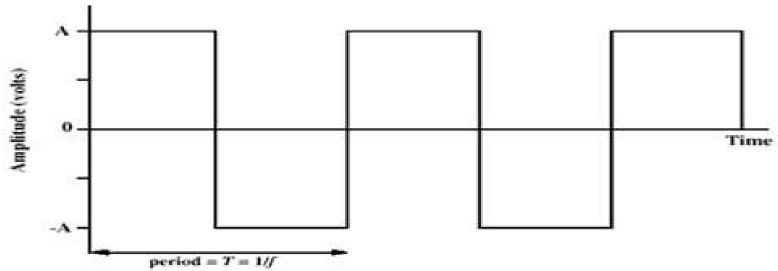
- Digital: discrete values; varies abruptly（数字信号）
  - Example: voltage representing 0s an 1s



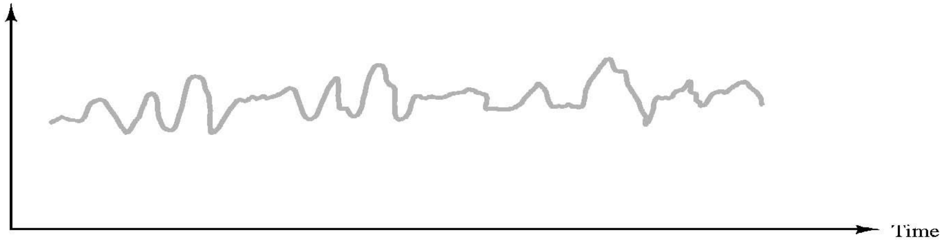
Signals: Periodic vs. Aperiodic

- Periodic（周期性）: repeat over and over again, once per period
  - 周期Period (*T*) is the time it takes to make one complete *cycle*
  - 频率 Frequency (*f*) is the inverse of period, *f = 1/T;* measured in *hertz(hz)*





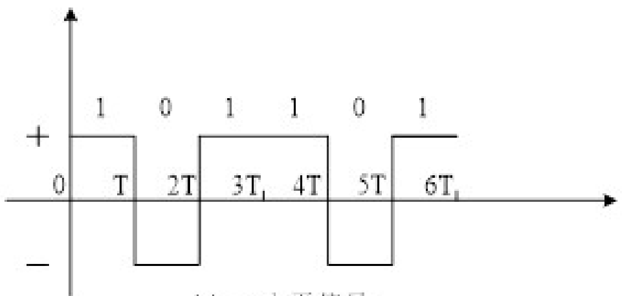
- Aperiodic（非周期性）: don't repeat according to any particular pattern



Bits vs. Signal units

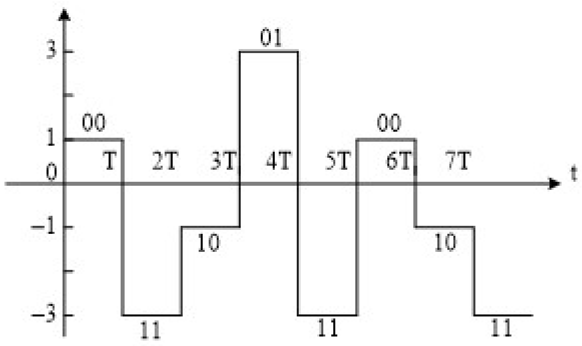
- Signal units(码元)：承载信息的基本信号单元，一个码元能承载的信息量**D**有多少，是由脉冲信号所能表示的有效状态个数**V**决定的。
- Bits (比特)：二进制表示的信息量的基本单位
- 一码元携带的信息量的计算公式:

  $D=\log_2 V$，V表示一个脉冲信号所能表示的有效状态数



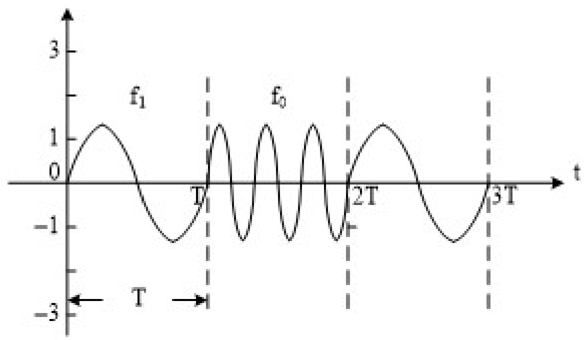
(a) 二电平信号

1码元携带1比特信息



(b) 四电平信号

1码元携带2比特信息



(c) 二状态调频波

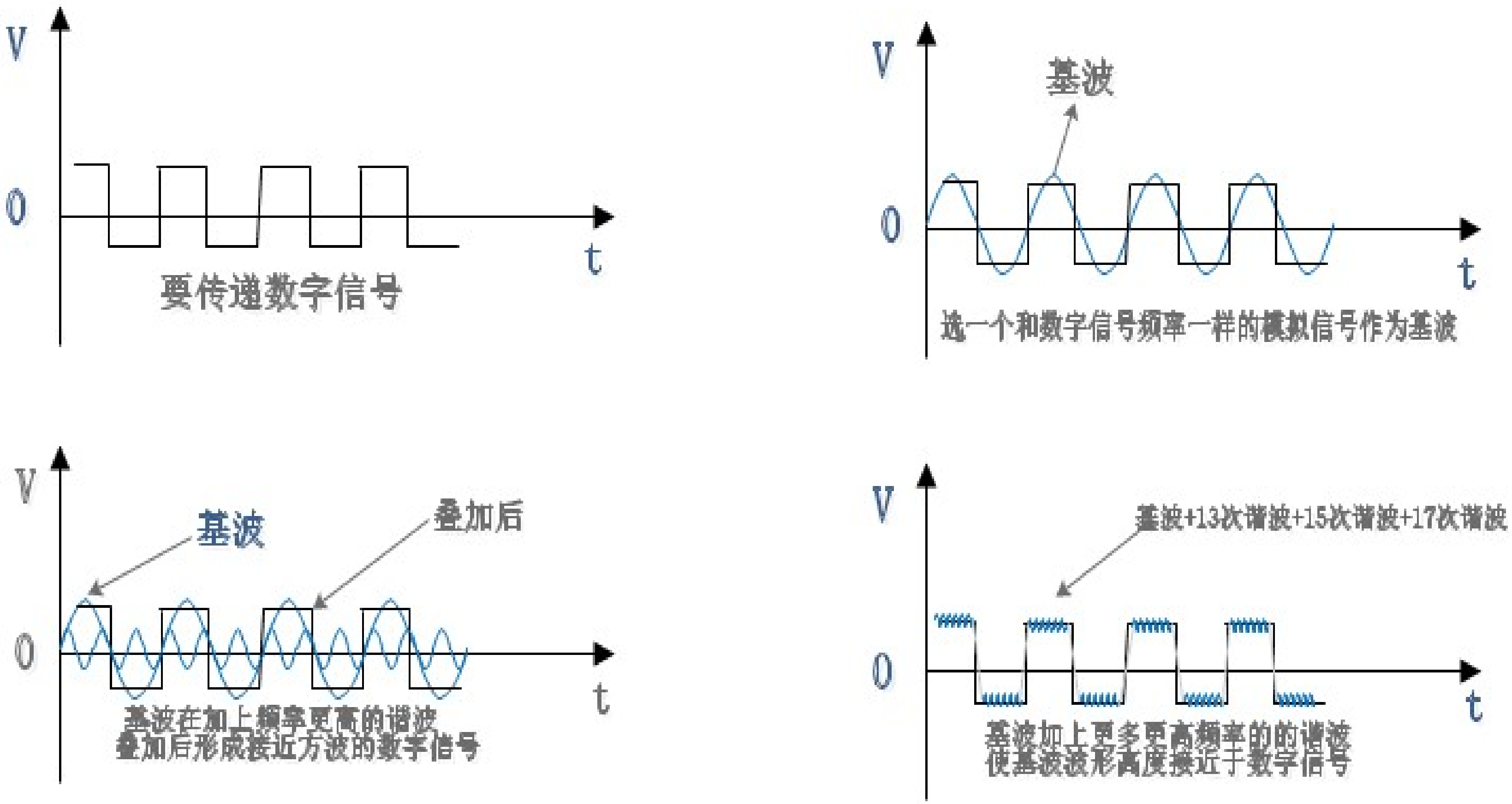
1码元携带1比特信息

推导带宽和最大数据率的关系：Fourier Transform（傅里叶变换）

- Any periodic signal g(t) with period T (=1/f) can be constructed by summing a (possibly infinite) number of sines and cosines

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- To construct signal g(t) we need to compute the values 振幅a_0, a_1, …, b_0, b_1, …, and 常数c
- Energy: proportional to root-mean-square amplitude $\sqrt{a_n^2 + b_n^2}$
- Often the magnitude of the a_n's and b_n's get smaller as the frequency ($2\pi nf$) gets higher (频率越高，能量越小)



数字信号由模拟信号谐波而成

Bandwidth-Limited Signals

- All transmission facilities diminish different Fourier components by different amounts
  - Physical media attenuate (reduce) different harmonics at different amounts
  - Any transmission system has a limited band of frequencies
    - The width of the frequency range transmitted without being strongly attenuated is called the **bandwidth**
    - The bandwidth is **a physical property** of the transmission medium.

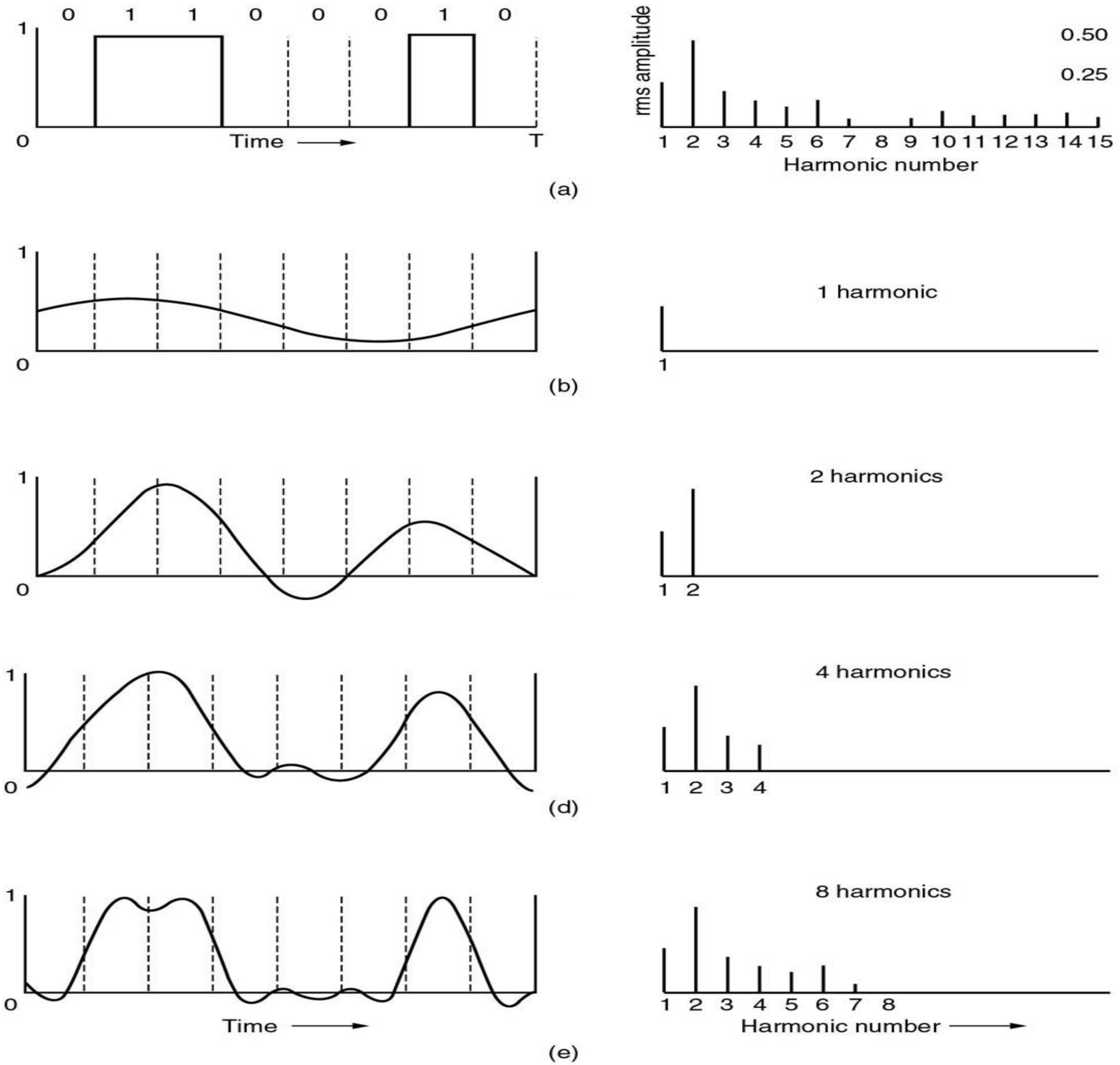
Bandwidth-Limited Signals

- **Key point: a "reasonable reconstruction" can be often be made from just the first few terms (harmonics谐波)** (用几个低次谐波就可以重构信号),
- Tough the more harmonics the better the reconstruction

- Example: Transmission of ASCII character "b" (0110 0010)
- Fourier analysis:

$$a_n = \frac{1}{\pi n}[\cos(\frac{\pi n}{4}) - \cos(\frac{3\pi n}{4}) + \cos(\frac{6\pi n}{4}) - \cos(\frac{7\pi n}{4})]$$

$$b_n = \frac{1}{\pi n}[\sin(\frac{3\pi n}{4}) - \sin(\frac{\pi n}{4}) + \sin(\frac{7\pi n}{4}) - \sin(\frac{6\pi n}{4})]$$



(a) A binary signal ('b') and its root-mean-square Fourier amplitudes

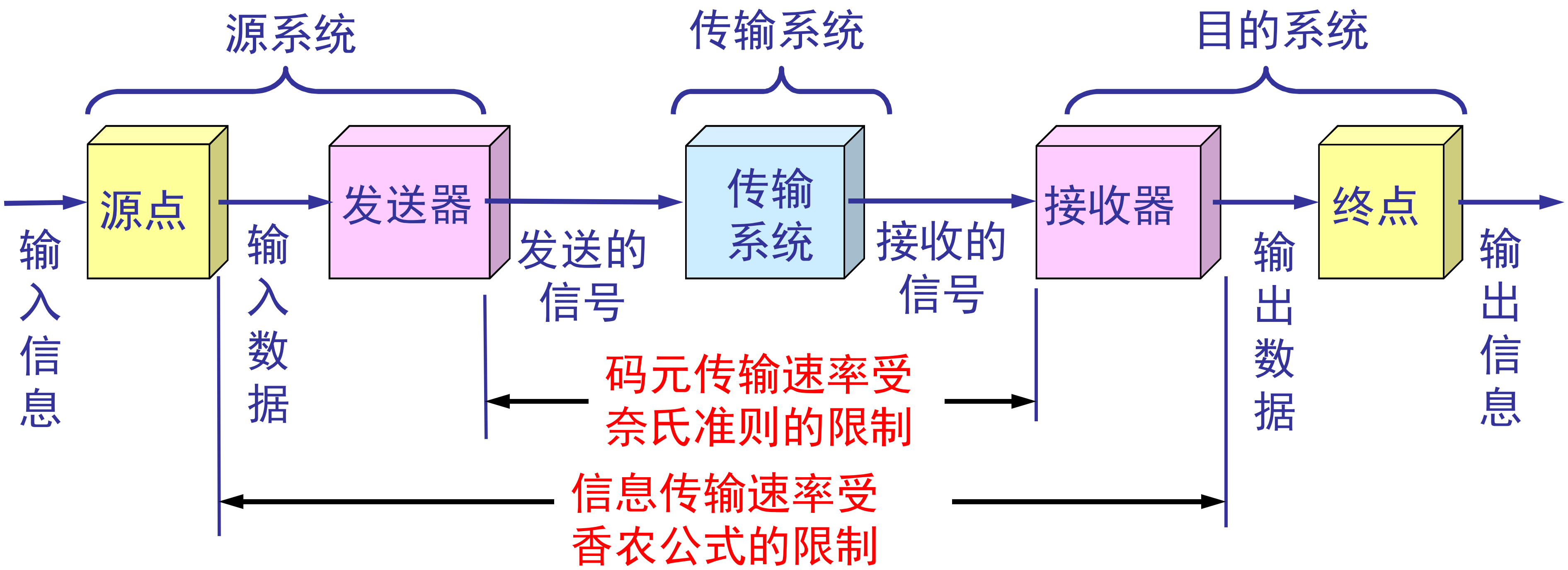
Relation between data rate and harmonics

b	T=8/b	f=1/T=b/8	3000/f
Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

- Voice grade channel: 3000Hz
- Conclusion: Higher data rates require more bandwidth. Limiting bandwidth limits data rate.

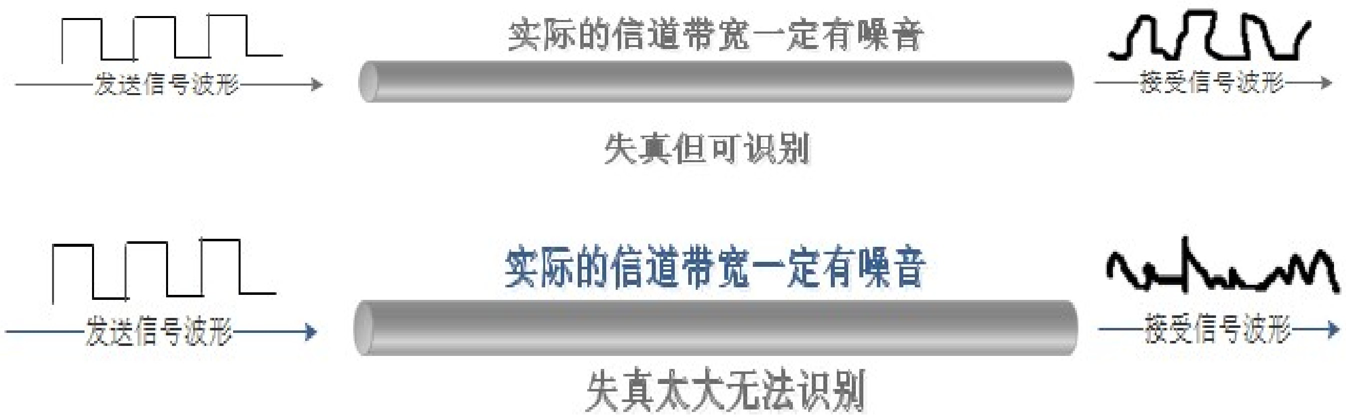
Usage of Shannon limit and Nyquist limit

- The Shannon capacity gives us the upper limit
- The Nyquist formula tells us how many signal levels we need



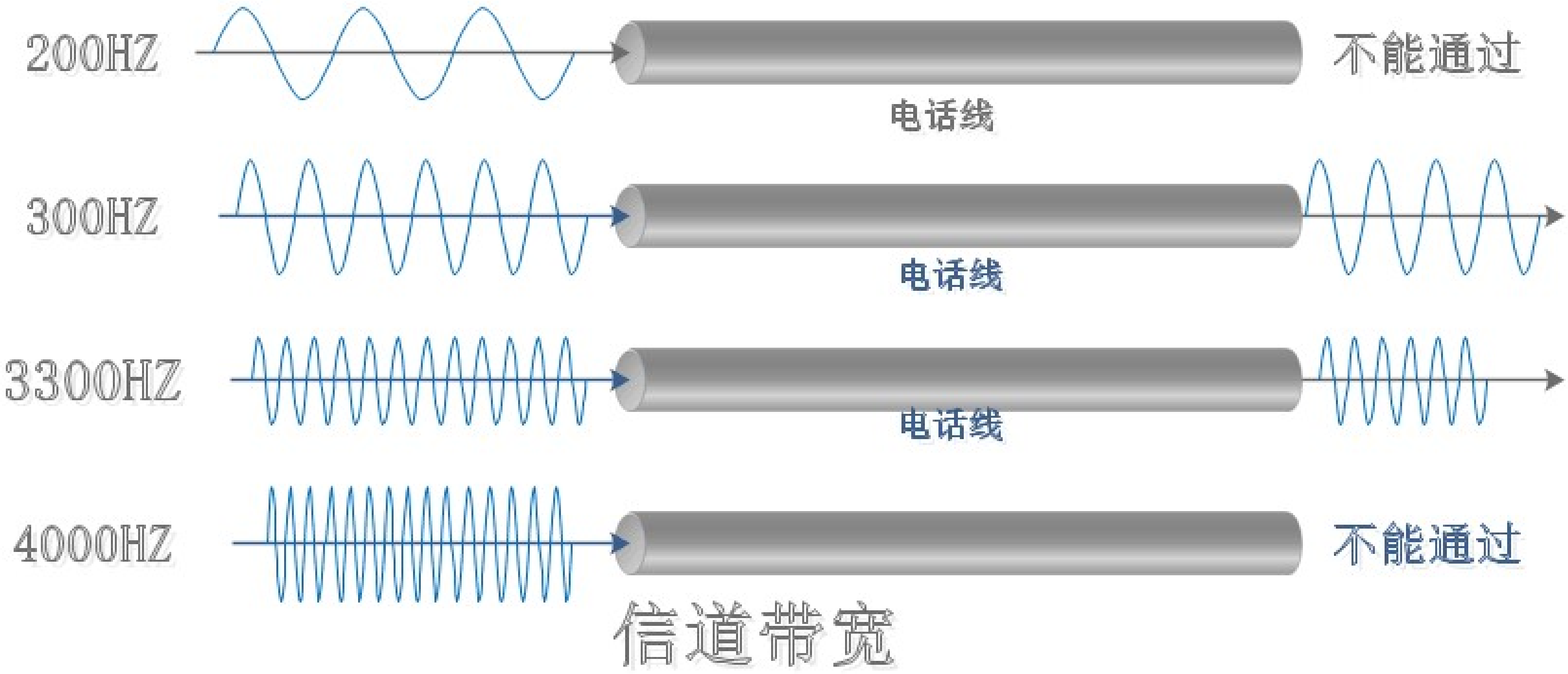
数字信道极限容量

- Digital communication advantages: 接收端只要能够从失真的波形中识别出原来的信号，则该失真对通信质量就没有影响。



- **码元传输速度越高**、信号传输的距离越远、噪声干扰越大、传输媒体质量越差，在信道的接收端，波形的失真就越严重。

信道带宽的概念

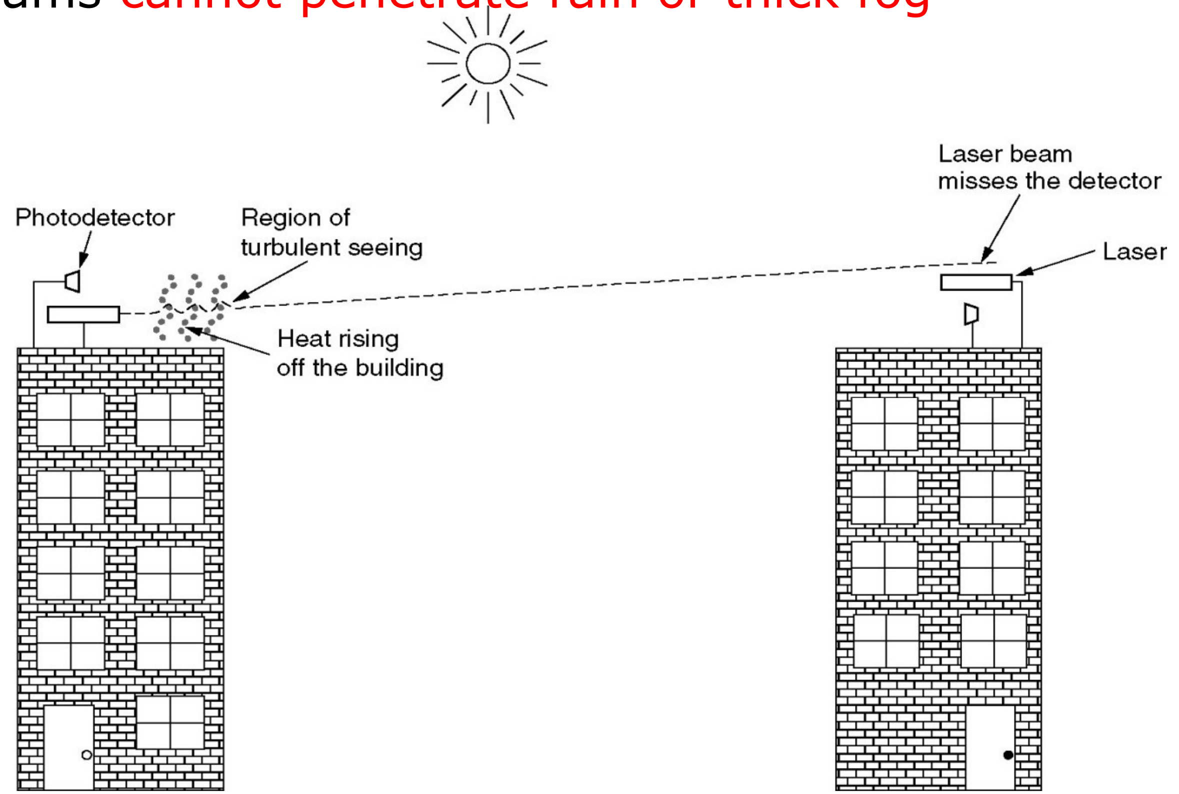

200HZ
不能通过
电话线
300HZ
电话线
3300HZ
电话线
4000HZ
不能通过
信道带宽


数字信号高额分量不能通过信道
接受端收到波形前沿和后沿不陡峭

Wireless communication: Lightwave Transmission

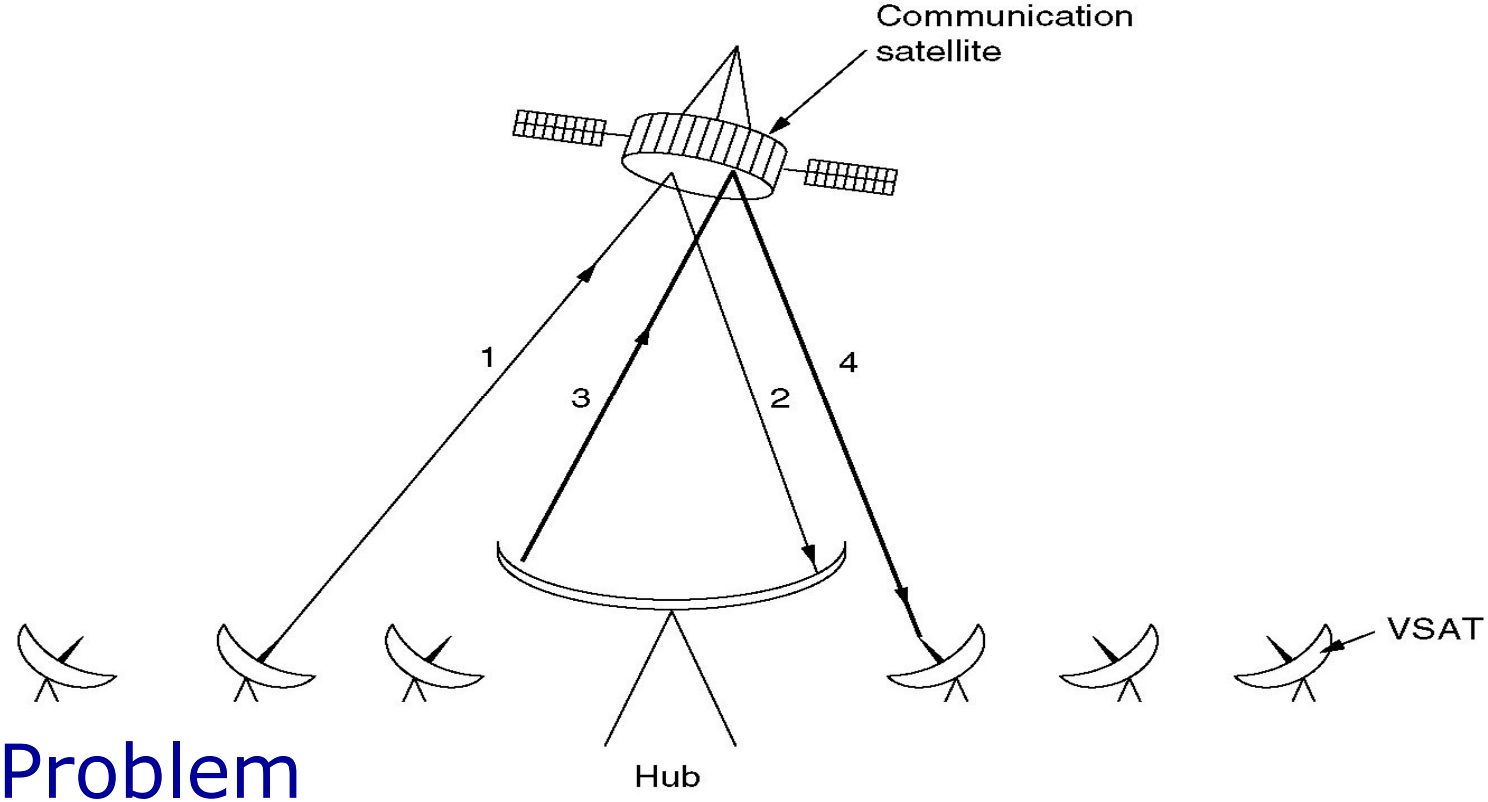
(Unguided optical signaling)

- Very narrow beam：Aiming a laser beam1mm wide at a target the size of a pin head 500m away
- Wind and temperature changes can distort the beam
- Laser beams cannot penetrate rain or thick fog



Convection currents interfere with laser communication systems

Example of Satellites：
GEO: VSATs Using a Hub

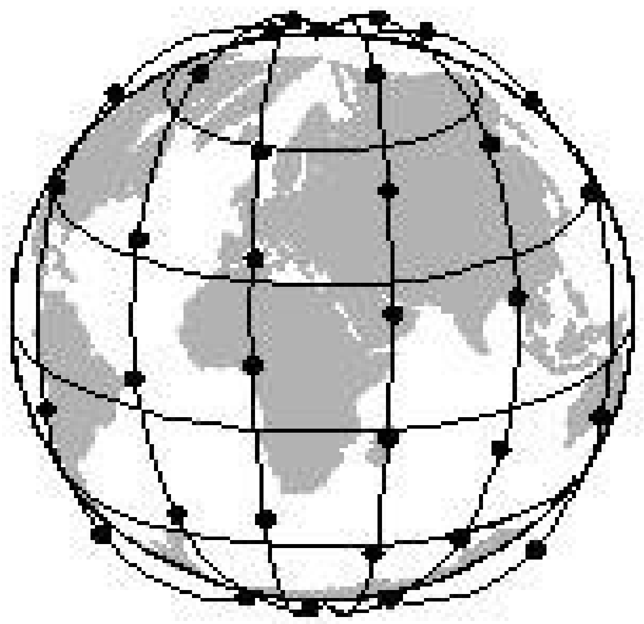
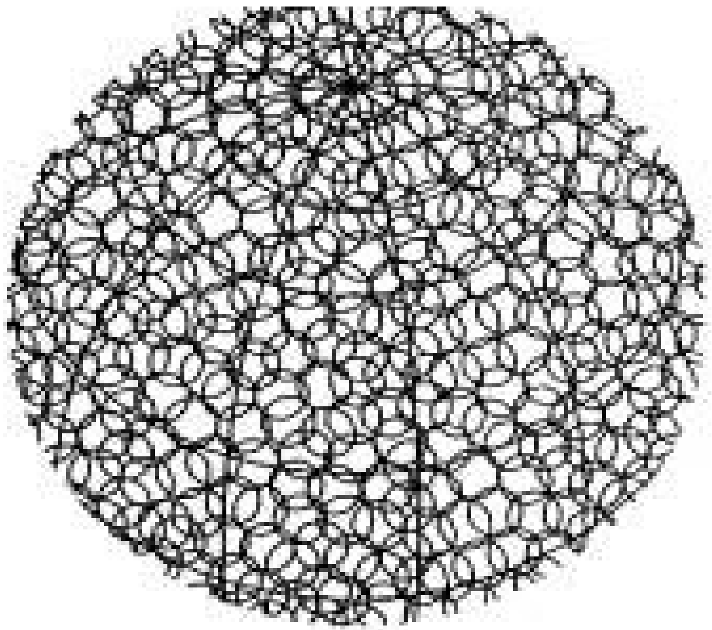


- **Problem**
  - Micro-station do not have enough power to communicate directly with one another via the satellite

LEO: Iridium

- LEO
  - The ground stations do not need much power
  - Round-trip delay is only a few ms
- Iridium
  - Altitude: 750km
  - 6 satellite necklaces, 11x6=66 satellites
  - Each satellite has a max of 48 cells (spot beams)
  - **Relaying in space**： Communication between distant customers takes place **in space.**
- 实施情况
  - 90年策划，97年发射，98年运营，99年破产，2001年重新开始提供服务

Low-Earth Orbit Satellites: Iridium（铱星）

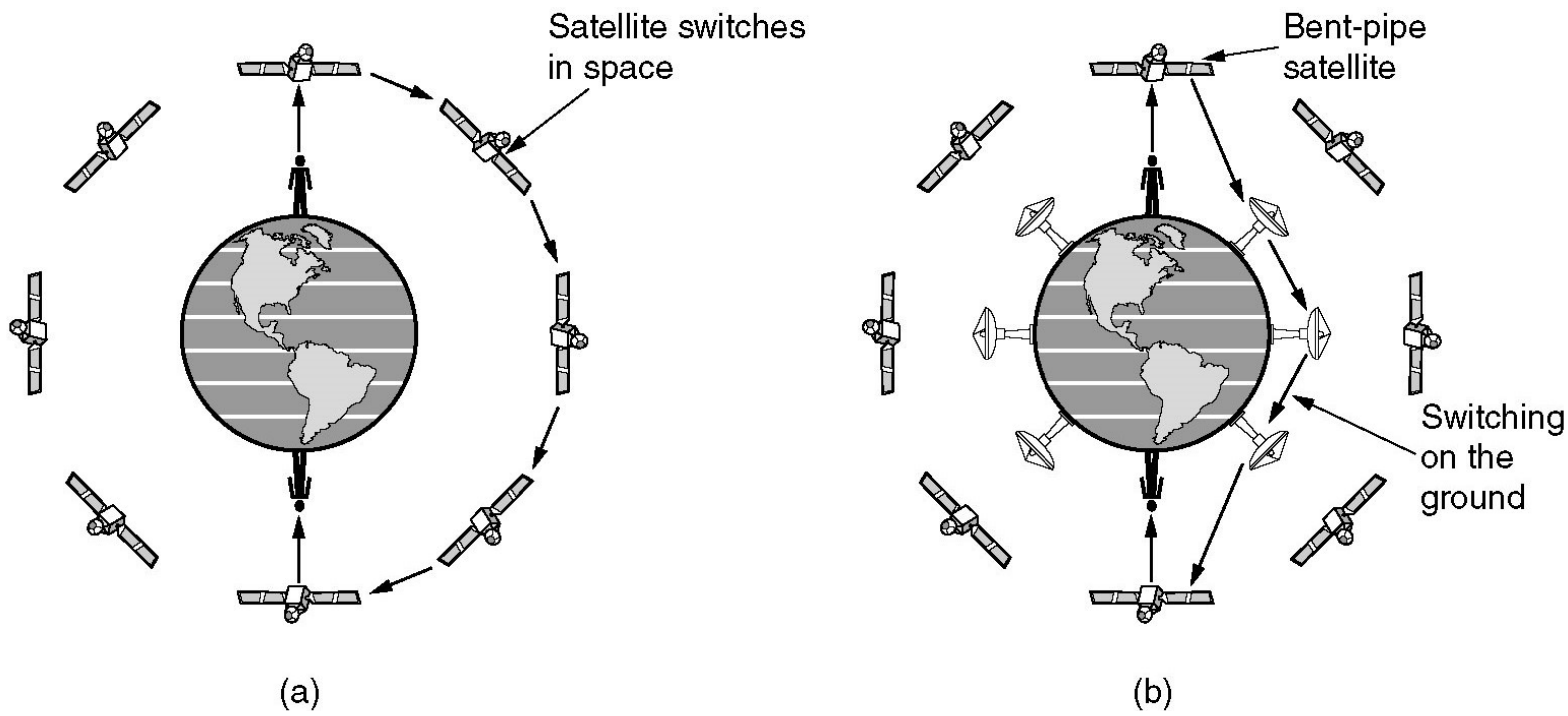
(a) The Iridium satellites form six necklaces around the earth. 6*11=66颗卫星

(b) 3840 channels

LEO: Globalstar

- 48 LEOs satellites
- Relaying on the ground
  - puts much of the complexity on the ground, ease to manage
  - lower-powered telephones can be used (a few milliwatts)

Low-Earth Orbit Satellites: Globalstar （全球星）



(a) Relaying in space.
(b) Relaying on the ground.

Line Encoding: 4B/5B

- Every 4bits is mapped into a 5bits pattern with a fixed translation table
- The five bit pattern are chosen so that there will never be a run of more than three consecutive 0s

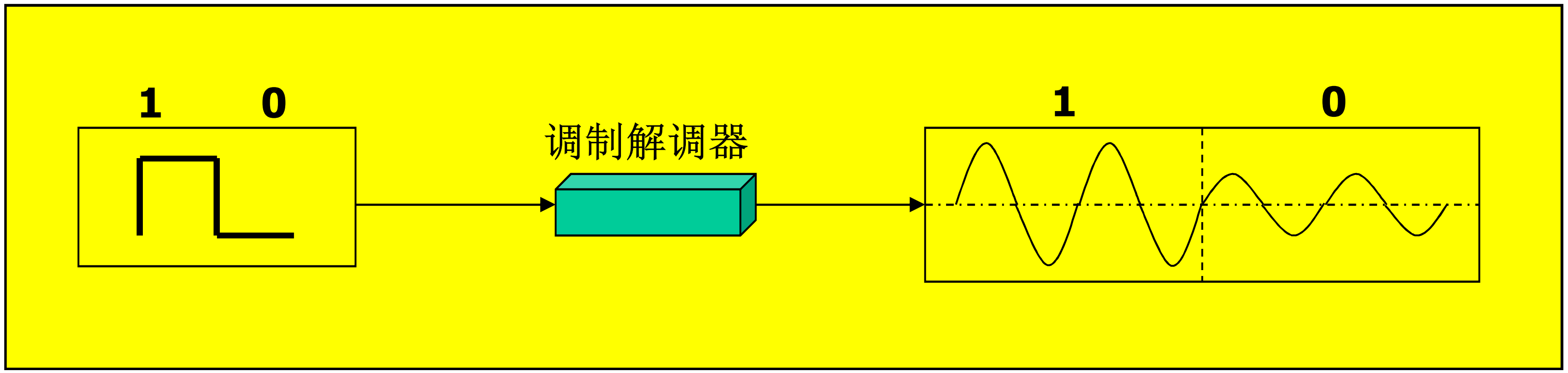
Data(4B)	Codeword(5B)	Data(4B)	Codeword(5B)
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

Line Encoding: 8B/10B

- The 8 bits are split into a group of 5 bits, which is mapped to 6 bits, and a group of 3 bits, which is mapped to 4 bits.
- The 6-bit and 4-bit symbols are concatenated as a group.

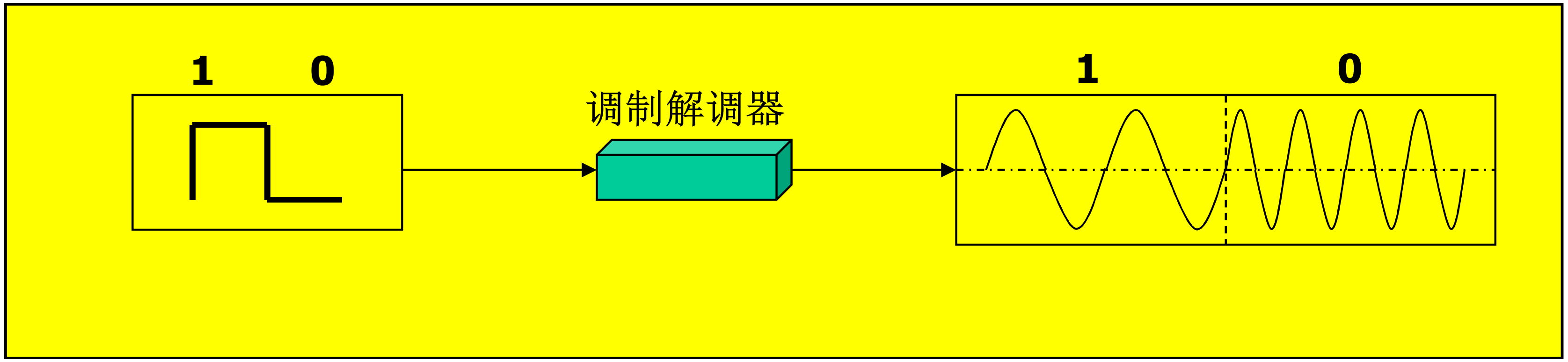
Modulation：ASK

- 2ASK基本原理
  - 正弦载波的幅度随二进制基带信号的变化而变化的数字调制，即源信号为“1”时，发送载波，源信号为“0”时，发送0电平
- 2ASK信号的时域波形



Modulation：FSK

- 2FSK基本原理
  - 正弦载波的频率随二进制基带信号在f1和f2两个频点间变化，产生2FSK信号
- 2FSK信号的时域波形



CDMA: Idea (1)

- Key of CDMA：able to extract the desired signals while rejecting everything else as random noise
- Each station is assigned given a *m-bit* unique code pattern, called chip sequence（码片序列）
  - To transit a 1 bit, a station sends its chip sequence
  - To transit a 0 bit, it sends the complement of its chip sequence
- Bipolar notation is used for convenience
  - Binary 1 and 0 are marked as +1 and -1 respectively

Idea of CDMA(2)

- All chip sequences are pairwise orthogonal （两两正交）

$$S \bullet T \equiv \frac{1}{m}\sum_{i=1}^{m} S_i T_i = 0$$

$$S \bullet S = 1$$

$$S \bullet \overline{S} = -1$$

- When 2 or more stations transmit simultaneously, their bipolar signals add linearly
- The receiver does the recovery by computing the normalized inner product of the received chip sequence $S \bullet C$

What sender and receiver do?

- They share same chip sequence s.
- The sender sends out chip sequences in bipolar way, s for "1", $\bar{s}$ for "0".
- All the chip sequences sent by different stations are added linearly, assuming the compounding sequence is c .

- The receiver computing $s \bullet c$
  - If the result is 1, then the received bit is 1;
  - If the result is -1, then the received bit is 0;
  - If the result is 0, then the sender did not send.

CDMA coding example

A: 0 0 0 1 1 0 1 1	A: (−1 −1 −1 +1 +1 −1 +1 +1)
B: 0 0 1 0 1 1 1 0	B: (−1 −1 +1 −1 +1 +1 +1 −1)
C: 0 1 0 1 1 1 0 0	$\boxed{\text{C: } (-1\ +1\ -1\ +1\ +1\ +1\ -1\ -1)}$
D: 0 1 0 0 0 0 1 0	D: (−1 +1 −1 −1 −1 −1 +1 −1)
Binary chip seq. for 4 stations	Bipolar chip sequences

Six examples:

— — 1 —	**C**	$S_1 = (-1\ +1\ -1\ +1\ +1\ +1\ -1\ -1)$
— 1 1 —	**B + C**	$\boxed{S_2 = (-2\ \ 0\ \ 0\ \ 0\ +2\ +2\ \ 0\ -2)}$
1 0 — —	**A + $\overline{\text{B}}$**	$S_3 = (\ 0\ \ 0\ -2\ +2\ \ 0\ -2\ \ 0\ +2)$
1 0 1 —	**A + $\overline{\text{B}}$ + C**	$S_4 = (-1\ +1\ -3\ +3\ +1\ -1\ -1\ +1)$
1 1 1 1	**A + B + C + D**	$S_5 = (-4\ \ 0\ -2\ \ 0\ +2\ \ 0\ +2\ -2)$
1 1 0 1	**A + B + $\overline{\text{C}}$ + D**	$S_6 = (-2\ -2\ \ 0\ -2\ \ 0\ -2\ +4\ \ 0)$

Six examples of transmissions

$S_1 \bullet C = (1\ +1\ +1\ +1\ +1\ +1\ +1\ +1)/8 = 1$

$\boxed{S_2 \bullet C = (2\ +0\ +0\ +0\ +2\ +2\ +0\ +2)/8 = 1}$

$S_3 \bullet C = (0\ +0\ +2\ +2\ +0\ -2\ +0\ -2)/8 = 0$

$S_4 \bullet C = (1\ +1\ +3\ +3\ +1\ -1\ +1\ -1)/8 = 1$

$S_5 \bullet C = (4\ +0\ +2\ +0\ +2\ +0\ -2\ +2)/8 = 1$

$S_6 \bullet C = (2\ -2\ +0\ -2\ +0\ -2\ -4\ +0)/8 = -1$

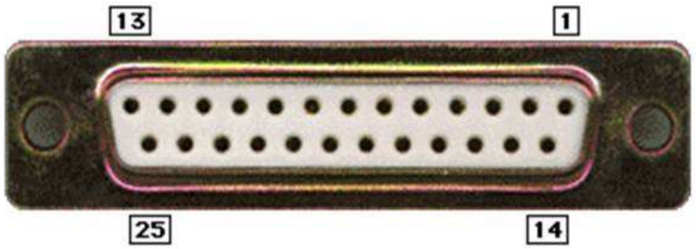
Recovery of station C's signal

Physical Interface: RS-232

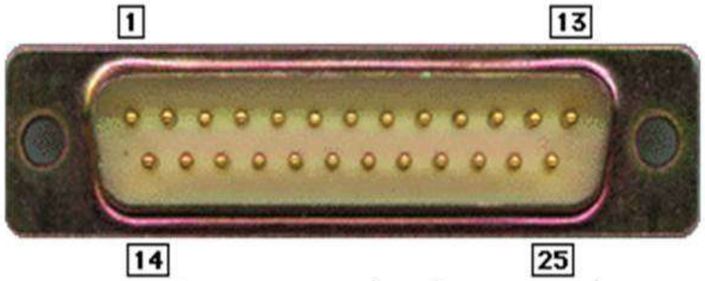
- EIA RS-232 （ITU-T V.24）
  - First issued by EIA in 1969
  - EIA-232-C EIA-232-D EIA-232-E
- Interface standard between DTE and DCE
  - DTE：Data Terminal Equipment, eg. computers
  - DCE：Data Communications Equipment, eg. modems
- Serial Transmission（串行传输）
- Asynchronous Transmission（异步传输）

RS-232 Specifications

- Mechanical features
  - 25-pin
- Electrical features
  - NRZL encoding
    - <-3V = binary 1， >3V = binary 0
  - Data rate: 20Kbps
  - cable length <15 m
- Functional features: pin definition
- Procedure features: protocol



DB-25 Female (DTE side)

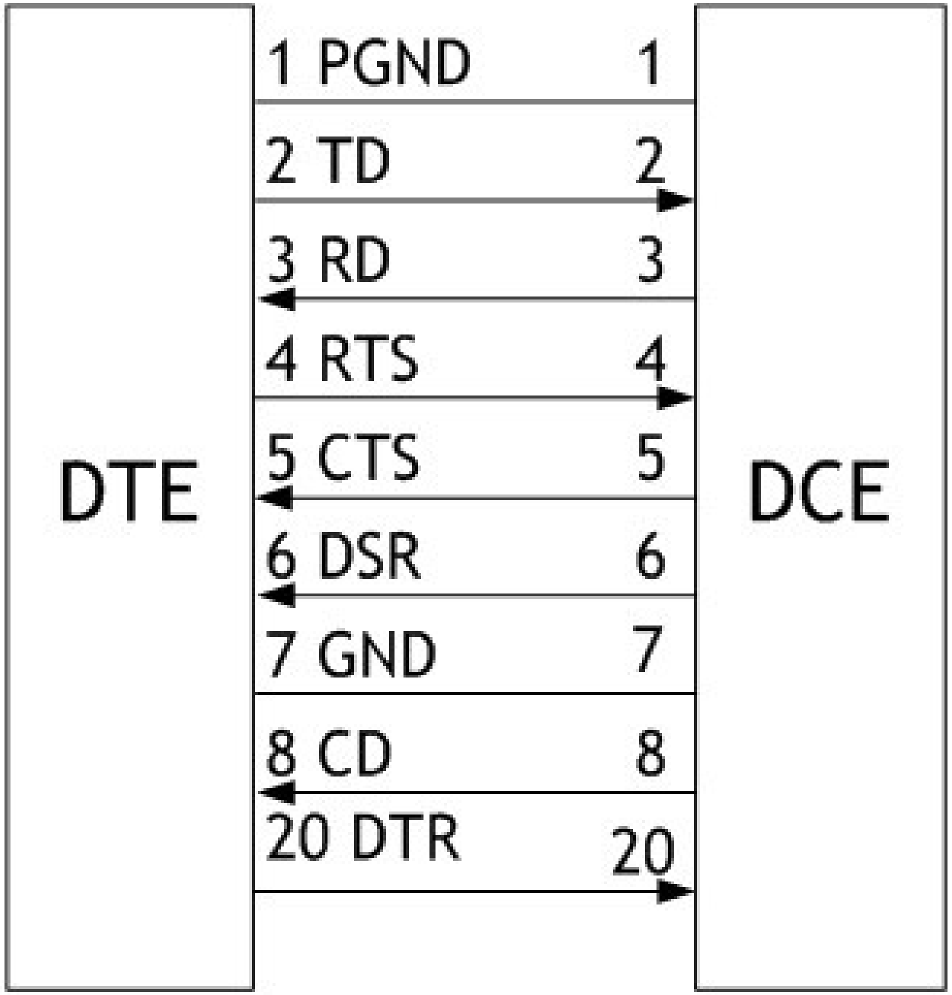


DB-25 Male (DCE side)

RS-232 DB-9 Connectors

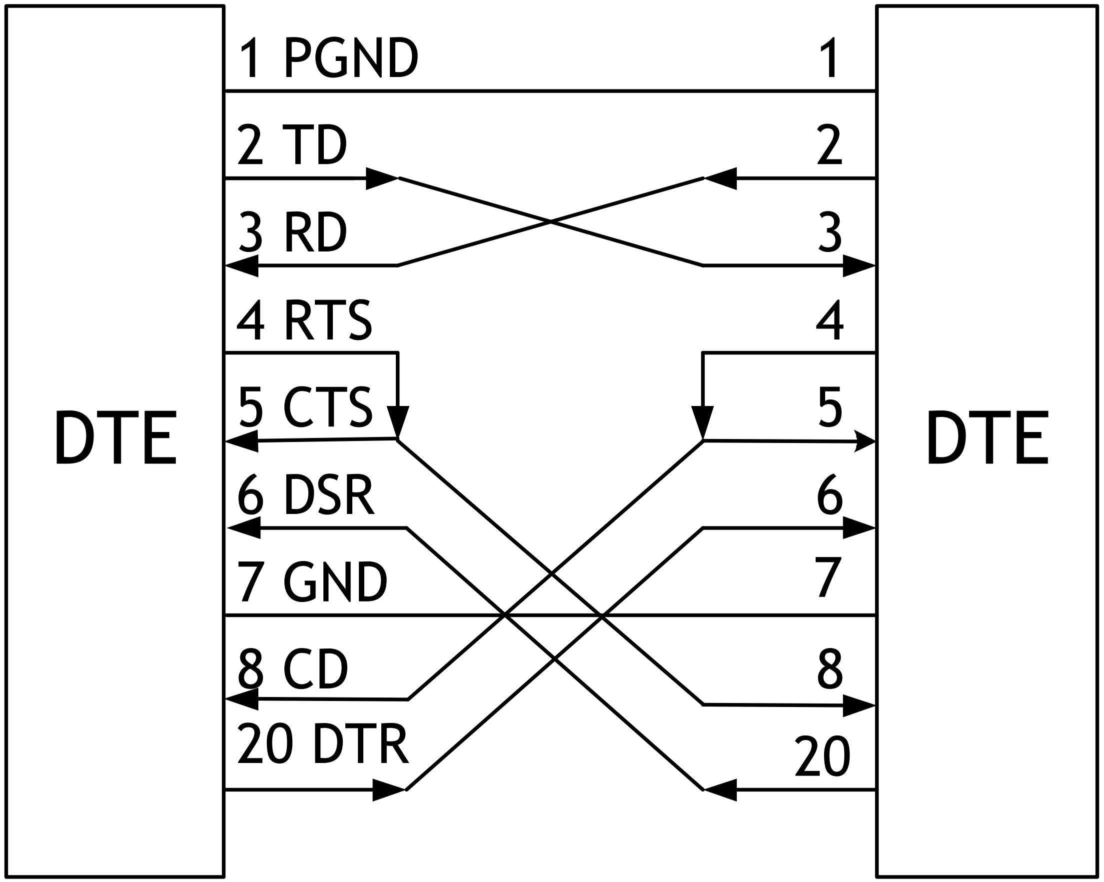
- Limited RS 232
- Used in practice

RS-232：Functional Specifications



- DTE: data terminal equipment
- DCE: data communications equipment
- TD: Transmitted data
- RD: Received data
- RTS: Request to send
- CTS: Clear to send
- DSR: data set ready
- CD: Data Carrier detection
- DTR: Data Terminal Ready

Virtual MODEM



Allows DTE to DTE direct communication

Virtual MOEM(2)